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基於隱藏式馬可夫模型之語音情緒辨識的初始
模型探討

Initial Model Study of Speech Emotion
Recognition Using Hidden Markov Model Based
System

研究生：黃俊修 Student : Jyun-Siou Huang

指導教授：邱瀝毅 Advisor : Lih-Yih Chiou

共同指導教授：雷曉方 Co-Advisor : Sheau-Fang Lei

Department of Electrical Engineering
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研究生：黃俊修

本論文業經審查及口試合格特此證明

論文考試委員：

郭致宏 姚書農

邱瀝毅 雷曉方

指導教授：雷曉方 邱瀝毅

系(所)主管：楊宏偉

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摘要

基於隱藏式馬可夫模型之語音情緒辨識的初始模型探討

黃俊修* 邱瀝毅** 雷曉方***

國立成功大學 電機工程研究所

語音情緒辨識在人機互動領域上是最重要的主題之一，可以應用在聊天機器人、心理檢測、安全提醒等等情境。近幾年，已嘗試過非常多不同的特徵及分類器，例如，音高、共振峰、梅爾倒頻譜係數這些特徵以及支援向量機、高斯混和模型、隱藏式馬可夫模型、類神經網路等等分類器。

語音情緒可視為一段語氣的狀態變化。在本研究中，基於此論點以高斯混和模型、離散隱藏式馬可夫模型、連續隱藏式馬可夫模型作為分類器並選用梅爾倒頻譜係數作為特徵的辨識率結果做出一個完整的比較，並得出使用狀態變化機率的隱藏式馬可夫模型優於使用統計資訊做分類的高斯混和模型，以及使用多維機率密度函數的連續隱藏式馬可夫模型優於使用離散機率的離散隱藏式馬可夫模型。

上述系統存在的下溢位及奇異性問題，也會在本論文討論及提出解決方法。再者，

*作者

*** 共同指導教授

**指導教授

關於隱藏式馬可夫模型中初始模型假設的討論也會在本論文提出。

最後，最高的辨識率分別以高斯混和模型、離散隱藏式馬可夫模型、連續隱藏式馬可夫模型作為分類器的結果分別為，58.07%、65.67%、89.20%。而最高的平均辨識率分別以高斯混和模型、離散隱藏式馬可夫模型、連續隱藏式馬可夫模型作為分類器的結果為，51.12%、53.20%、70.15%。辨識率結果顯示出連續隱藏式馬可夫模型這三者中辨識率最好的分類器。

關鍵字：情緒辨識、梅爾倒頻譜、高斯混和模型、隱藏式馬可夫模型。



Abstract

Initial Model Study of Speech Emotion Recognition Using Hidden Markov Model Based System

Jyun-Siou Huang*, Lih-Yih Chiou**, Sheau-Fang Lei***

Department of Electrical Engineering
National Cheng Kung University, Tainan, Taiwan, R.O.C

Emotion recognition from speech signal is one of the most important topics in human-machine interaction, it is used for the chatting bot, mental examination, safety warning ...etc. For the past few years, there has been tried for several speech features and classifier, e.g. pitch, formant, Mel-frequency Cepstral Coefficient (MFCC) features and Support Vector Machine (SVM), Gaussian Mixture Model (GMM), Hidden Markov Model (HMM), Artificial Neuron Network (ANN) classifiers.

Emotional speech can be regarded as a prosody state flow. In this work, the accuracy results among GMM, discrete HMM and continuous HMM are compared which are used the MFCC as speech features. There are also underflow and singularity problems among the above systems, it will be discussed and overcome. Moreover, the pre-processing of initial model hypothesis for HMM classifier will be discussed in this paper.

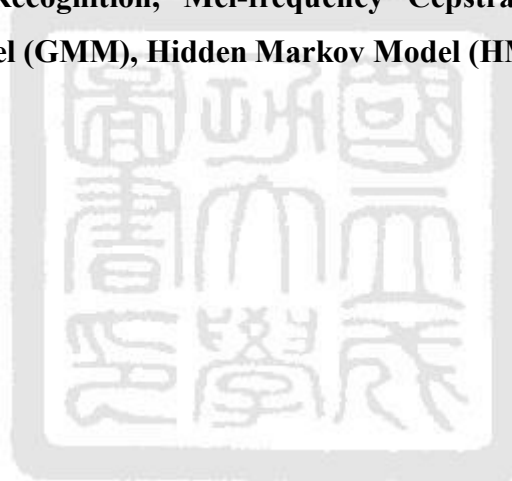
* The author

*** The co-advisor

** The advisor

Finally, the highest accuracy results of recognition are 58.07%, 65.67%, 89.20% for GMM, discrete HMM, continuous HMM classifiers respectively, and the highest average accuracy results of recognition are 51.12%, 53.20%, 70.15% for GMM, discrete HMM, continuous GMM classifiers respectively. The results show that continuous HMM are the best classifier among them. And it supports that the accuracy used HMM considering state flow outperforms GMM which is considering only statistical information. Moreover, it also supports that the continuous HMM which is used multivariant dimension probability density outperforms the discrete HMM used discrete probability.

Keywords: Emotion Recognition, Mel-frequency Cepstral Coefficient (MFCC), Gaussian Mixture Model (GMM), Hidden Markov Model (HMM)



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Chapter 1 INTRODUCTION

This introduction chapter is constructed of following sections: In section 1.1, describe the background of whole speech emotion topics described about the human-machine interaction, the human emotion and general recognition system in speech signal. In section 1.2, give some related works in history. In section 1.3, describe the motivation for this recognition works and why use hidden Markov model-based system. In the end of this chapter, list the organization of this thesis.

1.1 BACKGROUND

To comprehend the area of speech emotion recognition, it suggests the reader understand the topics that are the human-machine interaction (HMI), the human emotion and recognition system at first. Following subsections from 1.1.1 to 1.1.3 describe the topics.

1.1.1 Human-Machine Interaction (HMI)

For the several decades, researches kept trying to connect humans and machines. They intended to make machine understand what human saying, what human thinking and what is the human want and even more to make machine analog to the human's behavior such as the facial expression or speaking manner. We keep trying and studying in order to make human begins living convenient and easier because of the powerful and intellectual machine.

In the recent studies, we use the state-of-the-art artificial neural networks (ANNs) to establish the image recognition system to recognize the human's facial expression, it is intended to understand the human's thinking or human's affection via image information [1] [2]. Use the knowledge of natural language understanding (NLU) [3] and machine

comprehension methods to make machines understanding what human's saying, then we can use machines work via just speaking or natural languages instead of command codes. Another study is the chatting bot, these kinds of researches make machines analog to human's mind and human's thinking. These robots are built for interesting nowadays, but it also can be built for caring kids, lonely elders or someone needed for fun.

Now, back to the topic of emotion in the recent years. Machines can use the human emotion information to do more things in contrast to the simple ordering or chatting in semantic. We can build a machine that they can warn people's mental health, watch the people's safety in life or even chat with people contained more affections. Although there are several physiological signals can be detected or recognized the human's emotional states including facial express or electroencephalography (EEG) signals, it has indicated that some emotional states can only be recognized from speech signal [4]. Except for this reason, emotion recognition from speech has more advantages than using other physiological signals. Recognizing emotion from speech is more convenient and faster due to the communication environment. Human users can interact with machines by only speech information and convey the word meanings and emotion information simultaneously. There is no need to transfer image information even other bioinformation like EEG at the same time. Emotion recognition is more adapted for practical environment while using speech information.

1.1.2 Human Speech Emotions

The human emotion is the area of pure psychology and hardly labelled. But for more than a century studies, Darwin has indicated that a group of basic emotions, happy, sadness, angry, fear, disgust, surprise, and we call these primary emotions [5] [6]. Moreover, the more flexible method uses two dimensions circumplex model to label the human emotions which are constructed by activation as vertical axis and valence as horizontal axis as Fig. 1.1 [7]

[8].

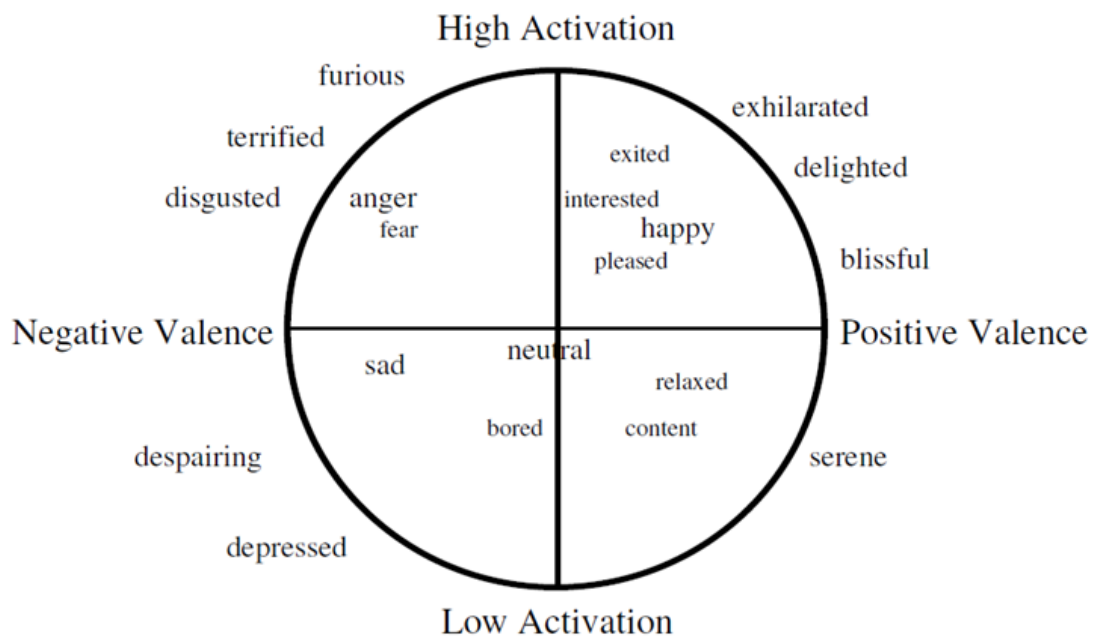


Figure 1.1 The two dimensions circumplex model of human emotions.

Unlike facial expression, recognizing emotion from speech is very different from via image. People speak the emotional utterance via the speech prosody, vocal tract variation and the energy variation that also called features in recognition terminology.

1.1.3 Recognition Systems

The recognition system is built using the pattern recognition technology that is to train a model θ makes $P(\mathbf{y}|\theta)$ optimal, where $\mathbf{y}$ is the pattern or observation vectors in recognition terminology. Every observation vector is set as features extracted from target utterance. The system is usually divided into two stages: feature extraction stage as front-end and classification stage as back-end.

For speech emotion recognition, there are many kinds of features are selected, features are generally separated into local features and global features [9]. The features extracted

from frame, the small segment of speech, are known as local feature. Otherwise, global features are the statistical calculations from all the frame in an utterance. Moreover, it has been claimed that global features can only distinguish the high-arousal, e.g. angry, fearful, happy, versus low-arousal emotional states, e.g. sadness. Therefore, using the local features is preferred due to considering temporary information [10].

Speech features can also be separated into prosodic type features, e.g. pitch, formant, voice quality and spectral or cepstral type features, e.g. linear predictive cepstral coefficient (LPCC), MFCC, log frequency power coefficient (LFPC) [10], sub-band based cepstral parameter (SBC) [11]. Although some researchers claim that prosodic type features are preferred because these prosodic features are easier extracted and carry large amount of emotional information [12]. However, there are several researchers work on cepstral type features [10] [11] [13] and had very effective results due to the sufficient information in cepstral coefficient.

The classifiers which are gaussian mixture model (GMM), support vector machine (SVM), discrete hidden Markov model (discrete HMM), continuous hidden Markov model (continuous HMM) are very regularly used in speech emotion recognition. GMM and SVM are proper used for the global features [12] [14], however, HMM is very suitable for the temporary local features. For the discrete HMM, the feature vector which is constituted of the selected features is encoded to discrete index by vector quantization method. Otherwise, continuous HMM is an extended version of discrete HMM which is usually combined with gaussian mixture distribution. There is no need for quantization in continuous HMM, so it has lesser distortion than discrete HMM, and uses gaussian mixture distribution to express the emission probability.

1.2 Related Works

In the past several decades, researches have tried out many different features with different classifiers. [15] uses several prosodic type global features including pitch, energy which are contented several their statistical parameters, and chooses GMM and continuous HMM as classifiers. They claimed using these type of features, global feature outperforms temporal feature. [16] uses 39 MFCCs including their first and second derivation as features and choose both continuous HMM, discrete HMM and SVM as classifiers and they claim continuous HMM are the best of above classifiers. In more detail of this work, they also try out different type HMM topologies such as left-to-right, backward, ergodic network, and they consider ergodic topology is not necessarily suitable for speech emotion recognized. [10] uses other spectral type feature which is LFPC with discrete HMM classifier, and they claim LFPC outperforms LPCC and MFCC. [11] considers the noise concept of speech data, they use the SBC extracted by wavelet transform and it outperform MFCC.

1.3 Motivation

Because the speech emotion signal is the state flow of the prosodic changing or spectral variation, it considers that the HMM is the most suitable classifier of all. The reason is that using global statistical features is just considered the observations in single statistical manner. However, speech emotion recognizing is not just the global statistical analysis in a single, it should also consider the state flow as a doubly stochastic analysis.

Take speech recognition as example, it uses the state of HMM to represent the phoneme of each word. And the state flow is represented as a word which is constructed by different combination of phonemes. In a similar concept, we use state of HMM to represent the different segment of an emotional utterance. However, unlike the speech recognition, the

state situation in an emotional utterance is implicit, i.e. it hardly knows what the exact meaning of the state in an emotional utterance and also hardly knows how many different states in an emotional utterance. So, it is necessary to find a way to hypothesize what kinds observation vectors in a state.

Otherwise, although the HMM classifier is better than the classifiers used global features, there are two important issues: underflow and singularity problems. There is a concise method to overcome underflow problem in discrete HMM by introducing the scaling coefficient [17], but the formulas in continuous HMM must be modified due to this scaling coefficient, it will be discussed in chapter 3. As for the singularity problem caused due to the intrinsic property of gaussian probability density, it is very hard to solved. To actually overcome this problem, it recommends the deep belief network (DBN) technique [18] because of the nonlinear property of neural networks. In this thesis, we use a simple way to overcome singularity problem, the system performs well even though the singularity problem is not completely solved.

Moreover, although the continuous HMM is the best classifier of above classifiers, it is hardly trained. The initial component parameters setting is one of reasons, bad initial settings causes zero probability, i.e. $P(\mathbf{y}_t|\theta) = 0$, causes singularity problem easier. It should add a constraint of component parameters to limit it.

1.4 Thesis Organization

The remainder of this thesis is organized as follows: In chapter 2, review all the algorithm of feature extraction, the classifiers we used and the details of underflow problem and singularity problem. In chapter 3, describe the proposed methodology, it starts from database to the whole proposed system and the estimation method. In chapter 4, show the

comparison results of all experiments, it includes the recognition accuracy comparison of different constraint of GMM components, different way for the initial state hypothesis, and the system with different classifiers. In the final chapter five, conclude and discuss the future works for the proposed system and the future directions for speech emotion recognition



Chapter 2 REVIEW

Before the proposed system, we should review the related feature extraction and classifiers that we used. This review chapter is constructed of following sections: In section 2.1, describe the feature extraction methodology, and the MFCCs that we used as feature in detail. In section 2.2, describe all the classifiers we used including GMM, discrete HMM, continuous HMM. In section 2.3, present the important issues, underflow and singularity problems in detail.

2.1 Feature Extraction

The most important thing of the pattern recognition is selection of which feature type. It directly affects the final accuracy results because the feature is what the machine observes and what the samples process. Although there are many types of speech emotion features that are selected such as prosodic features and spectral or cepstral features. LFPC is selected and indicated better than LPCC, MFCC [10]. SBC is selected and claimed better than MFCC due to the less sensitivity to noise [11]. Also, the combination of two types of features is also used which the feature is consisted of pitch, formant, zero crossing rate (ZCR), log energy and MFCC [19]. However, in this thesis, we focus on the comparison among GMM, discrete HMM and continuous HMM, so the feature is selected as the most widely used 13 number of MFCCs and ignored the redundant first and second derivation coefficients.

2.1.1 Cochlea Model

The human auditory perception is sensed by the outer ear, middle ear and inner ear. The sound conveys to ear drum and transform the sound into frequency domain. Then this frequency signal conveys to cochlea in the inner ear as Fig. 2.1 shown [20].

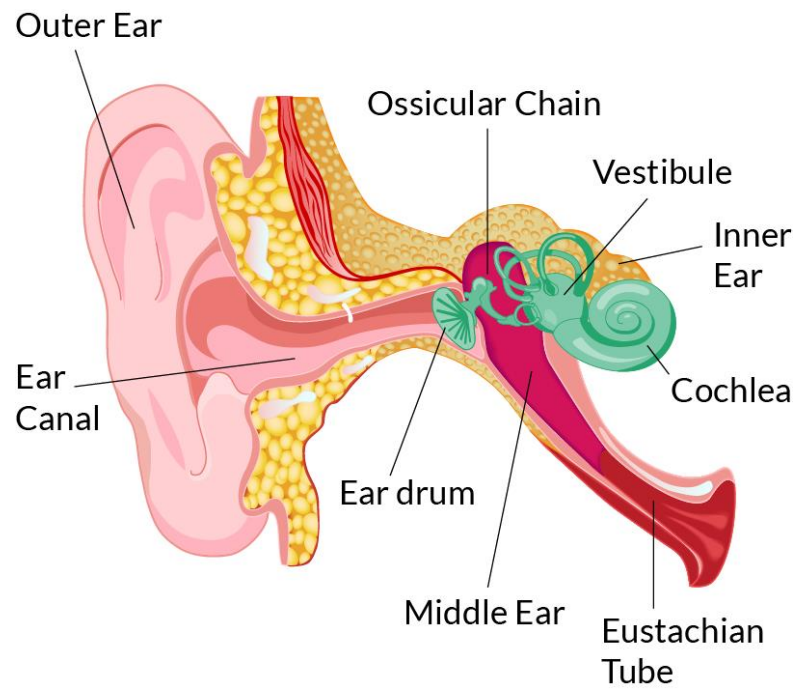


Figure 2.1 The whole human ear structure.

The main function of cochlea is the sound frequency analysis and the sound pressure level steering. The different frequency segments are shown as Fig. 2.2 [21]. As the figure illustration, the frequency segments are distributed at different length segments and the electrode senses the different frequency bands at different segments too. The numerical labels represent the critical band to different segments, and it is generally separated as 24 frequency segments. In engineering speaking, the cochlea can be seen as a 24 number of nonlinear filters with different critical bands. This property of cochlea leads to Mel-scale [22].

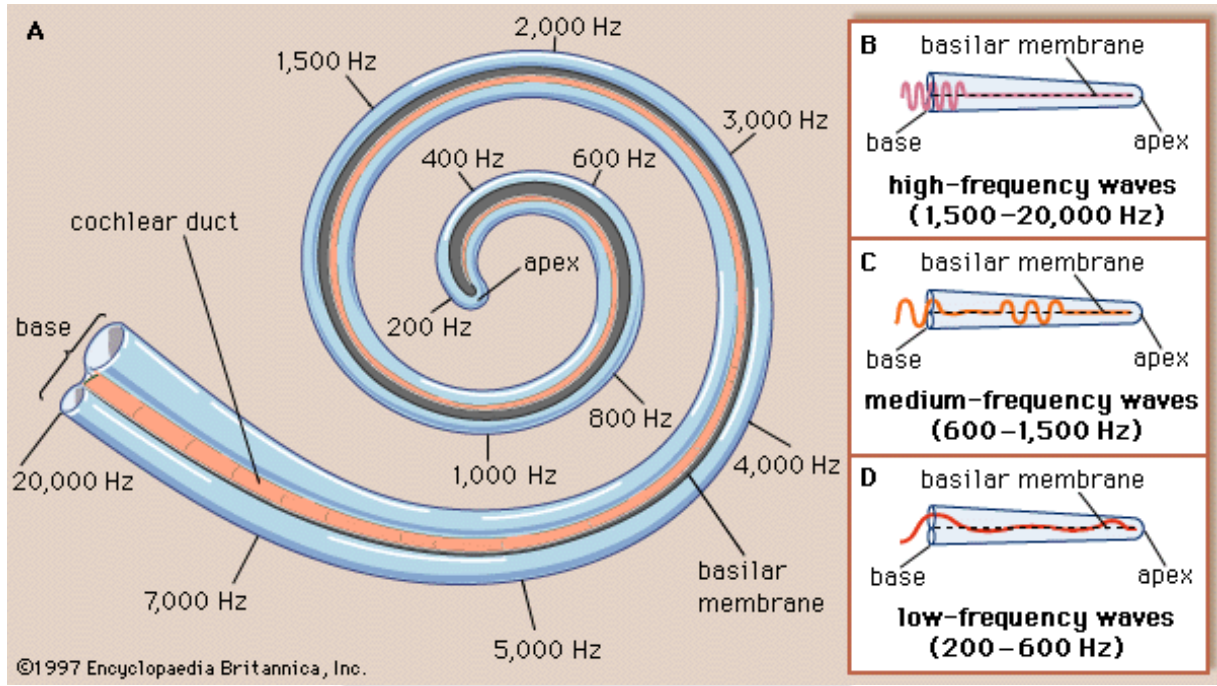


Figure 2.2 The electrode distribution in the cochlea.

2.1.2 Mel-frequency Cepstral Coefficient

According to psychophysical studies, human perception of the frequency of sound follows a nonlinear scale similar to cochlea which is called Mel-scale [22]. The Mel-scale is defined as,

$$f_{mel} = 2595 \log \left(1 + \frac{f}{700} \right) \quad (2.1)$$

where f_{mel} is the Mel-scale frequency corresponding to actual frequency f in hertz. Then, this Mel-scale conversion leads to the definition of MFCC which is calculated as follows.

Let $\tilde{y}[n]$ represent frame of the speech sample which is hamming windowed. At first, the $\tilde{y}[n]$ is converted to frequency domain by DFT and leads to spectrum, that is,

$$|Y[k]|^2 = \left| \sum_{n=1}^N \tilde{y}[n] W_N^{kn} \right|^2 \quad (2.2)$$

where $1 \leq k \leq N$, $W_N^{kn} = e^{-j\frac{2\pi k}{N}n}$. This spectrum is followed by the construction of a

triangular filter bank with Q numbers unity filters in Mel-scale frequency. The i-th filter

frequency response $\varphi_i[k]$ in the bank is defined as,

$$\varphi_i[k] = \begin{cases} 0, & \text{for } k \leq k_{b_{i-1}} \\ \frac{k - k_{b_{i-1}}}{k_{b_i} - k_{b_{i-1}}}, & \text{for } k_{b_{i-1}} \leq k \leq k_{b_i} \\ \frac{k_{b_{i+1}} - k}{k_{b_{i+1}} - k_{b_i}}, & \text{for } k_{b_i} \leq k \leq k_{b_{i+1}} \\ 0, & \text{for } k \geq k_{b_{i+1}} \end{cases} \quad (2.3)$$

where $1 \leq i \leq Q$, Q is the number of filters in the filter bank, k_{b_i} are the boundary points of filters and k denotes the index points in the filter $\varphi_i[k]$. The boundary points k_{b_i} are equally spaced in the Mel-scale and defined as,

$$k_{b_i} = \frac{N}{f_s} \cdot f_{mel}^{-1} \left\{ f_{mel}(f_{low}) + \frac{i[f_{mel}(f_{high}) - f_{mel}(f_{low})]}{Q+1} \right\} \quad (2.4)$$

where f_s is the sampling rate, f_{low} and f_{high} are the low and high frequency boundaries of the filterbank. The f_{low} , f_{high} and the sampling rate f_s are in Hertz, f_{mel} is in Mels. The Fig. 2.3 shows the frequency response of filter bank with 20 filters, 48kHz sampling rate, 24kHz upper frequency. Table 2.1 shows all the center frequencies of the 20 filters in Hertz.

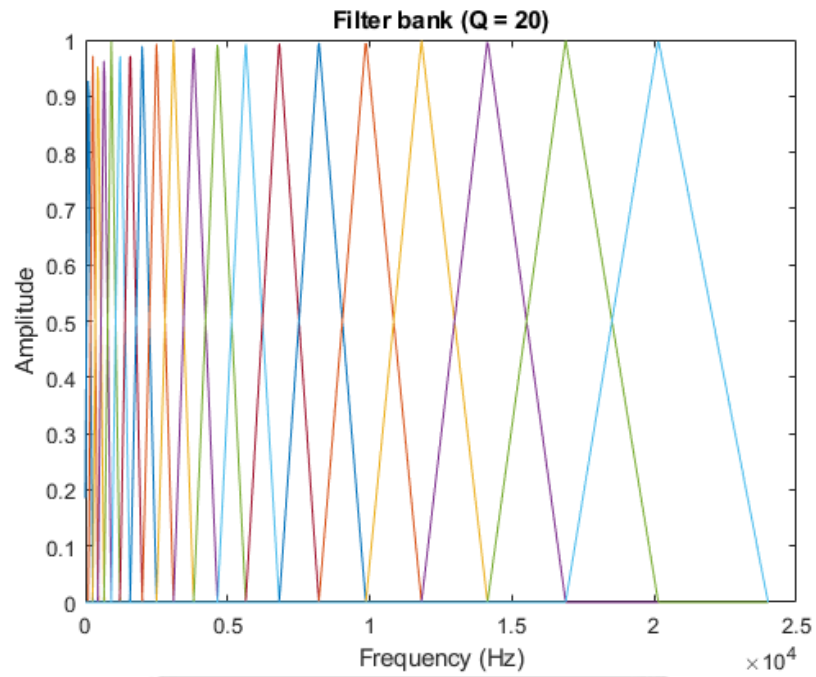


Figure 2.3 The frequency response of the filter bank with 20 filters, 48kHz sampling rate, 24kHz upper frequency in Hertz.

Filterbank Number	Center Frequency (Hz)
1	96.1
2	264.3
3	432.4
4	648.6
5	912.9
6	1225
7	1586
8	1994
9	2498
10	3099
11	3796

12	4637
13	5646
14	6823
15	8216
16	9850
17	11820
18	14150
19	16890
20	20130

Table 2.1 The center frequency of Mel-frequency filterbank in hertz.

Next, this filter bank is imposed on spectrum which is calculated in Eq. (2.2) and the output spectrum after filtered $\bar{Y}(i)$ is defined as,

$$\bar{Y}(i) = \sum_{k=1}^{\frac{N}{2}} |Y[k]|^2 \varphi_i[k] \quad (2.5)$$

At last, the DCT is taken on the log energies then MFCCs are finally calculated as,

$$C_m = \sqrt{\frac{2}{Q}} \sum_{l=0}^{Q-1} \log[\bar{Y}(i+1)] \cos \left[m \left(\frac{2l-1}{2} \right) \frac{\pi}{Q} \right] \quad (2.6)$$

where $0 \leq m \leq M-1$, M is the number of the coefficients which are taken 13 in general work. Fig. 2.4 shows the block diagram of MFCC algorithm.

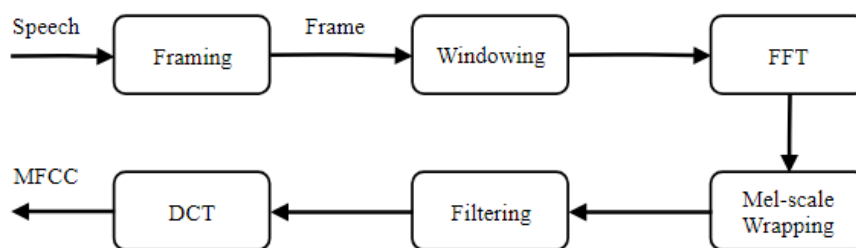


Figure 2.4 The block diagram of MFCC algorithm.

2.2 Classifiers

Classifiers generally are a statistical mechanism based on conditional probability and Bayes' theorem technology. Classifier established by a trainable model learns the optimal model iteratively and makes this optimal model fit the latent sample more, i.e. $P(\mathbf{y}|\theta^*) \geq P(\mathbf{y}|\theta)$. Several models of machine learning method are all based on this concept, e.g. SVM, HMM, ANNs, ..., etc. For speech emotion recognition system, due to the property considering temporary information, HMM based system is still one of the best classifiers even the state-of-the-art ANNs rapidly spring up in the recent years [4]. For the analysis and comparison in this thesis which focuses on HMM type classifier, the GMM, discrete HMM and continuous HMM are described as following sections in detail.

2.2.1 Gaussian Mixture Model [23]

GMM is represented as a mixture of gaussian probability density distributions (pdfs) which is defined as,

$$P(Y_i = y_i | \theta) = \sum_{j=1}^k w_j \phi(y_i | \mu_j, \Sigma_j) \quad (2.7)$$

where w_j is the weight value of each mixture of pdf and $0 \leq w_j$, $\sum_{j=1}^k w_j = 1$. Y_i is the random variable of specified sample vector, θ is the constructed gaussian mixture model and μ_j, Σ_j are the mean and covariance matrix of j-th multivariant gaussian distribution which is defined as,

$$\phi(y_i | \mu_j, \Sigma_j) = \frac{1}{(2\pi)^{\frac{n}{2}} [\det(\Sigma_j)]^{\frac{1}{2}}} \exp \left(-\frac{1}{2} (y_i - \mu_j)^T \Sigma_j^{-1} (y_i - \mu_j) \right) \quad (2.8)$$

To optimize by expectation-maximization (EM) algorithm, we first define $\gamma_{ij}^{(m)}$ be the probability that the i-th sample is generated by j-th gaussian density component at m-th

iteration optimization as,

$$\gamma_{ij}^{(m)} = P(Z_i = j | y_i, \theta^{(m)}) = \frac{w_j^{(m)} \phi(y_i | \mu_j^{(m)}, \Sigma_j^{(m)})}{\sum_{l=1}^k w_l^{(m)} \phi(y_i | \mu_l^{(m)}, \Sigma_l^{(m)})} \quad (2.9)$$

Which satisfies $\sum_{j=1}^k \gamma_{ij}^{(m)} = 1$. Then imply the EM algorithm and maximize the Q-function $Q(\theta | \theta^{(m)}) = E_{Z_i | y_i, \theta^{(m)}}[\log p(y_i, Z_i | \theta)]$, find the optimal parameters by partial derivative. We conclude the optimal equations.

$$\text{Weight: } w_j^{(m+1)} = \frac{n_j^{(m)}}{\sum_{l=1}^k n_l^{(m)}}, \quad j = 1, \dots, k. \quad (2.10)$$

$$\text{Mean: } \mu_j^{(m+1)} = \frac{1}{n_j^{(m)}} \sum_{i=1}^n \gamma_{ij}^{(m)} y_i, \quad j = 1, \dots, k. \quad (2.11)$$

$$\text{Covariance: } \Sigma_j^{(m+1)} = \frac{1}{n_j^{(m)}} \sum_{i=1}^n \gamma_{ij}^{(m)} (y_i - \mu_j^{(m+1)}) (y_i - \mu_j^{(m+1)})^T, \quad j = 1, \dots, k. \quad (2.12)$$

where m is the iteration times, $n_j^{(m)} = \sum_{i=1}^n \gamma_{ij}^{(m)}$. More details about the derivation of GMM is referred to [23].

Now, after the iteration constraint is reached, the optimal model is adjusted more fit samples. Fig. 2.5 and Fig. 2.6 show the example for the random data generated with three different normal distributions with different color points and use initial GMM with three components.

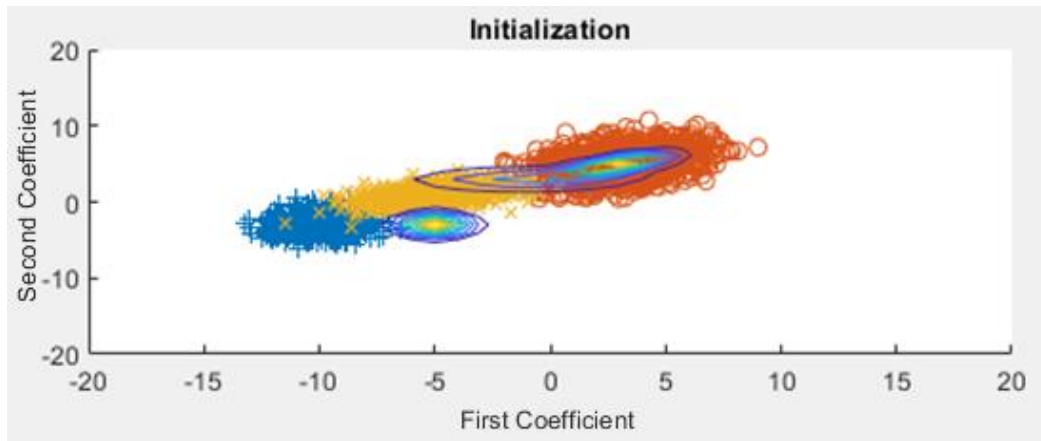


Figure 2.5 Random samples with every point and initial GMM density with three components.

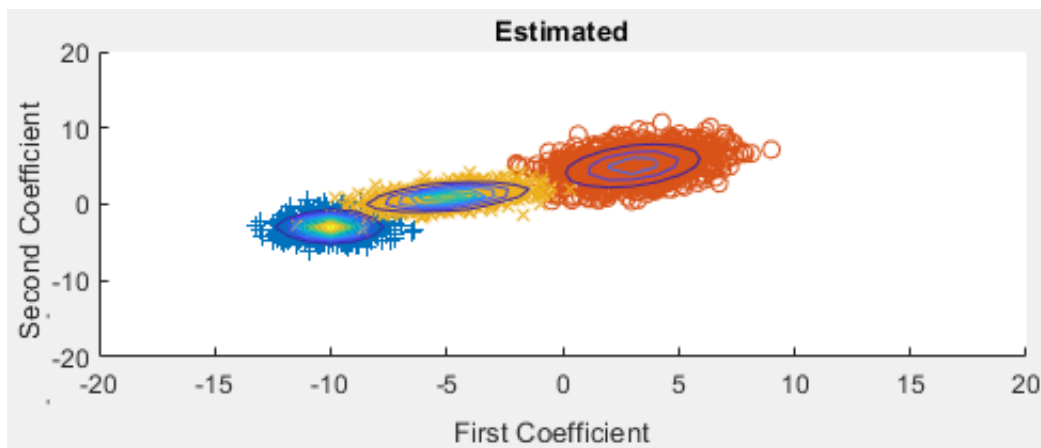


Figure 2.6 Random samples with every point and final optimal GMM density with three components.

Fig. 2.6 shows that the optimal GMM density fit the random samples at last, the center of GMM density indicates that there is the highest probability.

Moreover, it is necessary to show the similar example for practical case. Fig. 2.7 shows the initial GMM density with three almost components and the sample points which are the two MFCCs extracted from a random speech. After the iteration constraint is reached, Fig. 2.8 shows all the components of GMM density adjusted to fit all sample points, furthermore the GMM density indicates that there is the highest probability at upper right circle.

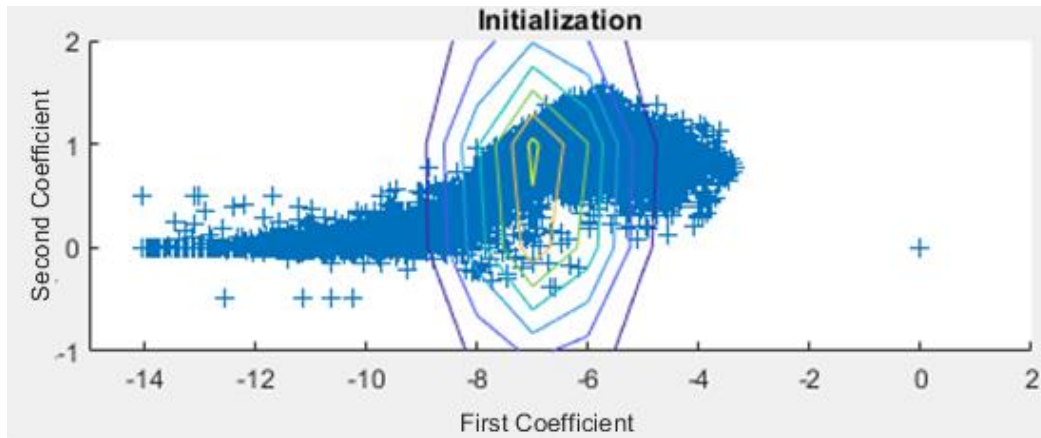


Figure 2.7 Random speech sample points extracted from two MFCCs and initial GMM density with three components.

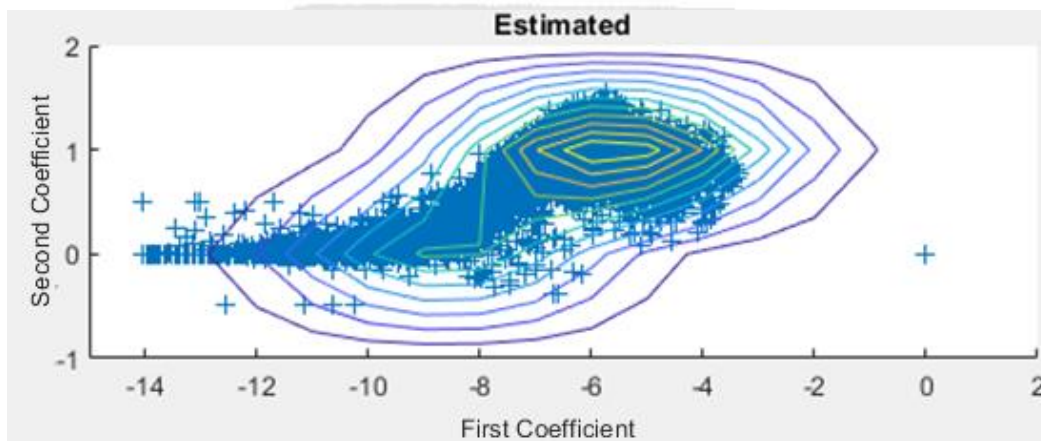


Figure 2.8 Random speech sample points extracted from two MFCCs and final optimal GMM density with three components.

For the practical comparison, Fig. 2.9 and Fig 2.10 show another speech sample. Note that the initial mean μ_j of GMM density is set as the mean of every sample point plus different biases. Also note that the final optimal GMM density is different from the example above and the highest probability locates at lower left which is also different from Fig. 2.8.

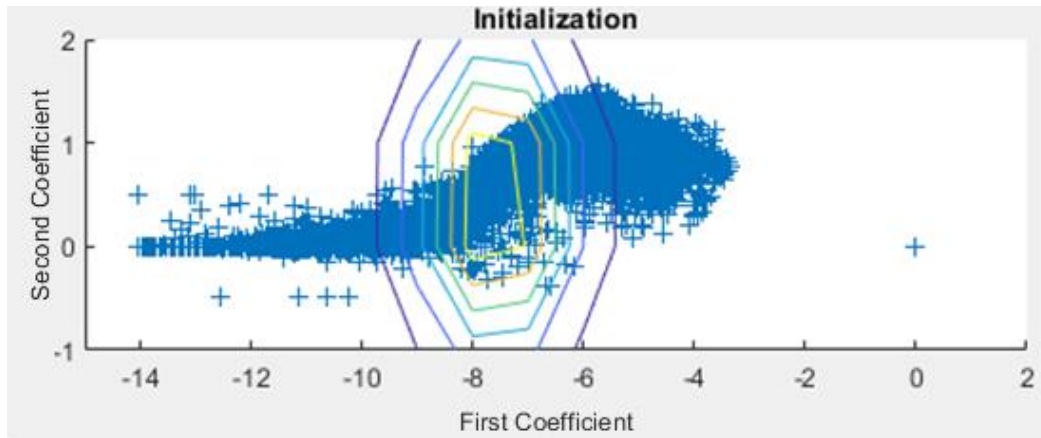


Figure 2.9 Another speech sample point extracted from two MFCCs and initial GMM density with three components.

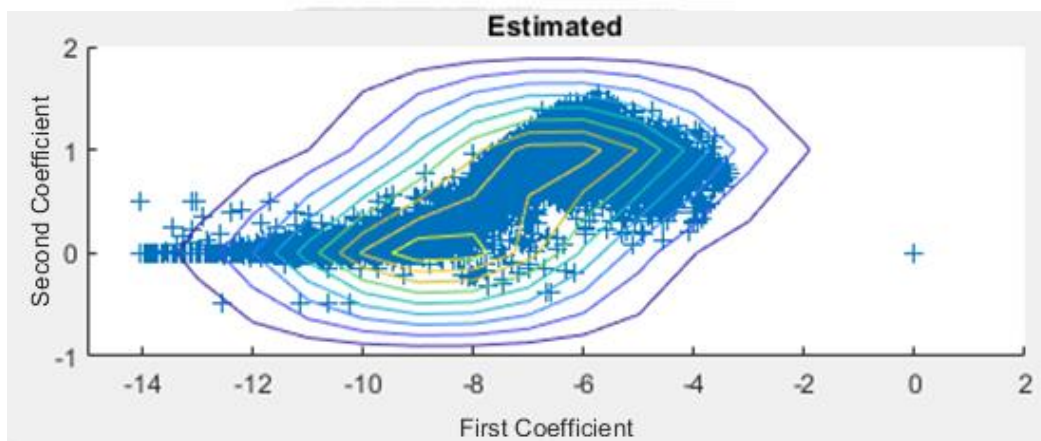


Figure 2.10 Random speech sample points extracted from two MFCCs and final optimal GMM density with three components.

2.2.2 Discrete Hidden Markov Model

HMM is a doubly stochastic process with the underlying states that are unobservable, so called hidden. Moreover, because the observation information is in discrete type, so called discrete. In detail, the mechanism of HMM can be regarded as a generation machine that it consecutively produces several observation information under a hidden state. Each observation can be produced by every hidden state according to the emission probability, so

which state produces the specific observation is implicit. To use this algorithm as a classifier, there are three basic problems of interest need to be solved. The three basic problems are described following:

- 1) **Basic Problem 1, Evaluation:** Given the observation sequence $O = O_1, O_2, \dots, O_T$, and the model $\lambda = (A, B, \pi)$, where A is the transition matrix, B is emission matrix and π is the initial state distribution. Then, how to compute the probability of this observation sequence $P(O|\lambda)$.
- 2) **Basic Problem 2, Estimation:** Given the observation sequence $O = O_1, O_2, \dots, O_T$. How to uncover all the likely hidden states with respect to this observation sequence, i.e. $q_t = i$, for $t = 1, \dots, T$ and state $i = 1, \dots, k$.
- 3) **Basic Problem 3, Training:** How to adjust the model $\lambda = (A, B, \pi)$ to optimize the probability of the observation probability $P(O|\lambda)$.

Now, conclude the solution algorithms for these important basic problems of interest as follows.

- 1) **Basic Problem 1, Evaluation:** Consider the forward variable,

$$\alpha_t(i) = P(O_1, O_2, \dots, O_T, q_t = i | \lambda) \quad (2.13)$$

where q_t is the state variable with respect to the observation sequence at time t , and $i, 1 \leq i \leq k$ is the number of states.

Initialize the first forward variable by the equation,

$$\alpha_1(i) = \pi_i b_i(O_1) \quad (2.14)$$

where π_i is the initial state probability of specific state i and $b_i \in B$ is the emission probability with respect to specific state i .

Now, consecutively compute the forward variable for $t = 1, 2, \dots, T - 1$,

$$\alpha_{t+1}(j) = \sum_{i=1}^N \alpha_t(i) a_{ij} b_j(O_{t+1}) \quad (2.15)$$

Finally, the observation probability $P(O|\lambda)$ can be computed by,

$$P(O|\lambda) = \sum_{i=1}^N \alpha_T(i) \quad (2.16)$$

Note that this computation processing is called forward algorithm which is faster than directly computing the observation probability $P(O|\lambda)$ and intuitively comprehensive. The forward algorithm is explained as Fig. 2.11 and Fig. 2.12 shown. The forward variable $\alpha_{t+1}(j)$ is a linear combination among all the previous states with the transition probability and considering the emission probability while the observation O_{t+1} is observed. The computation of the observation probability $P(O|\lambda)$ is the sum of all the forward variables at the end circled by the blue shade as Fig 2.12 shown.

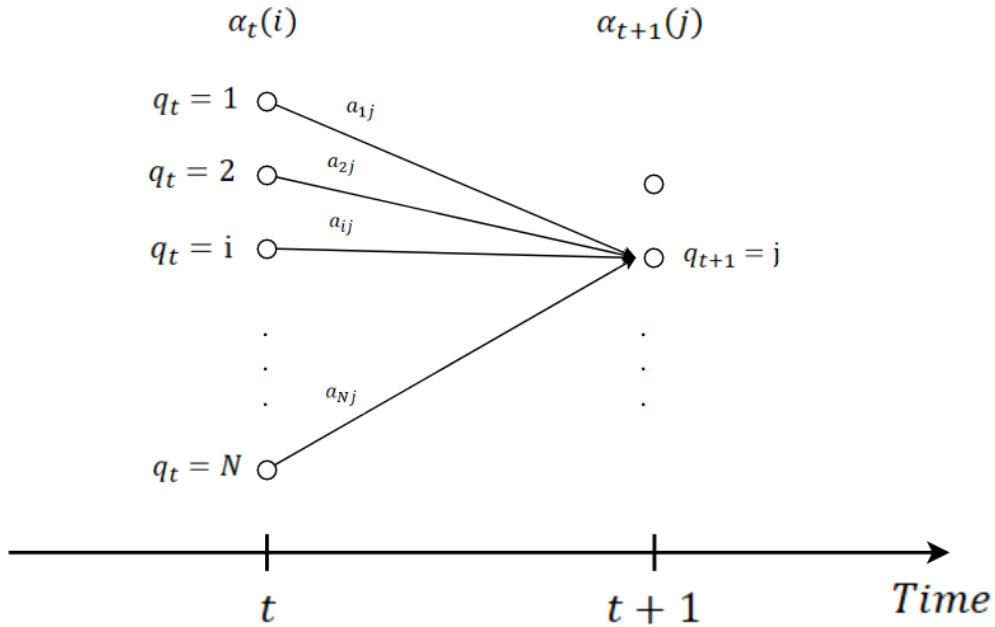


Figure 2.11 The computation of each particular forward variable at state j , $\alpha_{t+1}(j)$.

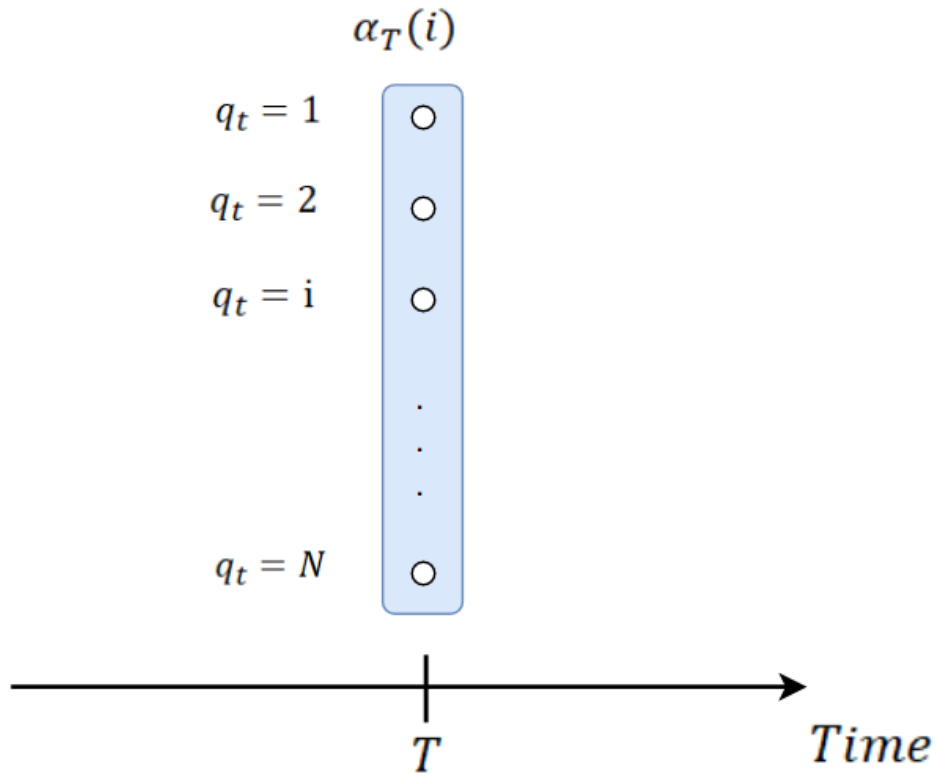


Figure 2.12 The computation of the observation probability $P(O|\lambda)$.

- 2) **Basic Problem 2, Estimation:** To solve this estimation problem, it has proposed a very fast and high-performance algorithm which is called Viterbi algorithm. Viterbi algorithm is a dynamic programming which leads to find the single best state instead of the probability of overall states. The algorithm is shown as following statements.

Initialize the first maximum forward variable $\delta_1(i)$ and first state $\Psi_1(i)$ as

$$\delta_1(i) = \pi_i b_i(O_1) \quad (2.17)$$

$$\Psi_1(i) = 0 \quad (2.18)$$

Then consecutively compute the maximum forward variable and state for

$t = 2, 3, \dots, T$.

$$\delta_t(j) = \max_{1 \leq i \leq N} [\delta_{t-1}(i) a_{ij}] b_j(O_t) \quad (2.19)$$

$$\psi_t(j) = \operatorname{argmax}_{1 \leq i \leq N} [\delta_{t-1}(i) a_{ij}] \quad (2.20)$$

Note that Viterbi algorithm is similar to forward algorithm except it only choose the maximal forward variable as $\delta_t(j)$ and it considers the state which has maximum likelihood to observe the observation at time t .

Finally, the state path is completely computed as the state variable $\psi_t(j)$, for $t = 1, 2, \dots, T$ as Fig. 2.13 shown.

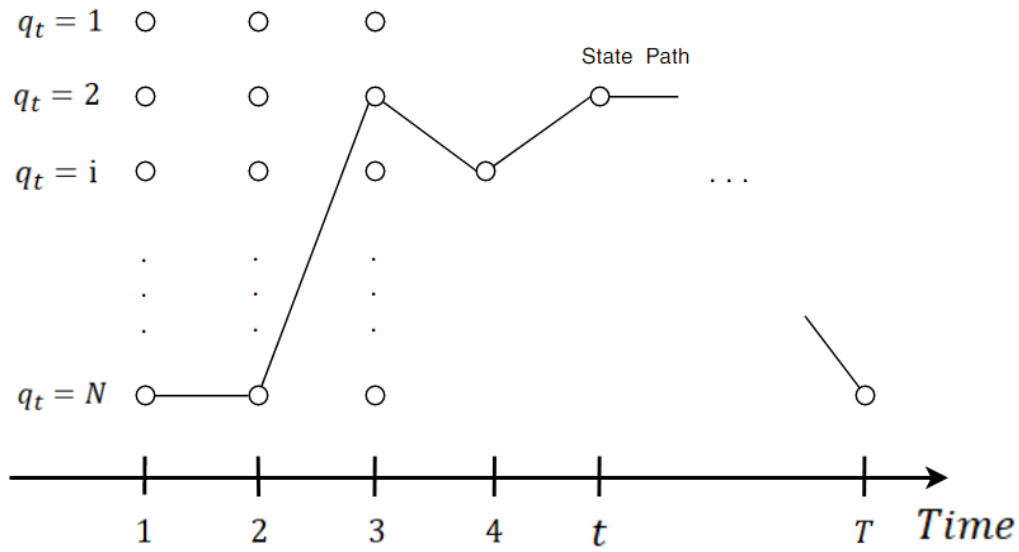


Figure 2.13 The state path uncovered using Viterbi algorithm.

3) **Basic Problem 3. Training:** To adjust the model $\lambda = (A, B, \pi)$ to fit observation sequence or to optimize the probability of the observation probability $P(O|\lambda)$, imply the EM algorithm which is familiar with the optimization to GMM. To estimate the optimal parameters of the discrete HMM, it takes the Q function maximization on different parameters and the estimation process is called Baum-Welch re-estimation. Following equations conclude the Baum-Welch re-estimation formulas,

a)

$$\pi_i^* = \gamma_1(1), \quad 1 \leq i \leq N \quad (2.21)$$

b)

$$a_{ij}^* = \frac{\sum_{t=1}^{T-1} \xi_t(i, j)}{\sum_{t=1}^{T-1} \gamma_t(i)} \quad (2.22)$$

c)

$$b_j^*(k) = \frac{\sum_{t=1, O_t=k}^T \gamma_t(j)}{\sum_{t=1}^T \gamma_t(j)} \quad (2.23)$$

where $\gamma_t(i) = \sum_{j=1}^N \xi_t(i, j)$ and $\xi_t(i, j)$ is defined as,

$$\xi_t(i, j) = \frac{\alpha_t(i) a_{ij} b_j(O_{t+1}) \beta_{t+1}(j)}{P(O|\lambda)} \quad (2.24)$$

and β_t is the backward variable which is defined as,

$$\beta_t(i) = \sum_{j=1}^N a_{ij} b_j(O_{t+1}) \beta_{t+1}(j) \quad (2.25)$$

for $t = T - 1, T - 2, \dots, 1, 1 \leq i \leq N$. Note that for the backward variable and variable $\xi_t(i, j)$, they are explained as Fig. 2.14 and Fig. 2.15 shown. As similar as forward variable, $\beta_t(i)$ is a linear combination among all the next states with the transition probability and considering the emission probability while the next observation O_{t+1} is observed. The computation of $\alpha_t(i) a_{ij} b_j(O_{t+1}) \beta_{t+1}(j)$ is the numerator of $\xi_t(i, j)$ corresponding to the algorithm called forward-backward algorithm. The re-estimation formulas are considering the variable $\xi_t(i, j)$ in whole time and normalized by each individual state.

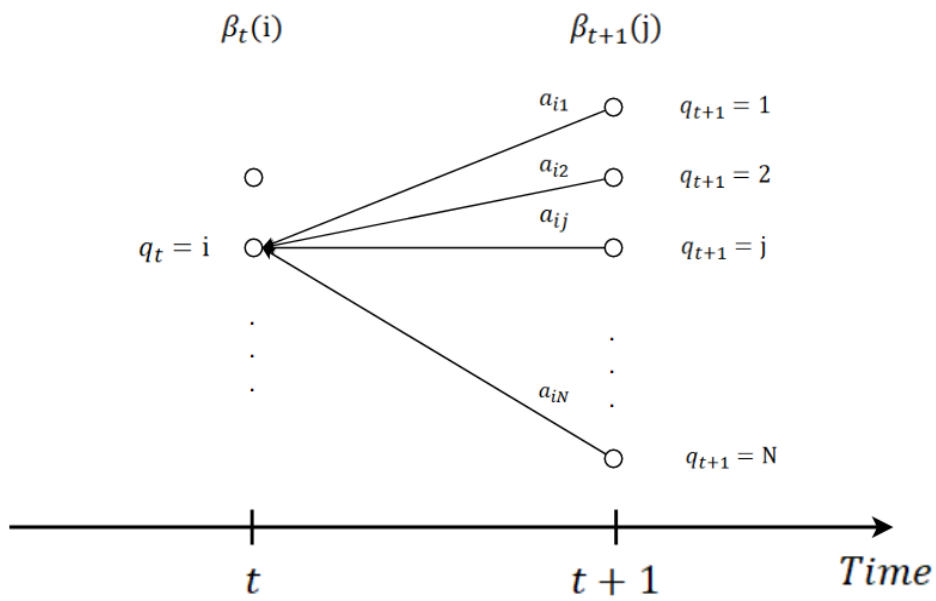


Figure 2.14 The computation of each particular backward variable $\beta_t(i)$ at state i .

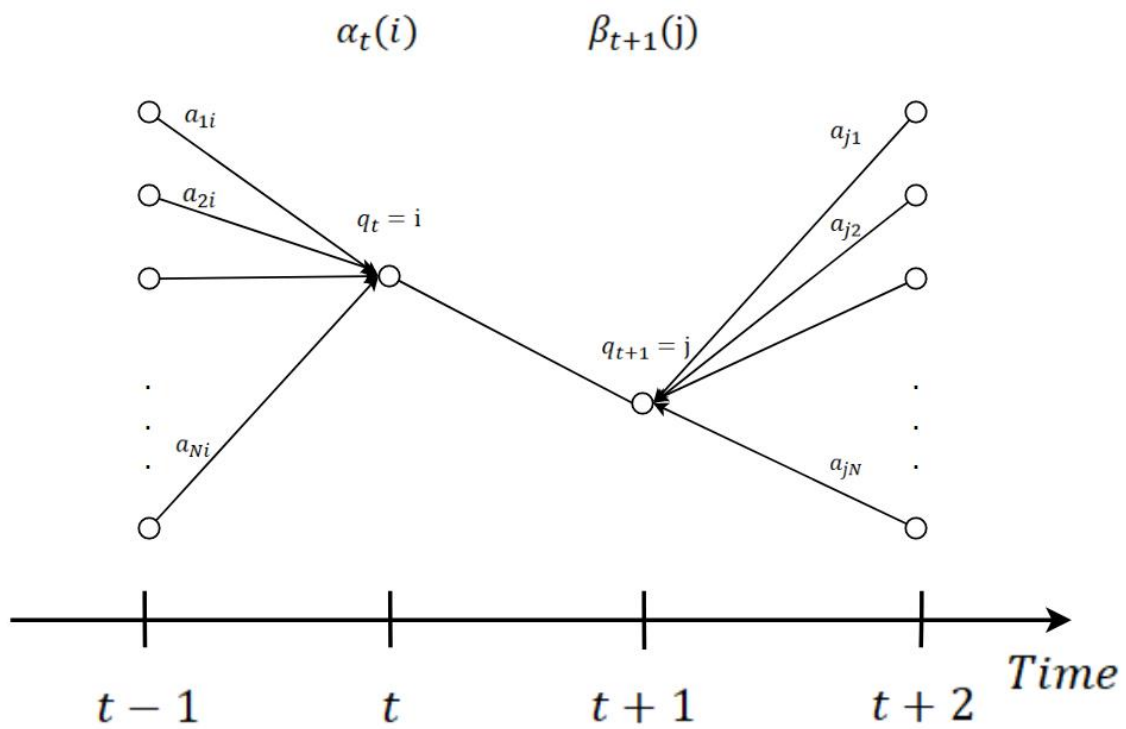


Figure 2.15 The computation of $\alpha_t(i)a_{ij}b_j(O_{t+1})\beta_{t+1}(j)$.

Note that more details about above algorithms are referred to [23] [24] [25]. Now, all the basic problems are solved. Since most of the observation information are not discrete number in practical case, the emission matrix B is impossible to represent the probability of the observations. For practical case, it is necessary to use quantization method to encode the observation information as an index sequence as the preprocessing. Before introducing the quantization methods, there are some notations and concepts must be reviewed.

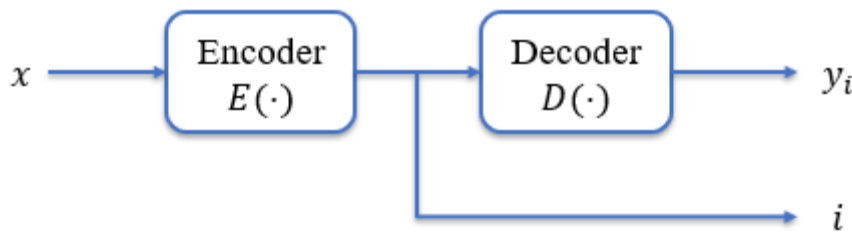


Figure 2.16 The basic block diagram for quantization.

As Fig. 2.16 shown, the basic block diagram for quantization is divided into encoder and decoder. Encoder encodes samples as an index i where the sample locates at quantization region R_i defined as,

$$R_i = \{x \in R: E(x) = i\} \quad (2.26)$$

Note that each region is mutually exclusive, i.e. $R_i \cap R_j = \emptyset$. In other words, the samples belong to which regions are explicit. The other block decoder decodes the index into representative y_i , so each sample in the region R_i is decoded into the same representative y_i , i.e. $D(x \in R_i) = y_i$. Every representative constructs the codebook C as,

$$C = \{y_0, y_1, \dots, y_{N-1}\} \quad (2.27)$$

where N represents the number of regions. At last, consider the mean square error (MSE) which is usually used for optimizing the quantizer $D(\cdot)$,

$$\text{MSE} = E \left[(x - D(x))^2 \right] \quad (2.28)$$

The scalar and vector quantization methods are introduced as follows.

a) Scalar Quantization

In general, the quantization is separated into two partial solutions:

- Given the quantization regions $R = \{R_0, R_1, \dots, R_{N-1}\}$, find the optimal codebook,

$$\{y_0^*, y_1^*, \dots, y_{N-1}^*\} = \underset{y_i}{\operatorname{argmin}} \int_{R_i} f_x(x) (x - y_i)^2 dx \quad (2.29)$$

- Given the codebook $C = \{y_0, y_1, \dots, y_{N-1}\}$, find the optimal quantization regions

$$\{R_0^*, R_1^*, \dots, R_{N-1}^*\} = \underset{R_i}{\operatorname{argmin}} \int_{-\infty}^{\infty} f_x(x) (x - D(x))^2 dx \quad (2.30)$$

$$\text{subject to } R_i \cap R_j = \emptyset, \quad \bigcup_{i=0}^{N-1} R_i = R \quad (2.31)$$

For the Eq. (2.29), to find the minimum by estimating the derivative with respect to the representative y_i , conclude the final equation as,

$$y_i^* = \frac{\int_{R_i} x f_x(x) dx}{\int_{R_i} f_x(x) dx} \quad (2.32)$$

Note that considering Eq. (2.32), the optimal particular representative y_i is exactly a conditional expected value for the $x \in R_i$. In other words, the optimal representative y_i is exactly the center of region R_i .

For the second partial solution, it can be rewritten as,

$$\int_{-\infty}^{\infty} f_x(x)(x - D(x))^2 dx \geq \int_{-\infty}^{\infty} f_x(x) \min_i (x - y_i)^2 dx \quad (2.33)$$

Since the codebook is given and fixed, the best region is choose enclosing a number of x closed y_i , that is,

$$R_i = \{x: (x - y_i)^2 \leq (x - y_j)^2, j \neq i\} \quad (2.34)$$

The Lloyd algorithm for quantizer design is iteratively computing two partial solutions above until the optimization constraint is reached. In general, the constraint set as every difference between the representative y_i, y_i^* lower than a constraint value ε , that is,

$$\max_i (y_i^{(k+1)} - y_i^{(k)})^2 \leq \varepsilon \quad (2.35)$$

A simple scalar quantization result is shown as Fig. 2.17, each segment of the line represents the quantization region R_i , the symbol i represents the index number after encoding with respect to specified region R_i .

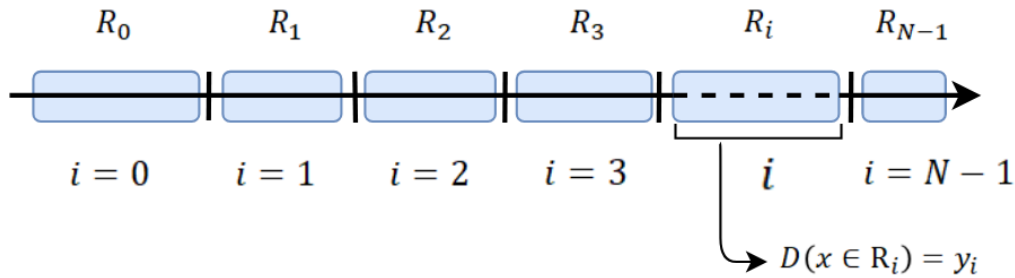


Figure 2.17 A result of scalar quantization.

b) Vector Quantization

In contrast to scalar quantization, samples $\mathbf{x}$ are a vector in R^n . The vector quantization is also separated into two partial solutions as similar as scalar

quantization, conclude the final equations as shown below,

$$\mathbf{y}_i^* = \frac{\int_{R_i} \mathbf{x} f_{\mathbf{x}}(\mathbf{x}) d\mathbf{x}}{\int_{R_i} f_{\mathbf{x}}(\mathbf{x}) d\mathbf{x}} \quad (2.36)$$

$$R_i = \{\mathbf{x}: \|\mathbf{x} - \mathbf{y}_i\|^2 \leq \|\mathbf{x} - \mathbf{y}_j\|^2, j \neq i\} \quad (2.37)$$

The Eq. (2.36) and Eq. (2.37) are as the same as Eq. (2.32) and Eq. (2.34) except samples $\mathbf{x}$ and the representatives $\mathbf{y}_i$ are vector in R^n . To design the vector quantizer, use Lloyd algorithm to iteratively find the optimal quantization regions $R^* = \{R_0^*, R_1^*, \dots, R_{N-1}^*\}$ and codebook $C^* = \{\mathbf{y}_0^*, \mathbf{y}_1^*, \dots, \mathbf{y}_{N-1}^*\}$. A two-dimensional example, i.e. $\mathbf{x} \in R^2$ is shown as Fig. 2.18, the representative $\mathbf{y}_i$ is the center of each region R_i .

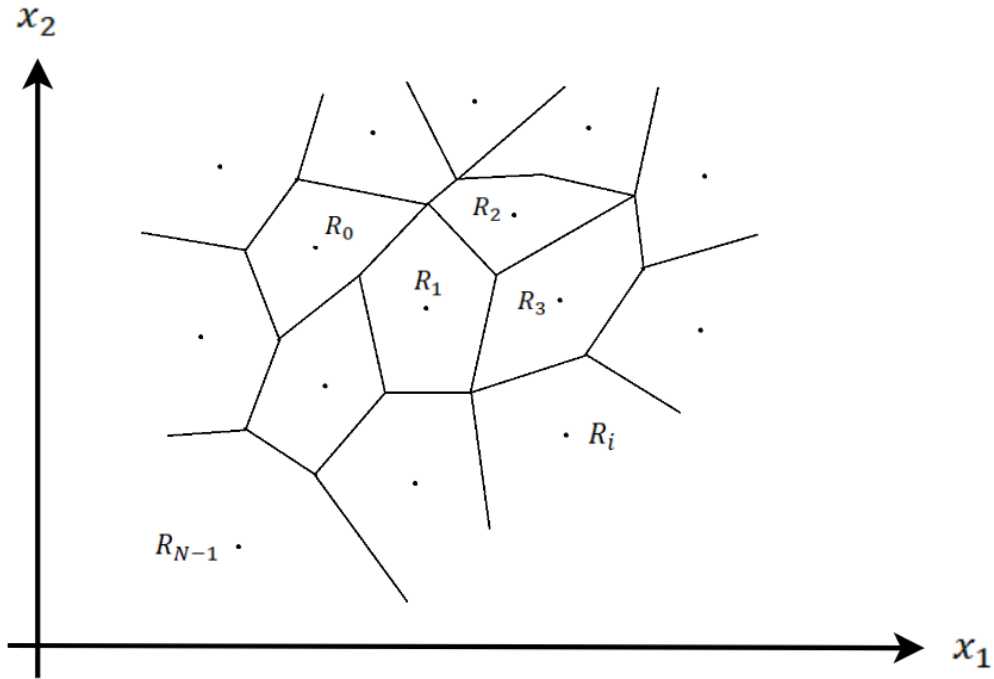


Figure 2.18 A result of vector quantization in R^2 .

By implying the quantization method, the observations can be encoded into a discrete index sequence to let the emission probability represent each practical observation, e.g. use

scalar quantization to encode pitch or energy which are one dimension information or use vector quantization to encode the MFCCs or a number of formants that the dimension is larger than one.

2.2.3 Continuous Hidden Markov Model

The continuous HMM is the extended version of discrete HMM by extending discrete observation sequences to continuous observation vector sequences. Substitute the emission matrix for the probability density distribution which the GMM is most widely used. Now, the solutions of basic problems are reserved except the re-estimation formulas, there are some parameters of GMM needed to be re-estimated [26] [27].

To estimate the optimal parameters of continuous HMM, it is similar to discrete HMM and GMM, it takes the Q function maximization on different parameters. But now there are additional weight, mean, covariance parameters need to be estimated because in continuous HMM, it uses GMM to represent the probability of observation $b_j(\mathbf{O}_t)$ instead of emission probability matrix, also note that the observation $\mathbf{O}_t$ is used as a vector instead of index in discrete HMM and probability of observation $b_j(\mathbf{O}_t)$ is the probability density without loss of generality. Using gaussian mixture distribution to represent the probability of observation $b_j(\mathbf{O}_t)$ is better than discrete because we have gaussian probability distribution function to compute continuous probability unlike discrete probability in discrete HMM. Another reason is that it has much distortion in discrete HMM when uses the observations which are quantized into index sequence. The re-estimation equations for the parameters weight, mean, covariance are concluded as follows,

$$\text{Weight: } c_{jk}^* = \frac{\sum_{t=1}^T \rho_t(j,k) \beta_t(j)}{\sum_{t=1}^T \alpha_t(j) \beta_t(j)} \quad (2.38)$$

$$\text{Mean: } \mu_{jk}^* = \frac{\sum_{t=1}^T \rho_t(j,k) \beta_t(j) \mathbf{o}_t}{\sum_{t=1}^T \rho_t(j,k) \beta_t(j)} \quad (2.39)$$

$$\text{Covariance: } \Sigma_{jk}^* = \frac{\sum_{t=1}^T \rho_t(j,k) \beta_t(j) (\mathbf{o}_t - \mu_{jk}) (\mathbf{o}_t - \mu_{jk})^T}{\sum_{t=1}^T \rho_t(j,k) \beta_t(j)} \quad (2.40)$$

where ρ_t is defined as,

$$\rho_t = \begin{cases} c_{jk} \frac{\partial b_j}{\partial c_{jk}} \Big|_{\mathbf{o}_1}, & t = 1 \\ \sum_{i=1}^N \alpha_{t-1}(i) a_{ij} c_{jk} \frac{\partial b_j}{\partial c_{jk}} \Big|_{\mathbf{o}_t}, & 1 \leq t \leq T \end{cases} \quad (2.41)$$

The simple concept of this re-estimation is that they are similar to the formulas of $b_j^*(\mathbf{o}_t)$ re-estimation except that it considers the probability of the observation $\mathbf{o}_t$ on different GMM components. The parameter ρ_t is the forward variable on different GMM components. Note that the transition probability re-estimation formula is reserved as Eq. (2.22).

2.2.4 Classifier Comparison

For the GMM classifier, it considers all the observations simultaneously as Fig. 2.7 to Fig. 2.10 shown. In speech emotion recognition topic, it improperly uses whole temporal observations as the feature to GMM classifier because this classifier considers observation without the time information and it also loses the time information while the emotional speech usually consists of the observation variation in time. Another problem in GMM is the singularity problem caused when lots of observations are almost stationary. It is frequently caused while using temporary information as features in GMM, because the feature is usually stationary within a narrow time. It properly uses statistical information as the global feature such as mean and maximum of pitch, mean and covariance of LPCC, so the deep

concept while using GMM as classifier is that we consider the global features are distinguishable in different emotional utterances and the point is what kind of feature should choose.

In contrast to GMM, the discrete HMM considers temporary information. HMM type classifiers consider different states which is producing corresponding temporary observations with the emission probability. Unlike GMM classifier, HMM recognizes emotions by considering the state transition probability and emission probability which compute forward variable when observes the different temporary observations. If someone thinks the statistical features are more distinguishable, it should use GMM as classifier. Otherwise, if the temporary information is much influence, it should use HMM as classifier.

However, the discrete HMM supports only one-dimension observation results from the discrete emission probability. It needs quantization at front stage and it causes the observation distorted and loses some distinguishable information. If we increase quantization level at very high number, it solves the distortion problem but needs more training data or it makes some emission probability untrained. For continuous HMM, there's not quantization at front and the quantization distortion problem. Instead of discrete emission probability, continuous HMM usually uses gaussian mixture probability density to compute the probability of specified observation $b_j(\mathbf{O}_t)$, so the continuous HMM is able to support multi-dimension observations while using multivariant gaussian mixture probability density such as LPCCs, MFCCs. But now the singularity problem exists in continuous HMM because the temporary observations are usually stationary in speech. So the continuous HMM is more hardly used than other classifiers.

2.3 Hidden Markov Model Topology

The structure of HMM state is variant, left-to-right model, ergodic model are the most used topologies [17]. Following figures show the above models. For the left-to-right model, there is probability only from left state to right state, that is, there isn't likelihood to transit back. For the ergodic model, every state is mutually connected with the transition probability.

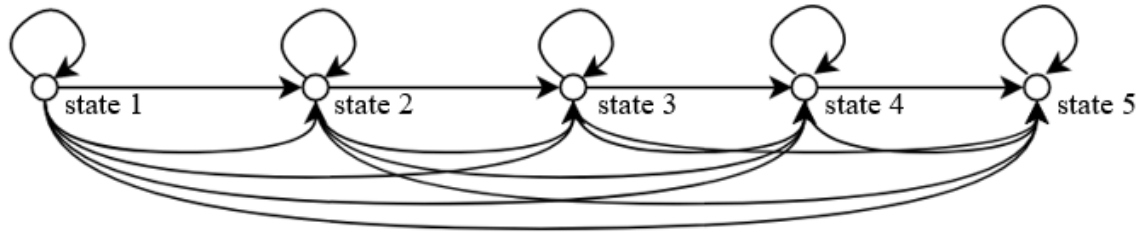


Figure 2.19 Left-to-right model.

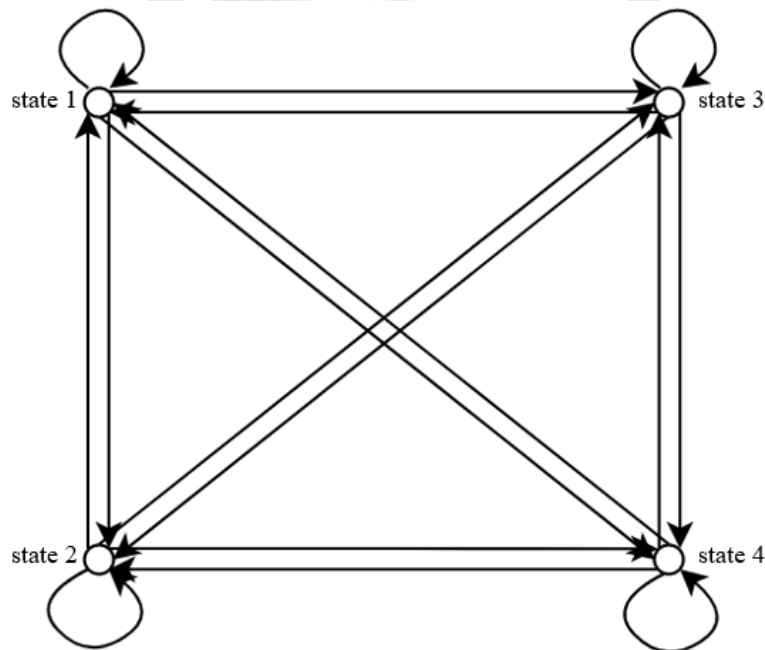


Figure 2.20 Ergodic model.

The left-to-right model is widely used for speech recognition, because the state represents phoneme for a word. A linguistic word consists of several basic unit called

phoneme and the phonemes in a word are usually arranged sequentially from beginning to the end of word. So left-to-right is suitable for speech recognition. However, in speech emotion recognition, the state represents the state in an emotion utterance. The different states in an emotion utterance are occurred repeatedly and implicit. For example, if we take pitch information in an angry emotion as feature, the lower pitch would be observed at beginning and end of the utterance or at some segments after a higher pitch segment, the higher pitch would be observed at some strong angry segments. So ergodic model is preferred while on the application of speech emotion recognition.

2.4 Hidden Markov Model Issues

In this section, we talk about the issues usually occurred when used HMM as classifier in practical cases that are underflow problem and singularity problem as following subsections. The solutions for these two problems are introduced at chapter three.

2.4.1 Underflow

Since HMM classifier recognizes the emotion by computing the probability of observation sequence $P(O|\lambda)$ which is computed equal to the summation of forward variable at the end $\sum_{i=1}^N \alpha_T(i)$. Because the probability is less than one, the forward variable is computed gradually lower and lower until reaches the limitation of computer, all rest forward variables become zero, that is underflow problem [17]. It occurs when the utterance duration is large, i.e. t is large which is most the cases in speech emotion recognition application.

2.4.2 Singularity

The singularity is caused because of the gaussian probability density function property. Review the optimal formulas Eq. (2.12) and Eq. (2.40) which contain the term $(\mathbf{O}_t -$

$\mu_{jk})(\mathbf{O}_t - \mu_{jk})^T$ at numerator, the mean μ_{jk} becomes more and more approached to the exact average of observations, after the iteratively optimization. The term $(\mathbf{O}_t - \mu_{jk})(\mathbf{O}_t - \mu_{jk})^T$ approaches zero gradually and makes covariance matrix singular, that is singularity problem. It seldom practically occurs while using GMM as classifier, because we use the statistical global feature. An amount of global feature is much lesser than temporary local feature, it takes much lesser iteration times to converge the optimization, so the singularity problem usually has not taken place. However, it perhaps occurs while using continuous HMM as classifier because the temporary observations are usually almost stationary even more within the similar state in emotion utterance.



Chapter 3

PROPOSED METHODOLOGY

The detail processing of the proposed speech emotion recognition from the database introduction and feature extraction at the front-end to the initial model hypothesis, emotion classification at back-end is described at this chapter. In section 3.1, introduce the speech emotion database in English language. In section 3.2, describe the feature extraction and what feature we select. In section 3.3, describe the initial model which is separated into initial model hypothesis and initial model constraint. In section 3.4, we introduce the solutions to overcome the issues in continuous HMM. In section 3.5, present the architecture for whole recognition system. In final section, shortly describe the accuracy estimation.

3.1 Database

We use the Surrey Audio-Visual Expressed Emotion (SAVEE) database [28] which consists of four male actors recording for seven different primary emotions, each emotion speech consists of 15 utterances except neutral emotion consists 30 utterances and totals 480 samples in British English. The primary emotions include neutral, happy, sadness, angry, fear, disgust and surprise. We choose all primary emotion except surprise emotion for comparison in this thesis experiment. The data were recorded in a visual media lab with high quality audio-visual equipment, processed and labeled. Each utterance is sample at 44.1kHz sampling rate about three second durations. The different actors are labelled DC, JE, JK, KL, we will notate it on the comparison figures at chapter four.

3.2 Feature Extraction

From now on, we have discussed about the statistical global feature and the temporary local feature. Due to the properties of above three classifiers, GMM, discrete HMM and continuous HMM, we choose the global feature for GMM classifier and local feature for HMM type classifier. Since we have mentioned several type features e.g. the prosodic feature and cepstral feature at chapter one, we choose 13 number of MFCCs as the feature because that's the most widely used feature. In this thesis, we focus on the discussions among the different classifier performances, so we don't use another feature such as LFPC, SBC that we had mentioned at chapter one, and use only 13 number coefficients ignoring the redundant velocity and acceleration coefficients. The selection of frame size and frame shift size influences the information of a speech, narrower frame makes more stationary and shorter temporal information of the specified utterance, wider frame selection contains sufficient information to extract feature but perhaps obtains the nonstationary information which is not desired while windowing a speech. For this trade-off reason, we decide to extract the 13 number of MFCCs with 16 millisecond frame size and 10 millisecond frame shift size, that is 706 points frame size and 265 points for frame shift size which is rounded at 44.1kHz sampling rate.

For GMM classifier, compute the statistical parameters that are mean, median, standard deviation, minimum and maximum combined as 65-dimension feature of all training samples for each emotion [11] as Fig. 3.1 and Fig. 3.2 shown. Fig. 3.1 shows all the feature sample points in blue, the statistical parameters mean in red, median in green, standard deviation in yellow, minimum in cyan, maximum in black closed to mean. Note that this plot only shows first two-dimension of the 13 number of MFCCs, the exact distance is computed by the norm of the 13 number of MFCCs. Fig. 3.2 shows the feature samples and the

statistical samples as similar to Fig. 3.1 in happy emotion. Although it only plots two-dimension points, the difference is pretty explicit.

For HMM type classifier, only compute 13 number of MFCCs as the observation sequence $\mathbf{O}_t = \{\mathbf{O}_1, \mathbf{O}_2, \dots, \mathbf{O}_T\}$, the notation t means t-th frame. The observation sequence is shown as Fig. 3.3 to Fig. 3.4.

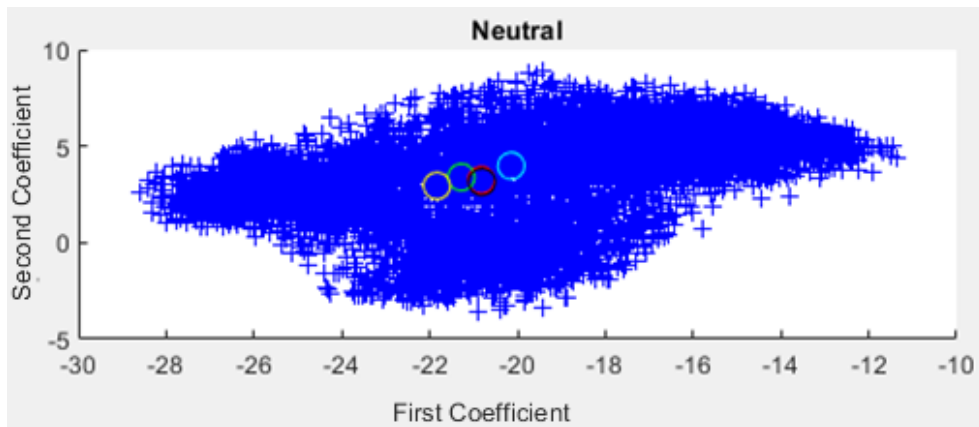


Figure 3.1 The feature samples and the statistical parameters for the neutral emotion.

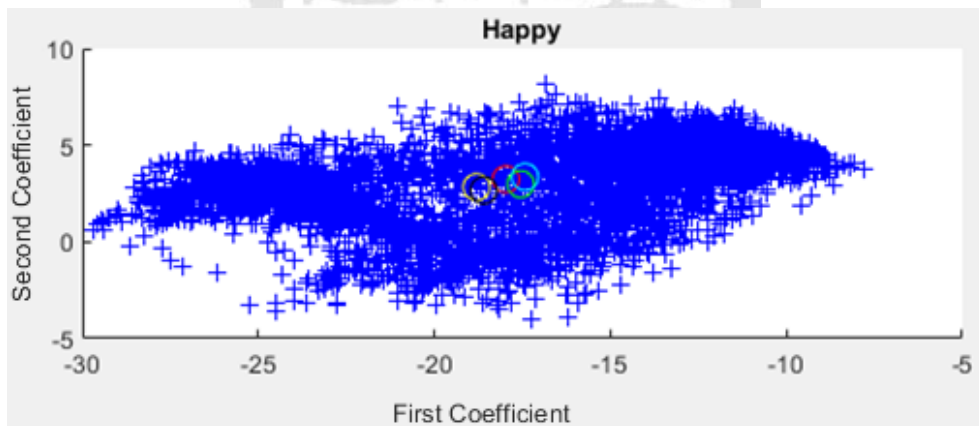


Figure 3.2 The feature samples and the statistical parameters for the happy emotion.

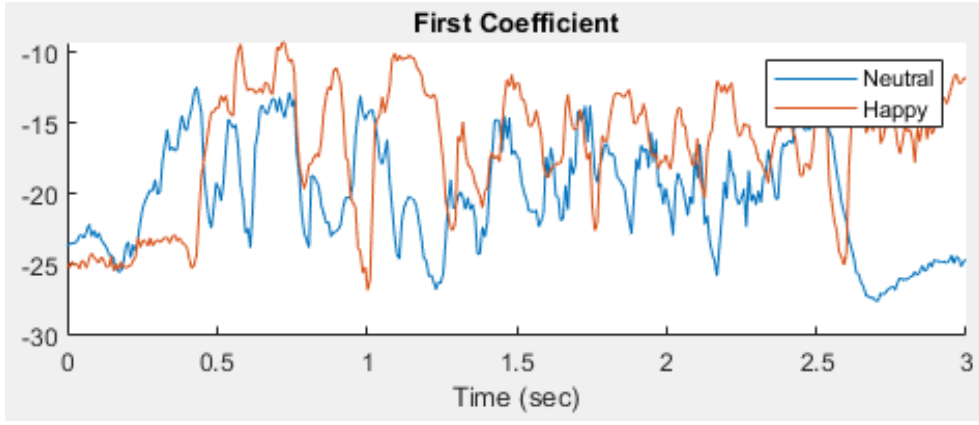


Figure 3.3 The first coefficient of MFCCs align time between neutral and happy emotions.

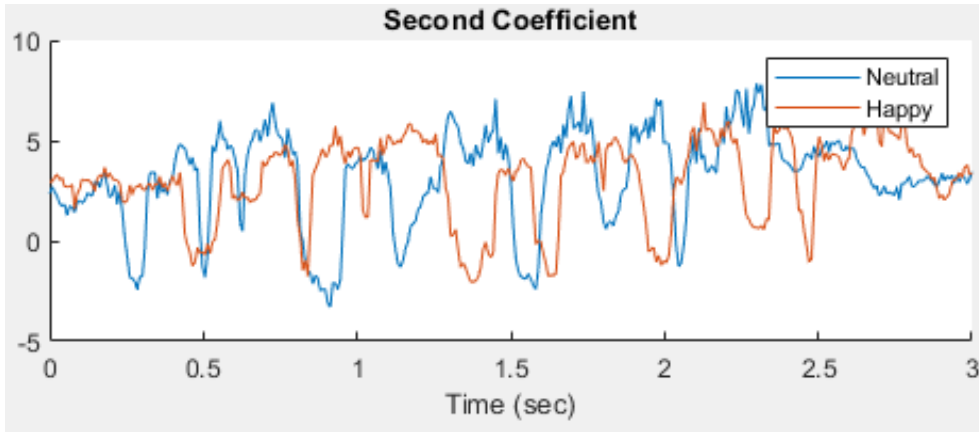


Figure 3.4 The second coefficient of MFCCs align time between neutral and happy emotions.

General speaking, it is hardly distinguishing the emotions between these MFCCs align time information by human's eyes even though it indeed exists some differences between them.

Note that since the observation must set one-dimension for discrete HMM, we use the vector quantization method which is Lloyd algorithm to encode the feature as an index sequence. The optimization constraint set as Eq. (3.1) with $\varepsilon = 1 \times 10^{-7}$, and note that it also uses the same codebook trained from training samples to encode testing samples.

$$\max_i \left(y_i^{(k+1)} - y_i^{(k)} \right)^2 \leq \varepsilon \quad (3.1)$$

The observation sequence after encoded is shown as Fig. 3.5, the observation after

encoded in red color with 32 quantization levels, the blue color is the first coefficient of MFCCs referred to corresponding emotional utterance.

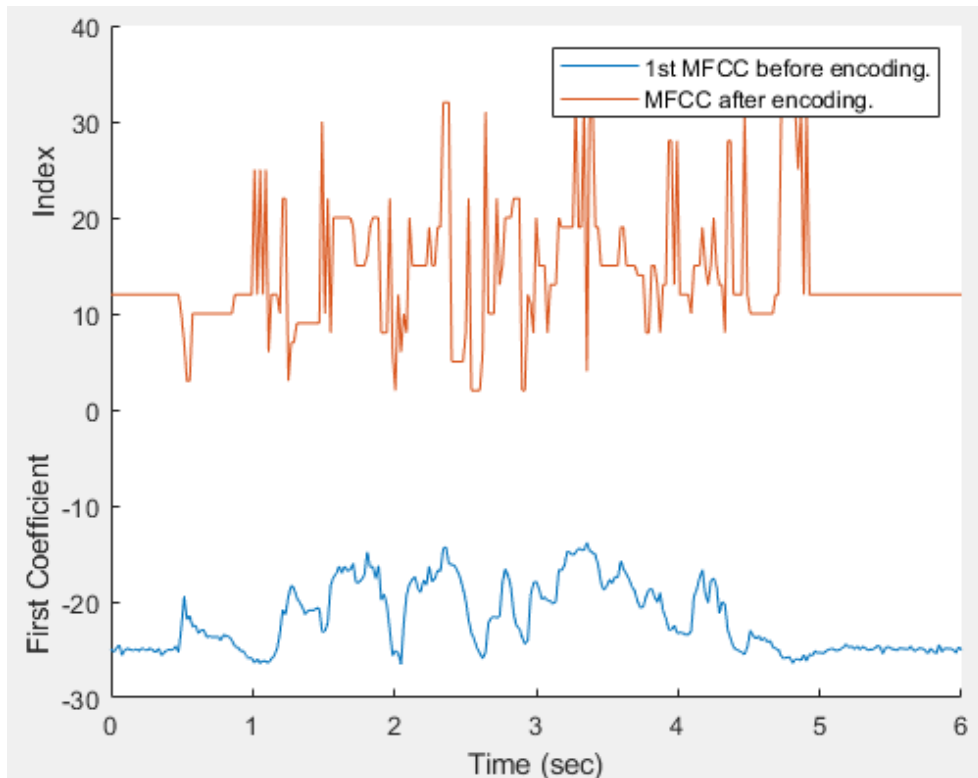


Figure 3.5 The observation sequence after encoded.

3.3 Initial Model Construction

Before train the emotion models for above three classifiers, we must offer an initial model to train. For the classifier using artificial neural networks (ANNs), many researches construct initial model with random weight value and some selections of other variant parameters, afterward choose the best trained model. However, above three classifiers are extended from EM algorithm which are very sensitive to initial model according to the property of EM algorithm especially the mean parameter [23] [29]. Due to this property, we consider it could train a best emotion model if the pretrained initial model set with a good hypothesis. A good emotion model means that the model fit the observation sequence with

corresponding emotional utterance, that is makes the probability of observation sequence $P(O|\lambda)$ as higher as it could.

Another reason for initial model construction is that if we randomly set the parameters of gaussian mixture distribution, the probability of observation $b_j(\mathbf{O}_t)$ perhaps produces zero probability. Unlike in a mathematical sense, if the probability density as Eq. (3.2) is computed reaching to the computer limitation, it produces zero and causes some issues while computing the optimization.

$$\phi(y_i|\mu_j, \Sigma_j) = \frac{1}{(2\pi)^{\frac{n}{2}}[\det(\Sigma_j)]^{\frac{1}{2}}} \exp\left(\frac{-1}{2}(y_i - \mu_j)^T \Sigma_j^{-1}(y_i - \mu_j)\right) \quad (3.2)$$

3.3.1 Initial State Hypothesis

As mentioned, we regard if find a way to hypothesis the state representation of an emotional utterance, it could train a better emotion model than constructing initial model randomly. The ways are described as follows,

1) Hypothesis by uniformly segmented

For this hypothesis idea, predict that the state representation of an emotion is uniformly segmented. In this way, an utterance is segmented for the number as the number of HMM states used as Fig. 3.6 to Fig. 3.8 shown.

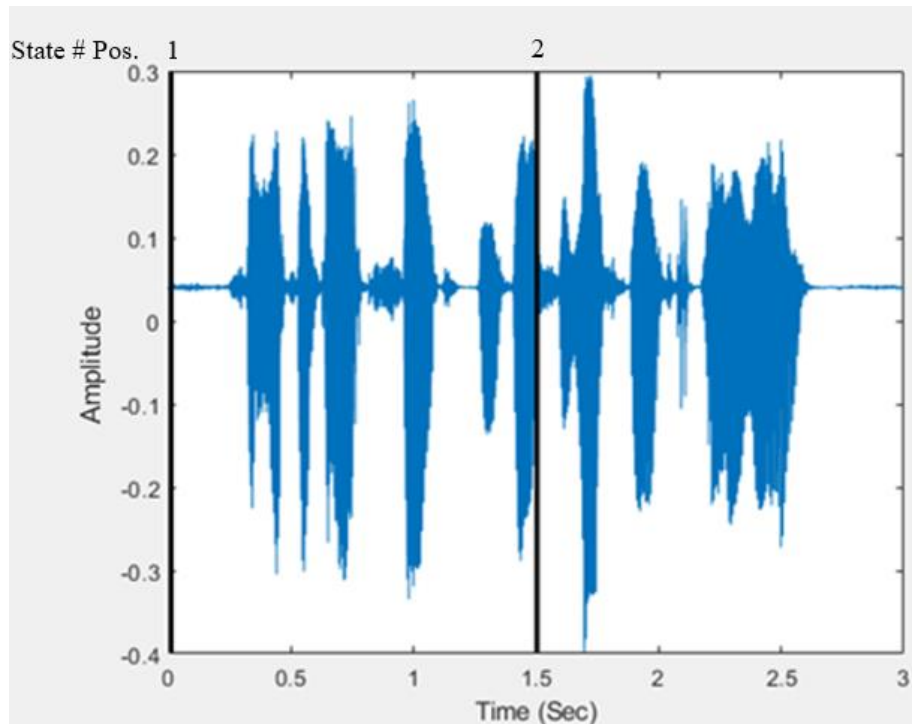


Figure 3.6 Prediction of two states representation located at the black line positions in time domain.

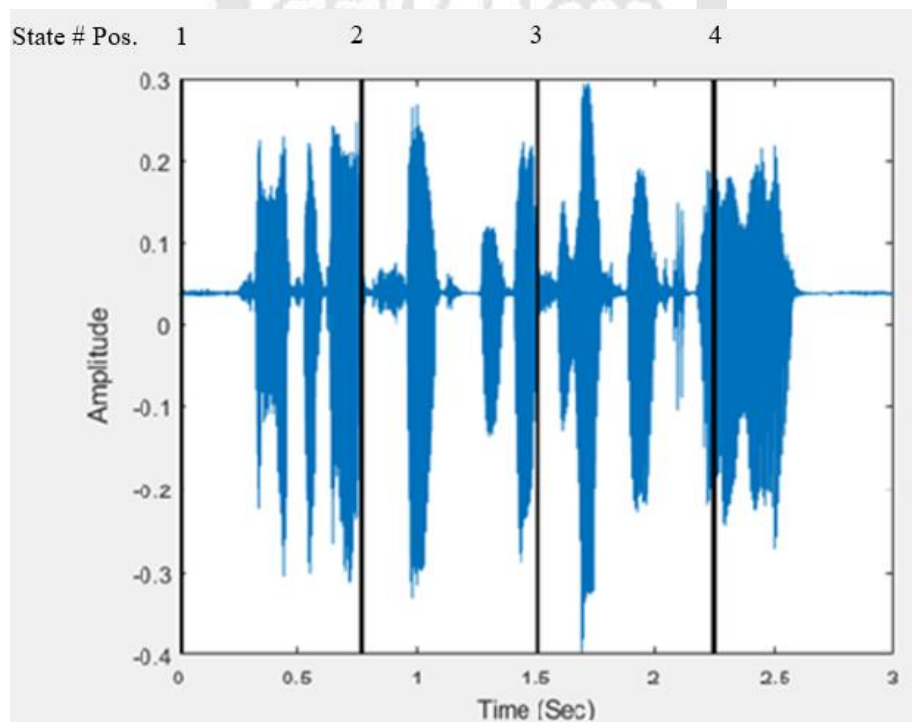


Figure 3.7 Prediction of four states representation located at the black line positions in time domain.

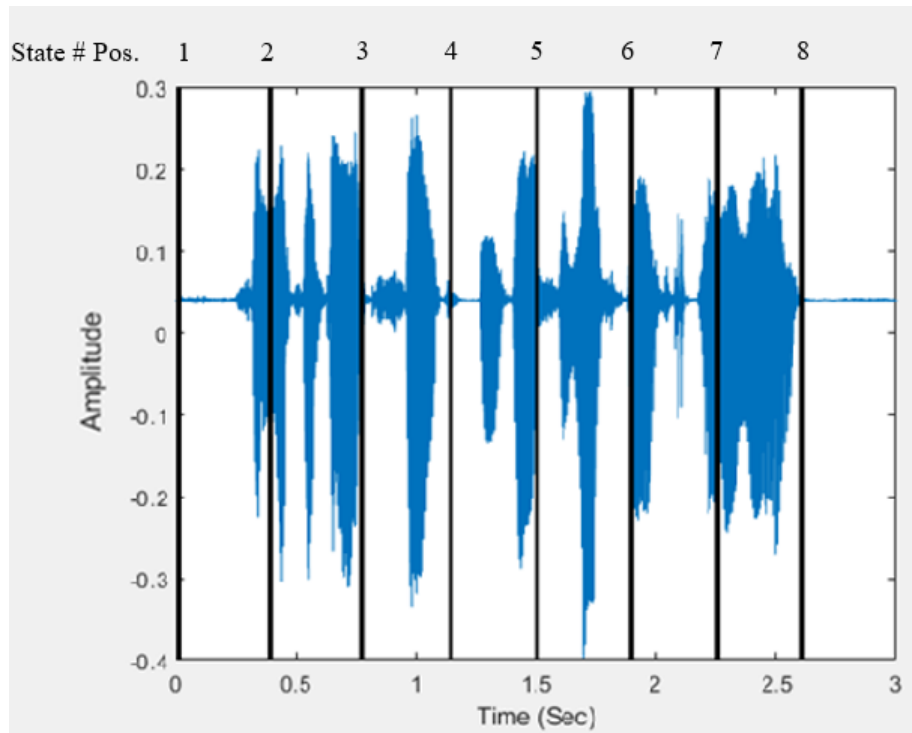


Figure 3.8 Prediction eight states representation located at the black line positions in time domain.

As these figures shown, if it used with lesser states, it perhaps has insufficient number of states to represent different states of an emotion. Otherwise, if it uses too many states to represent an emotional utterance, some state settings would be similar to others and it perhaps makes the final trained emotion model lesser distinguishable. Moreover, using too many numbers of states also means too many trainable parameters, it much increases training time and affects some issues like overfitting or insufficient training samples.

2) Hypothesis by clustering method

In contrast to first hypothesis method, this method predicts the state representation of an emotion by clustering method which is used vector quantization as described at chapter two. In this way, the state settings are predicted by the center of clusters of all observation sequence in an emotional

utterance which considers the probability sense as Eq. (3.3) represented, and the number of states we used is represented as the number of clusters. Fig. 3.9 to Fig. 3.11 show the practical examples with 2 to 8 states hypothesis using this method.

$$y_i^* = \frac{\int_{R_i} x f_x(x) dx}{\int_{R_i} f_x(x) dx} \quad (3.3)$$

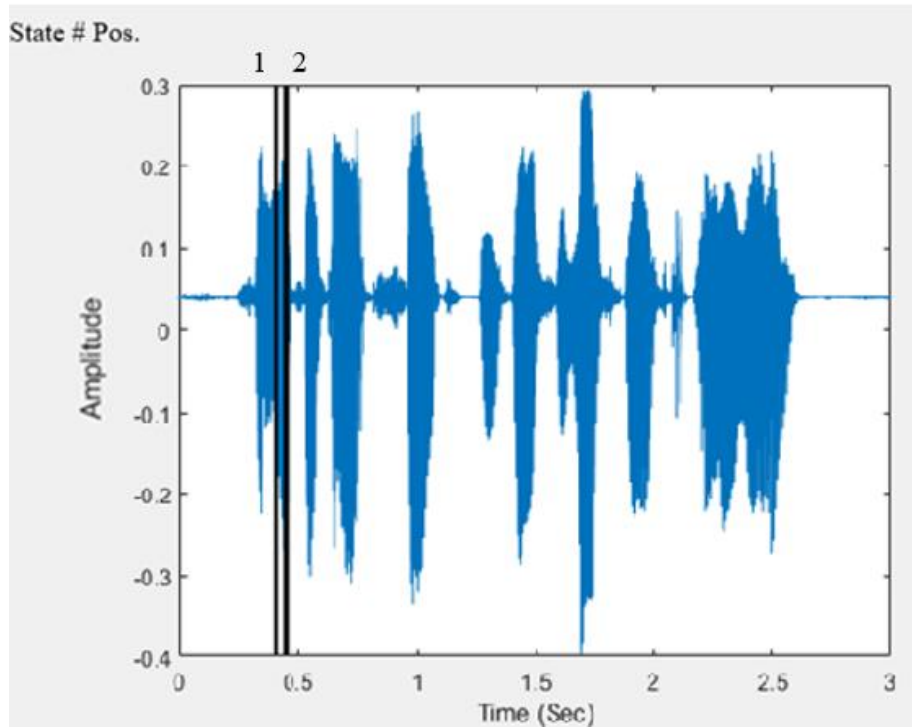


Figure 3.9 Prediction of two states representation located at the black line positions in time domain.

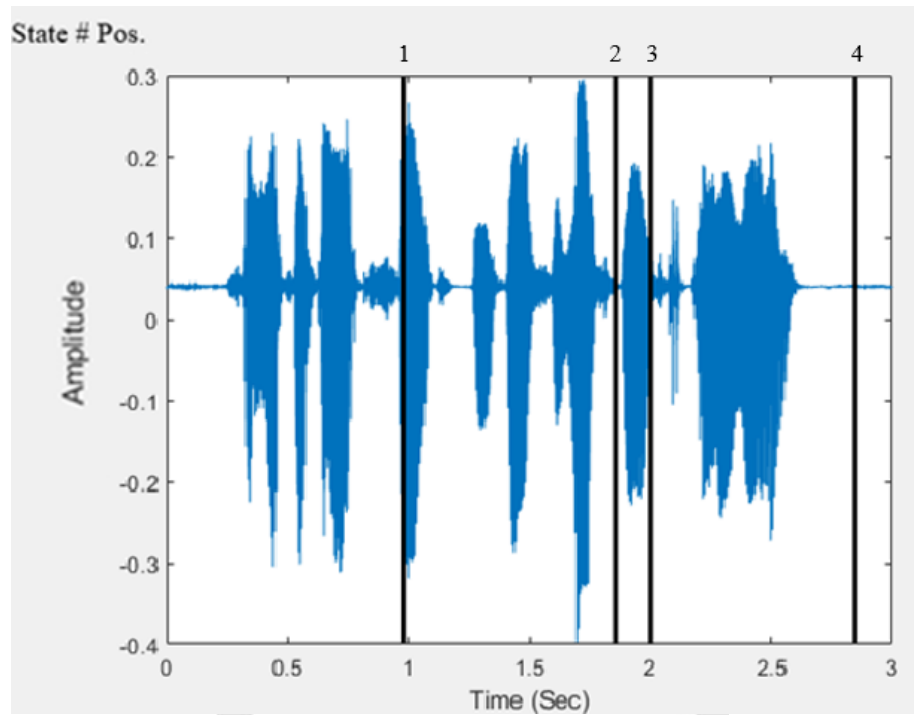


Figure 3.10 Prediction of four states representation located at the black line positions in time domain.

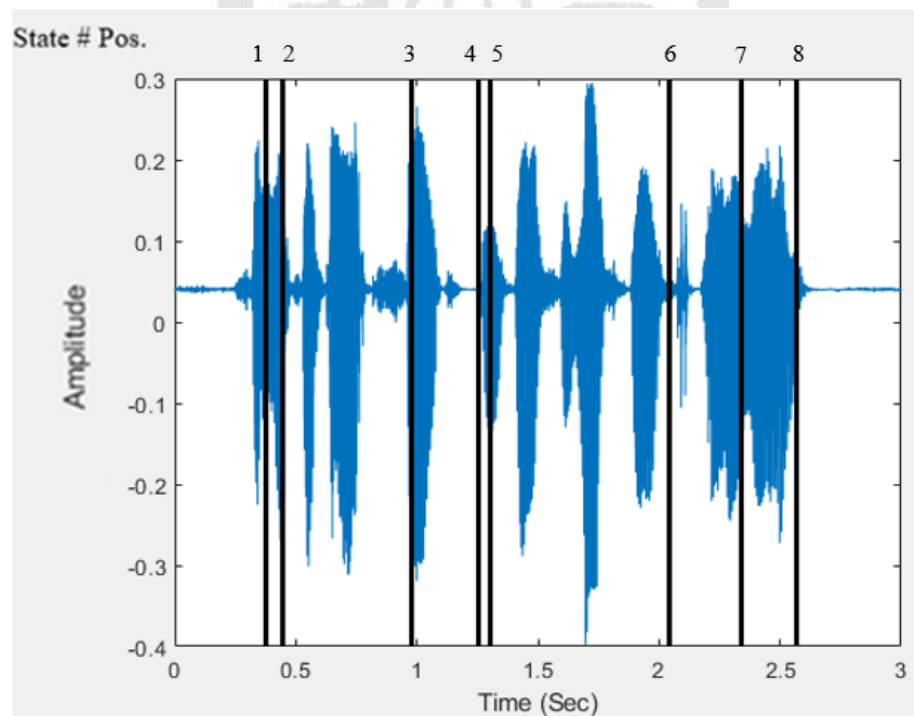


Figure 3.11 Prediction of eight states representation located at the black line positions in time domain.

Unlike uniformly segmented, this method could find the states more different. For the

four states, the state representation at very different positions from two states using clustering method. Although it predicts very different state representations by clustering method, some states are mutually similar while using too many numbers of states as Fig. 3.11 shown.

3.3.2 Initial GMM Constraint

The propose to set GMM constraint is to limit the range of gaussian mixture components or it produces zero probability and causes some issues as mentioned. We introduce a perturbation range η to limit the mean parameter within a range on the gaussian components. In detail, the initial model set by the hypothesized state which is described above, then decide each component's mean within the range η on each state. For instance, if there is a state represented as,

$$[-23.83 \ 2.30 \ 0.92 \ 0.70 \ 0.35 \ 0.03 \ -0.17 \ 0.09 \ -0.17 \ -0.03 \ 0.01 \ -0.01 \ 0.11]$$

Choose two number of components with the perturbation range $\eta = 0.2$. Then the state vector is multiplied by the range $[0.8 \ 1.2]$ which the results are as follows,

$$[-19.07 \ 1.84 \ 0.74 \ 0.56 \ 0.28 \ 0.02 \ -0.14 \ 0.07 \ -0.14 \ -0.02 \ 0.1 \ -0.01 \ 0.09]$$

$$[-28.61 \ 2.77 \ 1.11 \ 0.84 \ 0.42 \ 0.03 \ -0.20 \ 0.11 \ -0.20 \ -0.03 \ 0.01 \ -0.02 \ 0.13]$$

If choose four number of components, the state vector is multiplied by the range $[-0.8 \ 0.93 \ 1.07 \ 1.2]$ and so on. The choice of the perturbation range affects the accuracy performance, we discuss about it at chapter four.

3.3.3 Initial Model for Different Classifier

Since we compare the accuracy performance among three classifiers, GMM, discrete HMM, continuous HMM, we conclude all the initial model constructions.

For GMM classifier, compute the mean of corresponding emotion training samples,

note that it should be a 65-dimension vector, then multiplied by the perturbation range vector with respect to the number of components. Set the initial weight value uniform and covariance matrix as 65-dimension unit diagonal matrix.

For discrete HMM classifier, set the initial emission matrix uniform and plus some zeros to distinguish states. For example, for two states with eight quantization levels emission matrix, it set as,

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0.25 & 0.25 & 0.25 & 0.25 \\ 0.25 & 0.25 & 0.25 & 0.25 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The initial transition matrix is set as uniform probability.

For continuous HMM classifier, the initial transition matrix is set uniform like discrete HMM settings. The initial mean parameter set as the hypothesized state with corresponding emotion training samples and is multiplied by perturbation range vector as described at last subsection. The initial weight value is set uniform and covariance set as 13-dimension unit diagonal matrix.

3.4 Issues Solution

The main issues which are underflow and singularity problem have discussed at chapter two. These issues usually take place in practical case, because the length of practical observation sequence is usually too large which results in the forward variable reaching limitation of the computer limitation. Another reason is that the speech information is usually stationary, and it causes the singularity problem at the covariance matrix.

Following subsections describe how to solve those issues in this thesis.

3.4.1 Solution for Underflow

- Since the forward variable uses the continued multiplication of probability value, it reaches under limitation quickly. We use the log probability value

instead of original probability value, the continued multiplication becomes the continued addition as Eq. (3.4) [17],

$$\log \alpha_{t+1} = \left(\sum_{i=1}^N \log \alpha_t(i) + \log a_{ij} \right) + \log b_j(\mathbf{O}_t) \quad (3.4)$$

This approach solves most practical cases except the observation sequence length is really large or the observation probability is really small when the specified observation is less likely with respect to the gaussian mixture probability density function.

- Except for the log probability, it introduces a scaling coefficient to normalize every instant forward variable [17]. The scaling coefficient c_t is defined as,

$$c_t = \frac{1}{\sum_{i=1}^N \alpha_t(i)} \quad (3.5)$$

It normalizes instant forward variable as,

$$\hat{\alpha}_t(i) = \frac{\alpha_t(i)}{\sum_{i=1}^N \alpha_t(i)} \quad (3.6)$$

So as the backward variable. This way avoids the computation of both the probability of this observation sequence $P(\mathbf{O}|\lambda)$ and re-estimation formulas from underflow problem.

Fortunately, the probability of observation sequence $P(\mathbf{O}|\lambda)$ becomes a very elegant equation,

$$P(\mathbf{O}|\lambda) = \frac{1}{\prod_{t=1}^T c_t} \quad (3.7)$$

Other parameters in re-estimation in discrete HMM are reserved due to the current time independence of $\prod_{t=1}^T c_t$. But there are some parameters in

continuous HMM must be modified. After introducing this scaling coefficients, it produces extra scaling coefficients in the denominator of re-estimation formulas Eq. (2.22) and Eq. (2.38), so the extra scaling coefficients must be cancelled out for the correction of re-estimation formulas. Eq. (3.8) and Eq (3.9) show the optimal transition probability and weight value with extra scaling coefficient in dominator.

$$\hat{a}_{ij}^* = \frac{\sum_{t=1}^{T-1} C_T \alpha_t(i) a_{ij} b_j(\mathbf{o}_{t+1}) \beta_{t+1}(j)}{\sum_{t=1}^{T-1} C_T \cdot c_t \cdot \alpha_t(i) \beta_t(i)} \quad (3.8)$$

$$\hat{c}_{ij}^* = \frac{\sum_{t=1}^T C_T \sum_{i=1}^N \alpha_{t-1}(i) a_{ij} c_{jk} \left. \frac{\partial b_j}{\partial c_{jk}} \right|_{\mathbf{o}_t} \beta_t(j)}{\sum_{t=1}^{T-1} C_T \cdot c_t \cdot \alpha_t(j) \beta_t(j)} \quad (3.9)$$

where c_t is the instance scaling coefficient, C_T is multiplicated scaling coefficient defined as $C_T = \prod_{t=1}^T c_t$ which is independent on instant time t . Moreover, the mean and covariance parameter are reserved as Eq. (2.39) and Eq. (2.40).

3.4.2 Solution for Singularity

It is very hard to actually solve singularity problem, due to the intrinsic property of the re-estimation of gaussian pdf. As Eq. (3.10) and Eq. (3.11) which refer to Eq. (2.12) and Eq. (2.40) at chapter two.

$$\Sigma_j^* = \frac{1}{n_j} \sum_{i=1}^n r_{ij} (y_i - \mu_j)(y_i - \mu_j)^T, \quad j = 1, \dots, k. \quad (3.10)$$

$$\Sigma_{jk}^* = \frac{\sum_{t=1}^T \rho_t(j, k) \beta_t(j) (\mathbf{o}_t - \mu_{jk})(\mathbf{o}_t - \mu_{jk})^T}{\sum_{t=1}^T \rho_t(j, k) \beta_t(j)} \quad (3.11)$$

The mean parameter μ_j , μ_{jk} are iteratively computed approach to average of observations,

it causes the nominator term $(y_i - \mu_j)(y_i - \mu_j)^T$ and $(\mathbf{O}_t - \mu_{jk})(\mathbf{O}_t - \mu_{jk})^T$ smaller, the pdf extremely concentrates on the mean position and reaches singularity afterward as Fig. 3.12 to Fig. 3.14 shown. As Fig. 3.14, after the re-estimation converges, the gaussian mixture distribution is extremely concentrate on the mean position, and the covariance matrix almost reaches singularity.

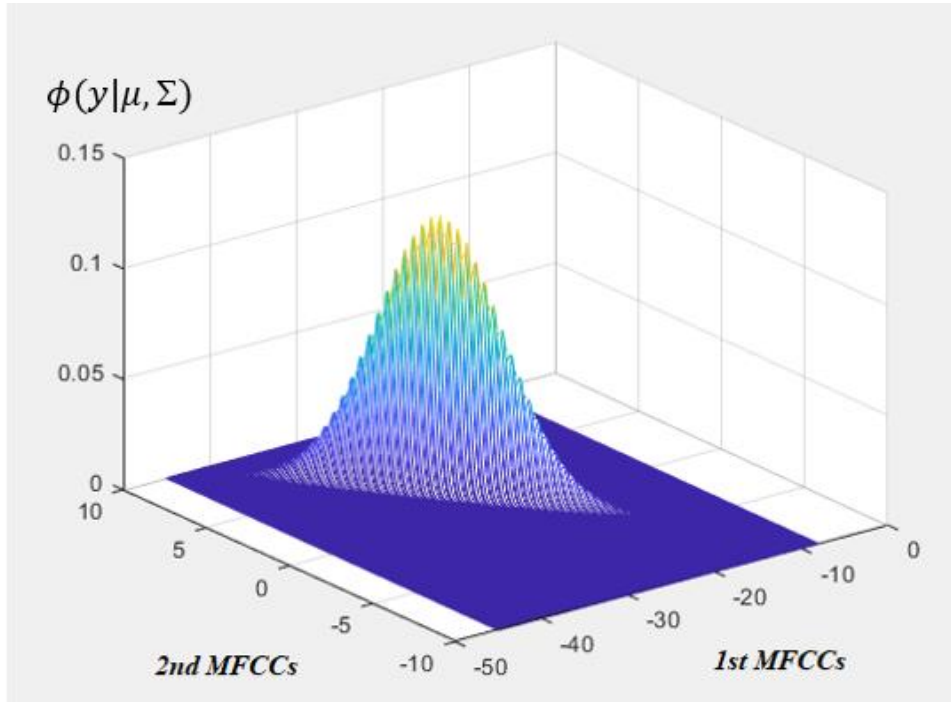


Figure 3.12 The first component of gaussian mixture distribution for first and second MFCCs before re-estimation.

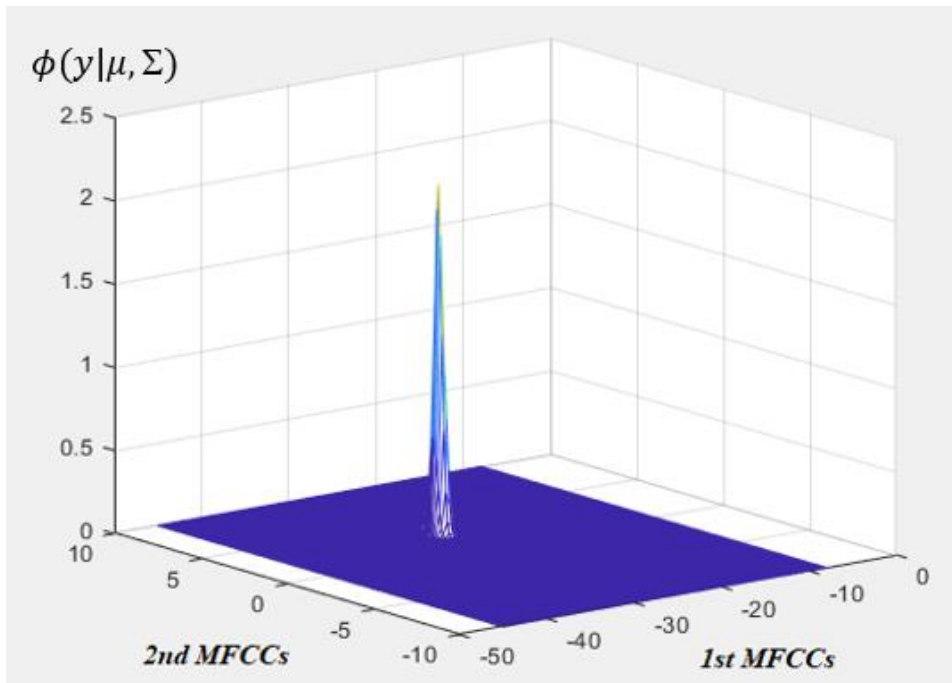


Figure 3.13 The first component of gaussian mixture distribution for first and second MFCCs after times re-estimation iteration.

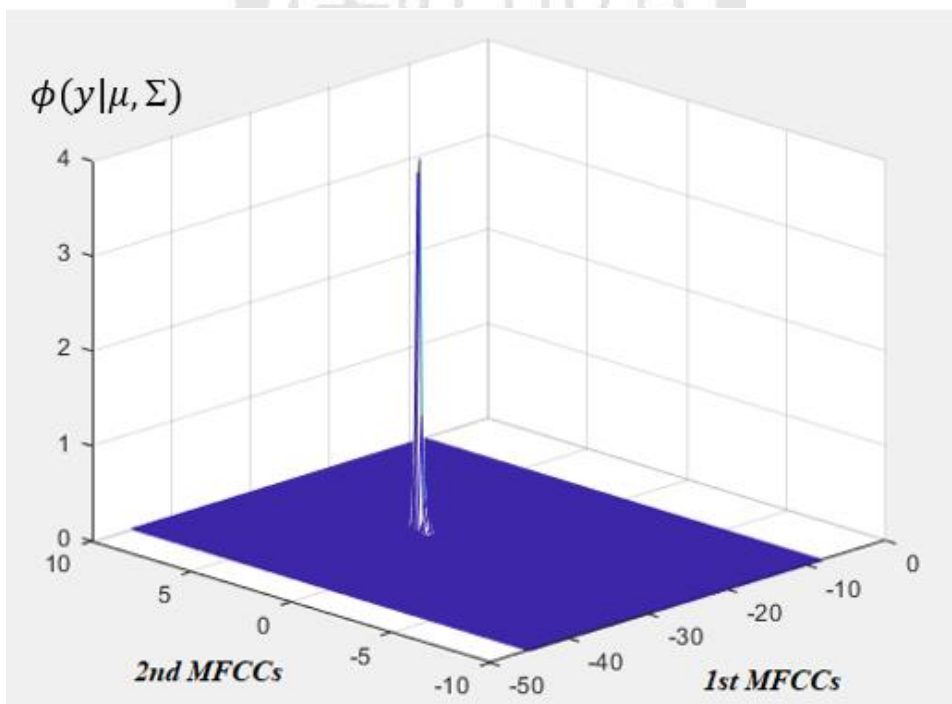


Figure 3.14 The first component of gaussian mixture distribution for first and second MFCCs after re-estimation converges.

To solve this problem, it recommends the DBN technique [18] because of the nonlinear property of neural networks. In this thesis, we use another approach. In this approach, set a singular value constraint 1×10^{-150} . If the determinant of the optimal covariance matrix is lesser than the constraint, reserve the covariance matrix and ignore the optimization afterward. It limits the covariance matrix to actual singularity and doesn't break the model after optimizing.

3.5 System Architecture

The recognition system architectures compared in this thesis are shown as Fig. 3.15 to Fig. 3.17. Figure top is the training stage, all training samples are fed to feature extraction as front-end, then process the initial model by different algorithms on three different recognition systems and train each emotion model prepared to recognize. Figure bottom is the testing stage, each testing sample is fed to feature extraction as similar to training stage, then process the observation sequence except the system using continuous HMM classifier. At last, fed to classification stage to recognize emotions by the collective models which are done training at training stage. Note that the notation $\{c_m\}_1^L$ represents the collection of L numbers temporary MFCCs with 13 number of coefficients, $\{g_k\}$ represents the global feature with $5 \times 13 = 65$ numbers, and N is the number of emotions we compared.

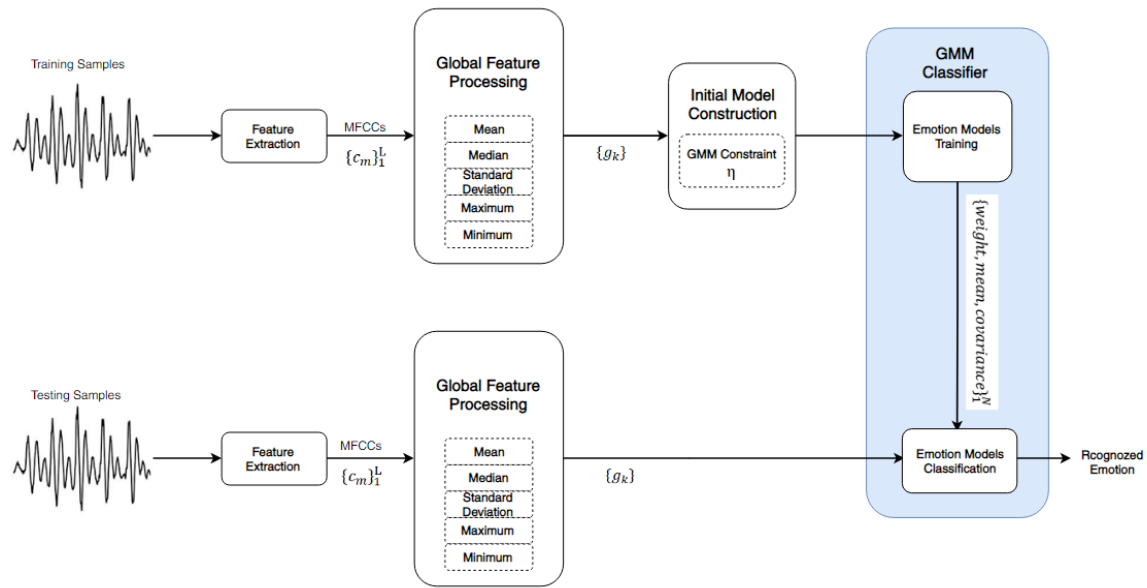


Figure 3.15 Recognition system with GMM as classifier.

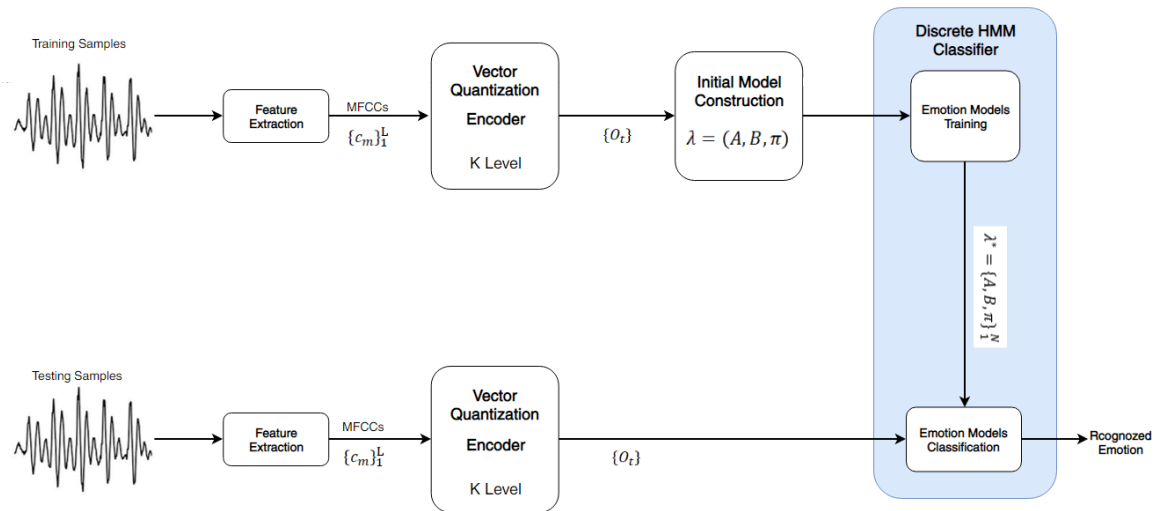


Figure 3.16 Recognition system with discrete HMM as classifier.

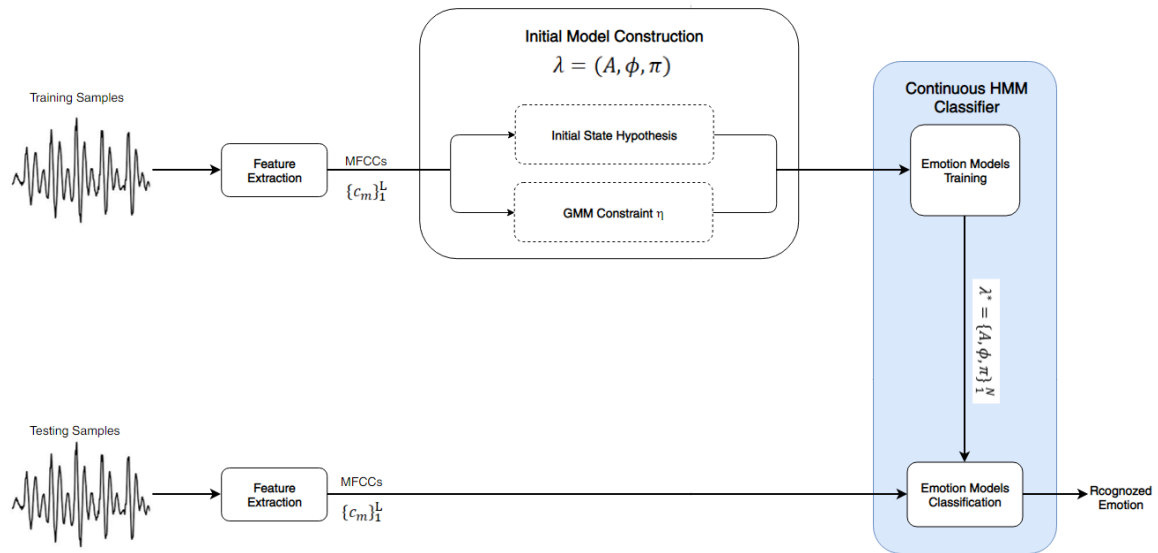


Figure 3.17 Recognition system with continuous HMM as classifier.

For the recognition system with GMM as classifier, there is a sub-processing to compute the global feature and concatenate to the initial model construction with GMM constraint. At the classifier stage, it outputs the emotion model with the weight, mean, covariance parameters. For the recognition system with discrete HMM as classifier, there is a sub-processing to encode the MFCCs feature to discrete observation index sequence and the initial model constructed with uniform transition matrix A , emission matrix B and initial probability π . At the classifier stage, it outputs the emotion model with the optimal transition matrix A , emission matrix B and uniform initial probability π . At last, the recognition system with continuous HMM as classifier, there is a sub-processing of the initial model constructed with initial state hypothesis and GMM constraint. At the classifier stage, it outputs the emotion model with the optimal transition matrix A , gaussian mixture pdf ϕ and uniform initial probability π .

3.6 Accuracy Estimation

Accuracy estimation is simple as Eq. (3.12).

$$Acc. = \frac{C}{N} \times 100\% \quad (3.12)$$

where C is the correct specified emotion, N is the number of total emotions that are compared. Note that if there are more than two probability of observation sequence $P(\mathbf{O}|\lambda)$ are the same, the system regards these cases as undistinguishable.

The accuracy result also shows the confusion table, it shows the percentage of which emotion the system recognizes as Eq. (3.13),

$$Conf. = \frac{c_{ij}}{N} \times 100\% \quad (3.13)$$

where i is the testing emotion, j is the emotion recognized. So $i = j$ is the case that testing emotion is recognized as correct emotion. Afterward, the testing emotion i at table first row, the recognized emotion j first at table column.

Chapter 4

EXPERIMENT RESULTS AND COMPARISONS

At this chapter, all the corresponding experiment results are compared including the number of GMM components, number of states, the constraint range η , number of quantization levels and the results with different initial state hypothesis approaches for three recognition systems with different classifiers. We also show some details in the mid-processing among the systems if the detail results affect accuracy results explicitly.

In section 4.1, describe how the experiment set. In section 4.2, compare the recognition system with GMM classifier by the number of components and different constraint ranges. In section 4.3, compare the recognition system with discrete HMM classifier by the number of states and number of quantization levels. In section 4.4, compare the recognition system with continuous HMM classifier by the number of components, number of states, different constraint ranges and different initial state hypothesis approaches. At last, in section 4.5, compare the best parameter settings among three recognition systems.

4.1 Experiment Settings

All training and testing samples are captured from the database SAVEE which is described at chapter three. We randomly choose five numbers of emotional utterances as training sample, the rest as testing sample. Every experiment has five random training

samples for each emotion, ten random samples for testing, except 25 random samples for neutral emotion testing.

The recognition experiment is set as speaker dependent training, so we have four different speakers labelled as DC, JE, JK, KL, then distinctly make the experiment with different speakers. We train each emotion with batch model, that is, the specified emotional training samples are concatenated as an utterance batch.

Every accuracy result is taken the average of five experiments with corresponding speaker, the confusion results are also taken the average of five experiments. All the average results are shown as charts.

4.2 Comparisons for GMM Classifier System

In this section, compare the recognition system with GMM classifier by the number of components and constraint range. Number of components is choose two, four, eight to compare. The constraint range η is choose 0.1, 0.2, 0.5 and 0.8, the difference among those choice will be seen at subsection.

At the processing, also show figures about gaussian mixture probability density with first and second coefficient of the global MFCCs, that is, the mean and median coefficients for gaussian mixture probability density.

4.2.1 Comparisons for Number of Components

For this comparison, choose constraint range $\eta = 0.2$, compare two, four, eight number of components separately. The initial models for neutral emotion and optimal models with different number of components are shown as Fig. 4.1 to Fig 4.3. Fig. 4.1 shows that there are two components gaussian probability density located at two different position with equal weight value 0.5. After optimization iteration, the position is changed to close the feature

position, the two components locate closely with different weight values, about 0.6 and 0.4. They are located too closely to become almost one component. Fig 4.2 shows that there are four components gaussian probability density located at four different positions with equal weight. After optimization iteration, the position is also changed to close the feature position, but the weight value almost distributes to second and third component. And Fig. 4.3 shows that there are eight components gaussian probability density located at eight different position with equal weight. After optimization iteration, the result is similar as Fig. 4.2. It could predict that the accuracy results are close when using four or eight number of components.

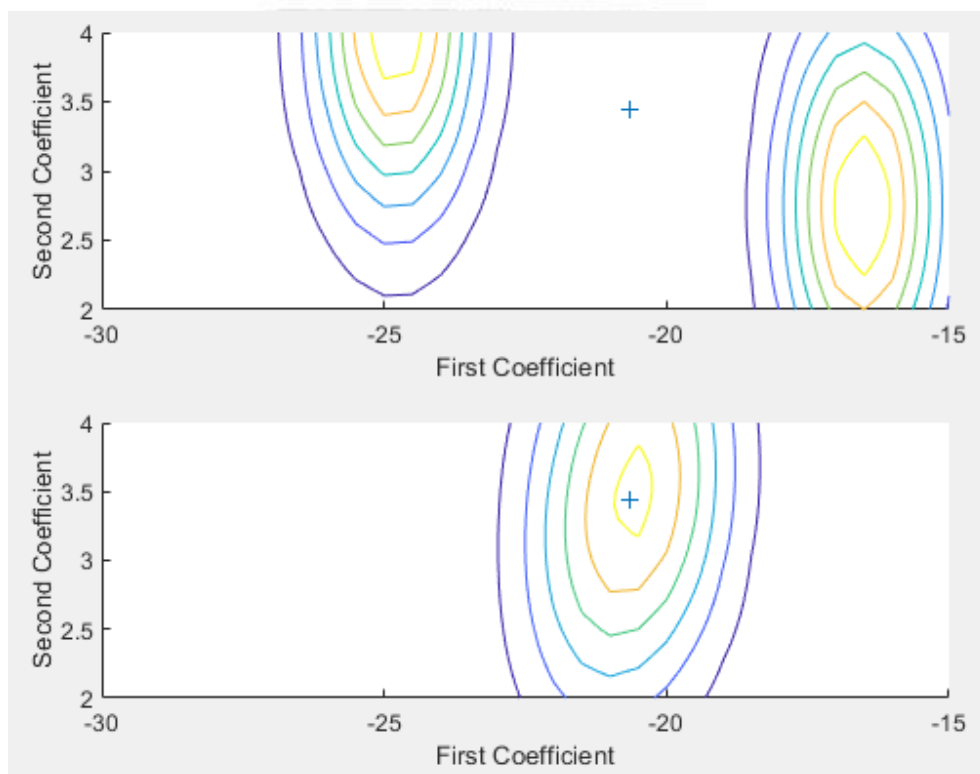


Figure 4.1 The initial model and optimal model with two number of components for neutral emotion.

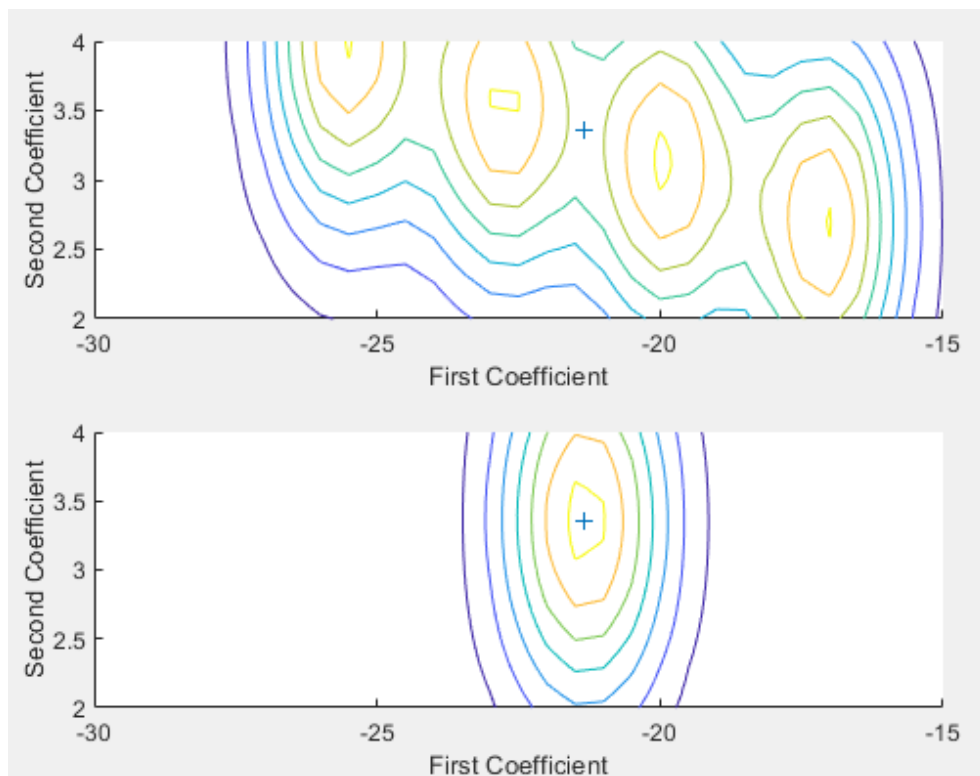


Figure 4.2 The initial model and optimal model with four number of components for neutral emotion.

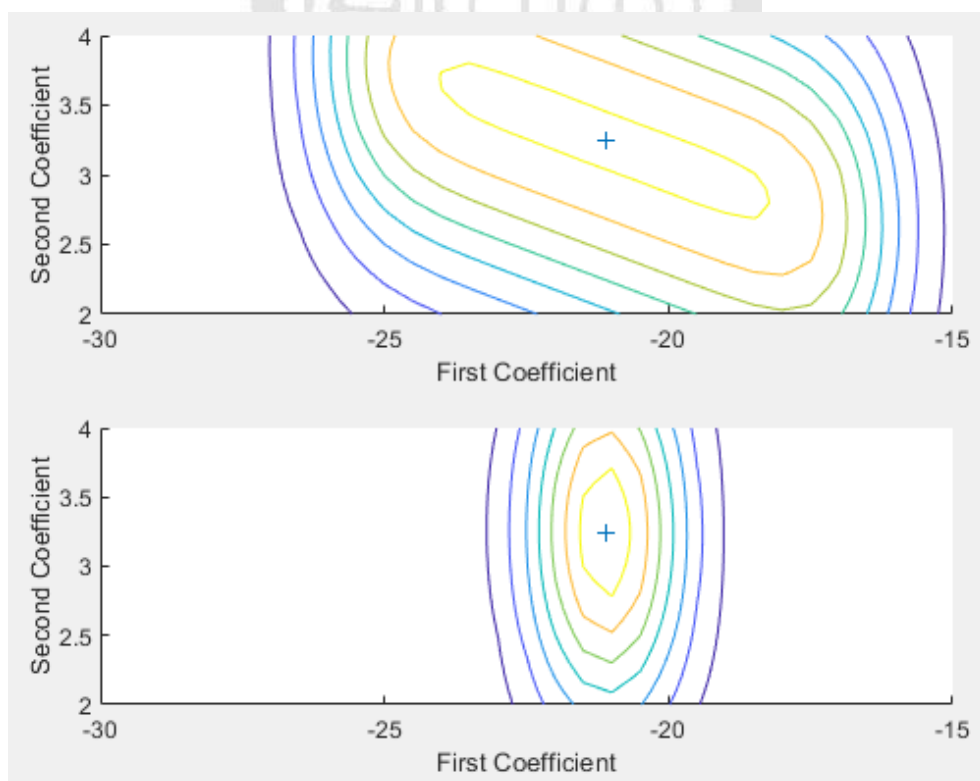


Figure 4.3 The initial model and optimal model with eight number of components for neutral emotion.

Now, Fig 4.4 shows accuracy results among different number of components, the highest accuracy is 58.07% for speaker KL at four number of components. Highest average accuracy is 50.25% at eight number of components.

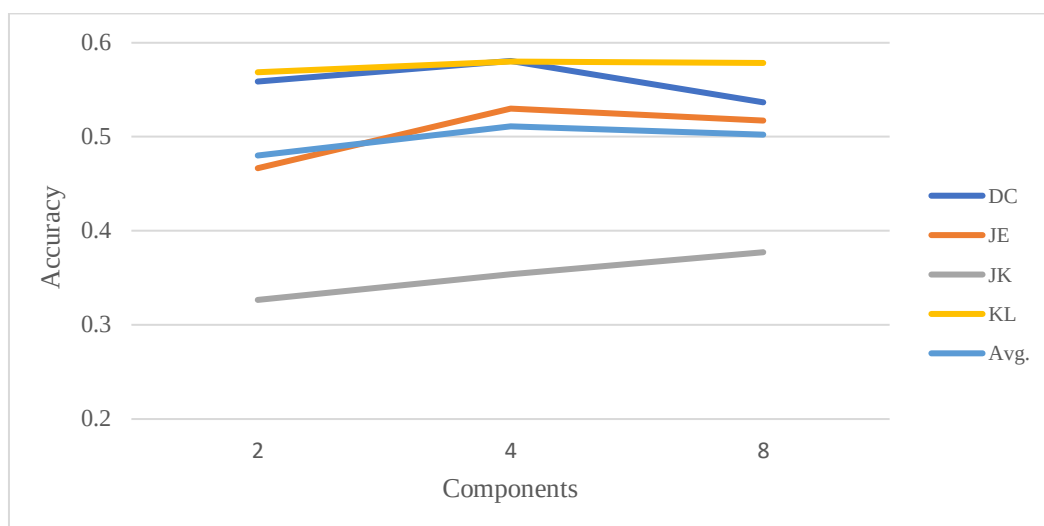


Figure 4.4 Accuracy results for different number of components.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	62%	2%	34%	0%	0%	0%
Disgust	0%	50%	24%	2%	6%	18%
Fear	0%	16%	48%	32%	0%	4%
Happy	12%	2%	32%	54%	0%	0%
Neutral	0%	4%	0%	0%	75.2%	20.8%
Sadness	0%	2%	0%	0%	34%	46%

Table 4.1 Confusion table with two number of components for speaker DC using GMM classifier system.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	72%	0%	6%	22%	0%	0%
Disgust	0%	48%	14%	10%	6%	22%
Fear	2%	10%	52%	32%	2%	2%
Happy	18%	0%	38%	44%	0%	0%
Neutral	0%	0.8%	0%	0%	78.4%	20.8%
Sadness	0%	6%	0%	0%	40%	54%

Table 4.2 Confusion table with four number of components for speaker DC using GMM classifier system.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	66%	0%	4%	30%	0%	0%
Disgust	0%	42%	20%	8%	8%	22%
Fear	0%	16%	42%	34%	0%	8%
Happy	28%	4%	16%	52%	0%	0%
Neutral	0%	1.6%	0%	0%	72%	26.4%
Sadness	0%	10%	0%	0%	42%	48%

Table 4.3 Confusion table with eight number of components for speaker DC using GMM classifier system.

4.2.2 Comparisons for GMM Constraint

After compared among different number of components, choose eight number of components for our setting due to the best average accuracy result. Compare the different constraint range parameter η with 0.1, 0.2, 0.5 and 0.8 separately. The initial models for neutral emotion and optimal models with different constraint range are shown as Fig. 4.5 to Fig 4.8. The initial gaussian probability density is enclosed within larger range, the gaussian

probability density after optimization is different while looking closely. To increase the range, the initial gaussian probability density is enclosed within much larger range. Afterward the initial gaussian probability density is enclosed within much larger range than Fig. 4.7 while choosing $\eta = 0.8$. However, the optimization model seems little different among four different ranges.

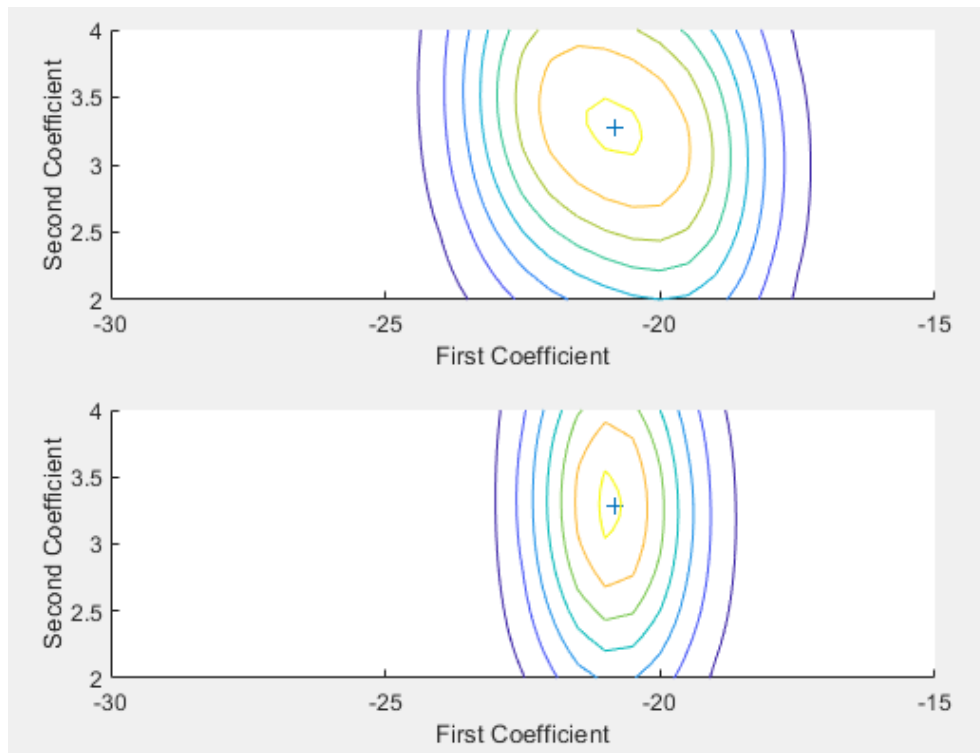


Figure 4.5 The initial model and optimal model with the constraint range $\eta = 0.1$ for neutral emotion.

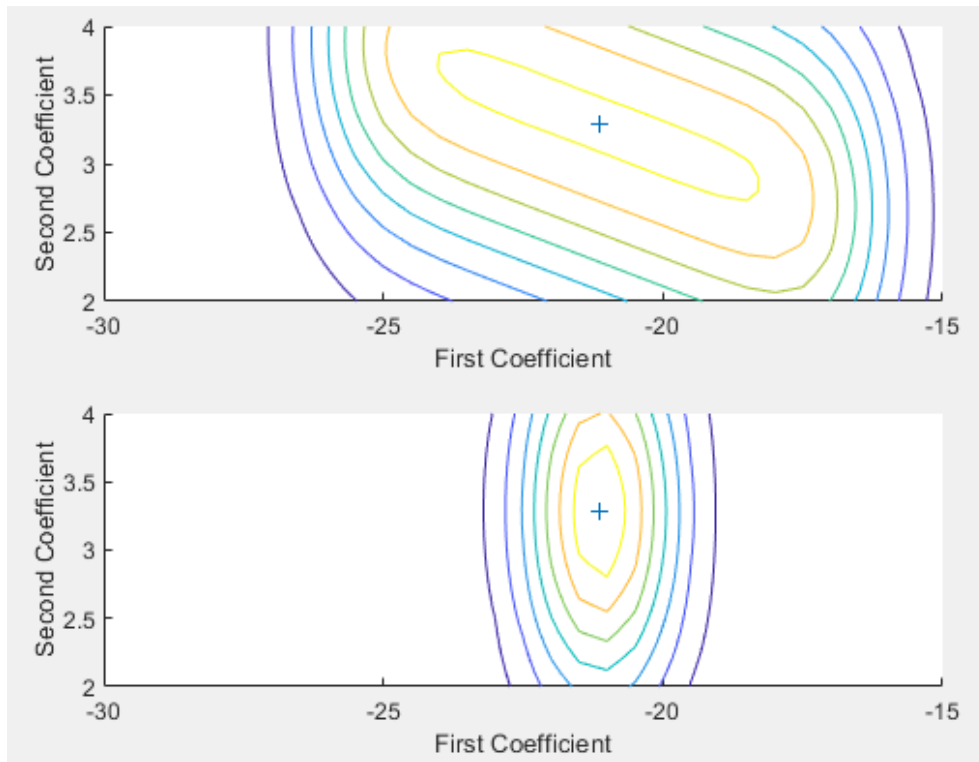


Figure 4.6 The initial model and optimal model with the constraint range $\eta = 0.2$ for neutral emotion.

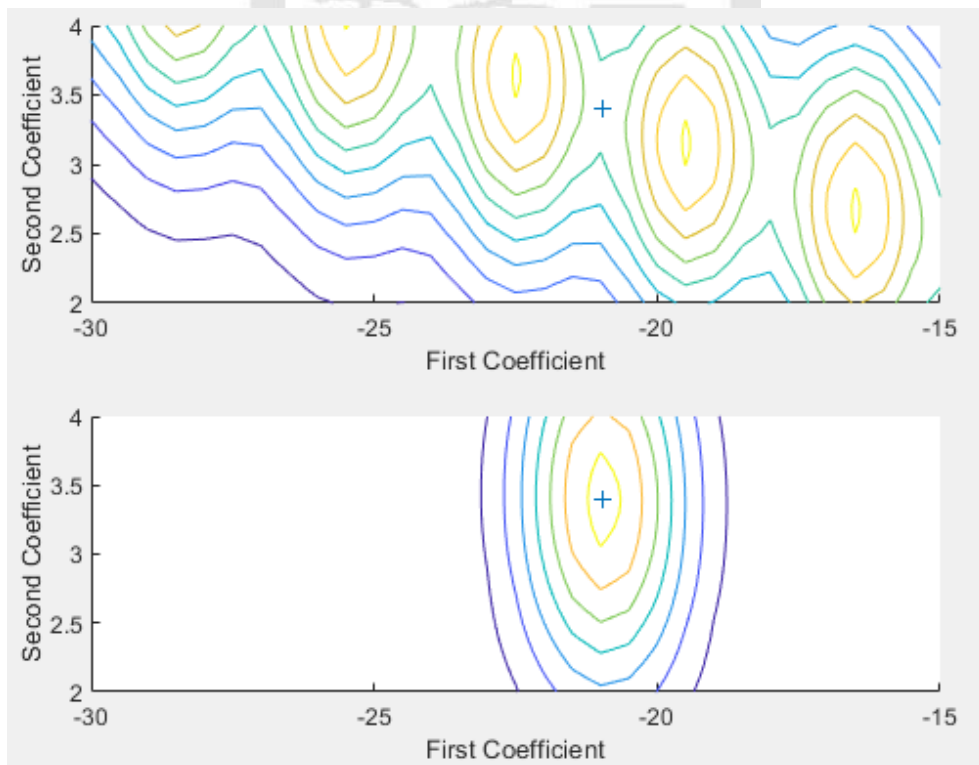


Figure 4.7 The initial model and optimal model with the constraint range $\eta = 0.5$ for neutral emotion.

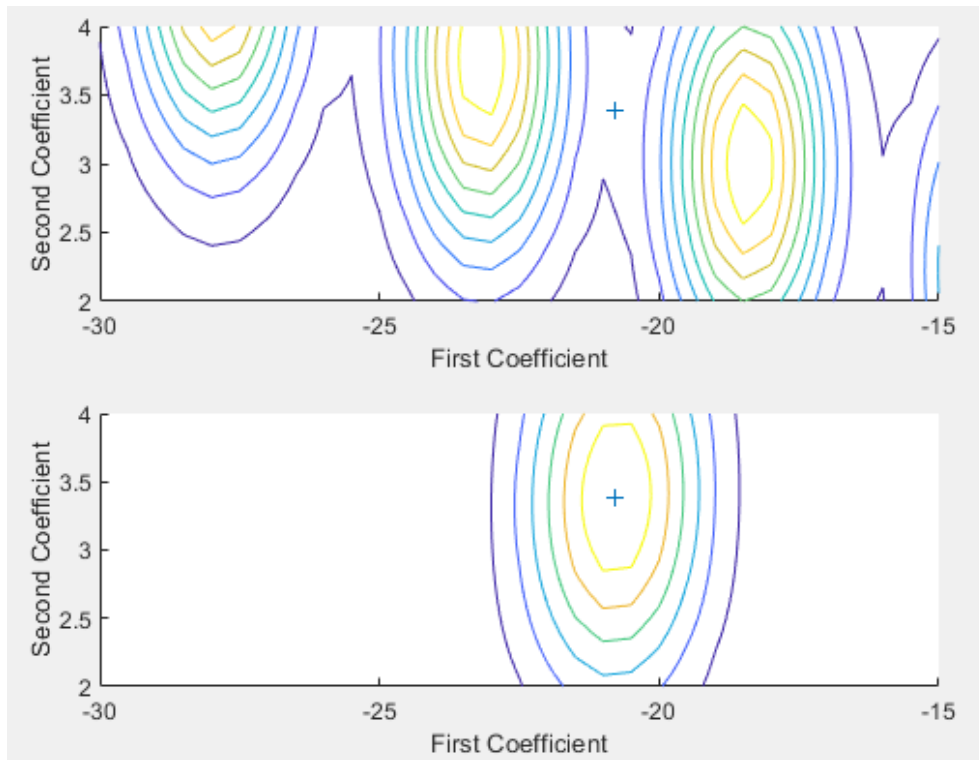


Figure 4.8 The initial model and optimal model with the constraint range $\eta = 0.8$ for neutral emotion.

Now, Fig 4.9 shows accuracy results among different constraint range. The highest accuracy is 59.13% for speaker KL at constraint range $\eta = 0.1$. Highest average accuracy is 50.32% at constraint range $\eta = 0.8$. As the average result shown, all average accuracies are about 50%, the constraint range doesn't much affect the GMM classifier recognition system.

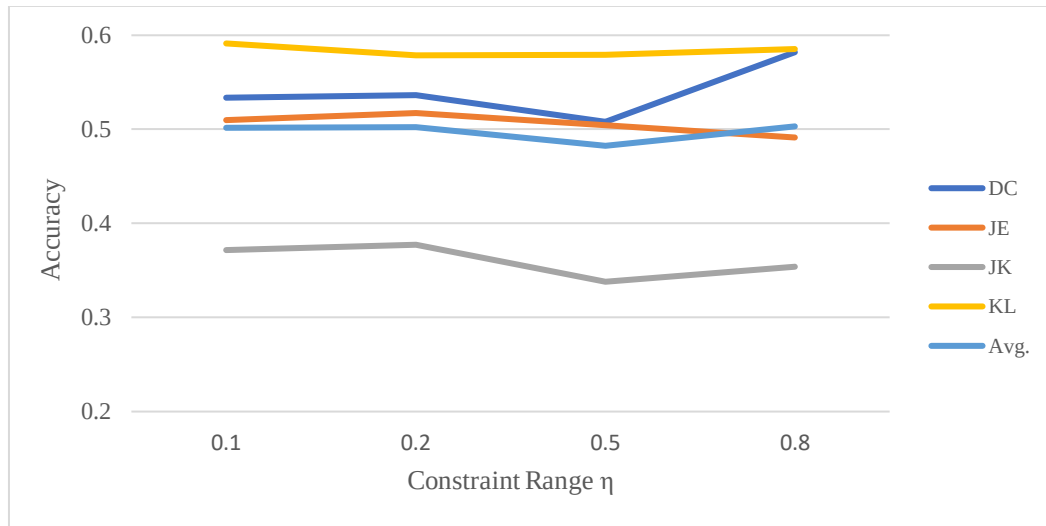


Figure 4.9 Accuracy results for different constraint range.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	66%	0%	2%	32%	0%	0%
Disgust	0%	40%	24%	0.08%	0.08%	20%
Fear	4%	16%	46%	28%	2%	4%
Happy	18%	0%	28%	54%	0%	0%
Neutral	0%	1.6%	0%	0%	72%	26.4%
Sadness	0%	20%	4%	0%	34%	42%

Table 4.4 Confusion table with constraint range $\eta = 0.1$ for speaker DC using GMM classifier system.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	66%	0%	4%	30%	0%	0%
Disgust	0%	42%	20%	8%	8%	22%
Fear	0%	16%	42%	34%	0%	8%
Happy	28%	4%	16%	52%	0%	0%
Neutral	0%	1.6%	0%	0%	72%	26.4%
Sadness	0%	10%	0%	0%	42%	48%

Table 4.5 Confusion table with constraint range $\eta = 0.2$ for speaker DC using GMM classifier system.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	60%	2%	4%	34%	0%	0%
Disgust	0%	32%	26%	2%	188%	22%
Fear	2%	18%	46%	30%	4%	0%
Happy	22%	2%	22%	54%	0%	0%
Neutral	0%	0%	0%	0%	68.8%	31.2%
Sadness	0%	2%	2%	0%	52%	44%

Table 4.6 Confusion table with constraint range $\eta = 0.5$ for speaker DC using GMM classifier system.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	66%	0%	4%	30%	0%	0%
Disgust	0%	50%	20%	6%	0%	24%
Fear	6%	14%	50%	28%	2%	0%
Happy	24%	2%	26%	48%	0%	0%
Neutral	0%	0%	0%	0%	75.2%	24.8%
Sadness	0%	8%	2%	0%	30%	60%

Table 4.7 Confusion table with constraint range $\eta = 0.8$ for speaker DC using GMM classifier system.

4.2.3 Comparisons for Feature Type

At this subsection, we show that the temporary local feature is more improper than global statistical feature via showing the plot of initial and optimal gaussian mixture density. In this experiment, the temporary feature is extracted without computing of statistical parameters and choose two number of components, constraint range $\eta = 0.2$. The initial and optimal gaussian mixture density for neutral emotion are shown as Fig. 4.10. After optimization iteration, the gaussian mixture density is too hard to fit all the temporary

features. because the features are considered simultaneously without the time information as mentioned at chapter two, the features scatter within large area which is caused gaussian mixture density unfit. We could consider the accuracy results are much very low as Fig. 4.11 shown. The average of accuracy using temporary feature is less than using global feature about 24.42% and highest difference is even less about 32.47%. Definitely, the GMM classifier recognition system is very improper to use temporary feature, both the mid-processing and accuracy results confirm it.

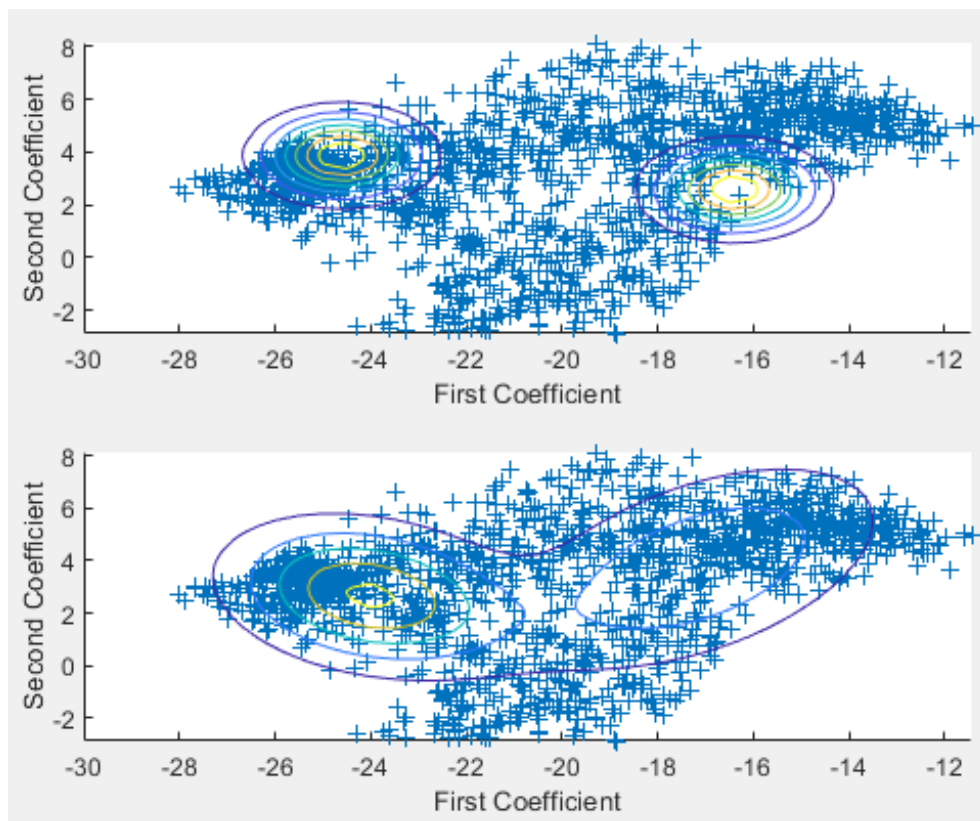


Figure 4.10 The initial and optimal gaussian mixture density for neutral emotion.

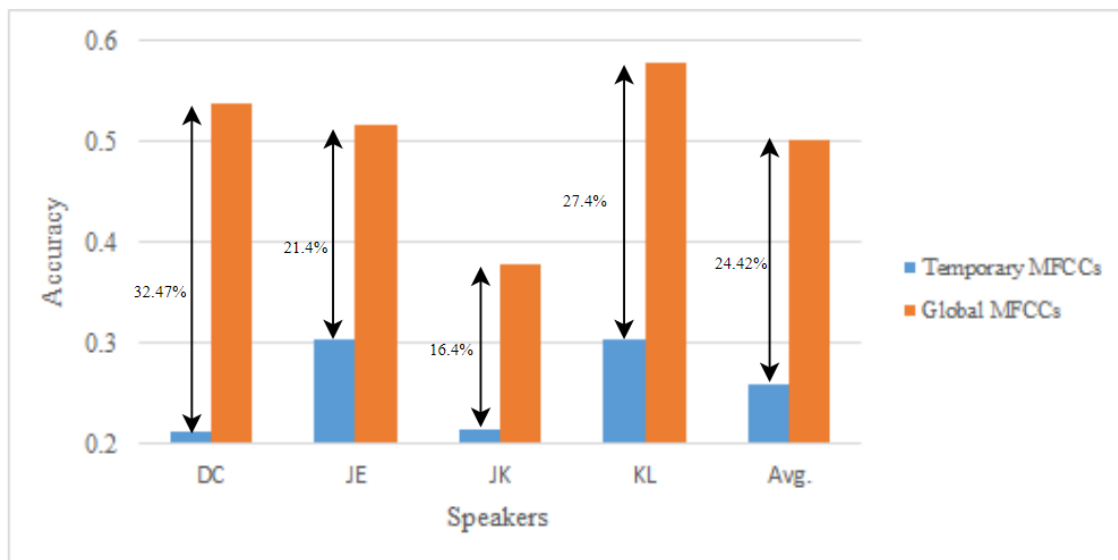


Figure 4.11 Accuracy results comparison between using temporary MFCCs as feature and using global MFCCs as feature.

4.3 Comparisons for Discrete HMM Classifier System

In this section, compare the recognition system with discrete HMM classifier by the number of states and different quantization levels. Number of states is chosen two, four, eight to compare. The quantization level is chosen 32, 64, 128. First compare the number of states and compare the different quantization levels at next subsection.

Show the change of emission matrix before and after optimization due to the importance, and show the affection caused at mid-processing in order to make the discrete HMM classifier clear.

4.3.1 Comparisons for Number of States

For this comparison, set quantization level as 32, as described at chapter three, the transition matrix and initial emission matrix are set uniform and every emotion model uses this initial model. At last, compare two, four, eight number of states. The initial emission matrix for four states is shown as Fig. 4.12. The sum of probability equal to one as the

probability law defined for any specified state. Initially, the zero probability makes the observation never happened in specified state to distinguish different states.

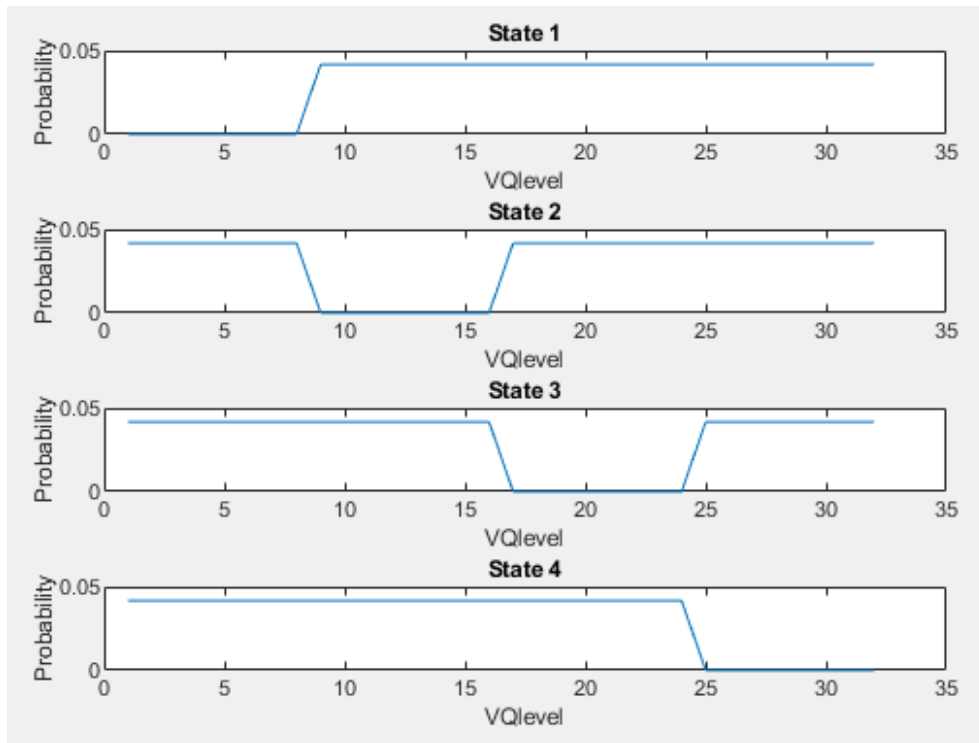


Figure 4.12 The discrete probability distribution of the emission matrix for four states.

After optimization, the optimal model for each emotion becomes different and distinguishable. Fig. 4.13 and Fig. 4.14 show the optimal neutral model and sadness model. The sum of probability also equal to one followed by the probability law. Now the likelihood that every observation occurs at state is clear. For example, the index ten observation never occurs but at state three. The index twelve observation both occurs at state one and state three, but obviously it is most probably occurring at state one. It must note that as Fig. 4.14 there is an index, index 23, the emission probabilities are all zero. That means for this optimal sadness model, this observation should never happen.

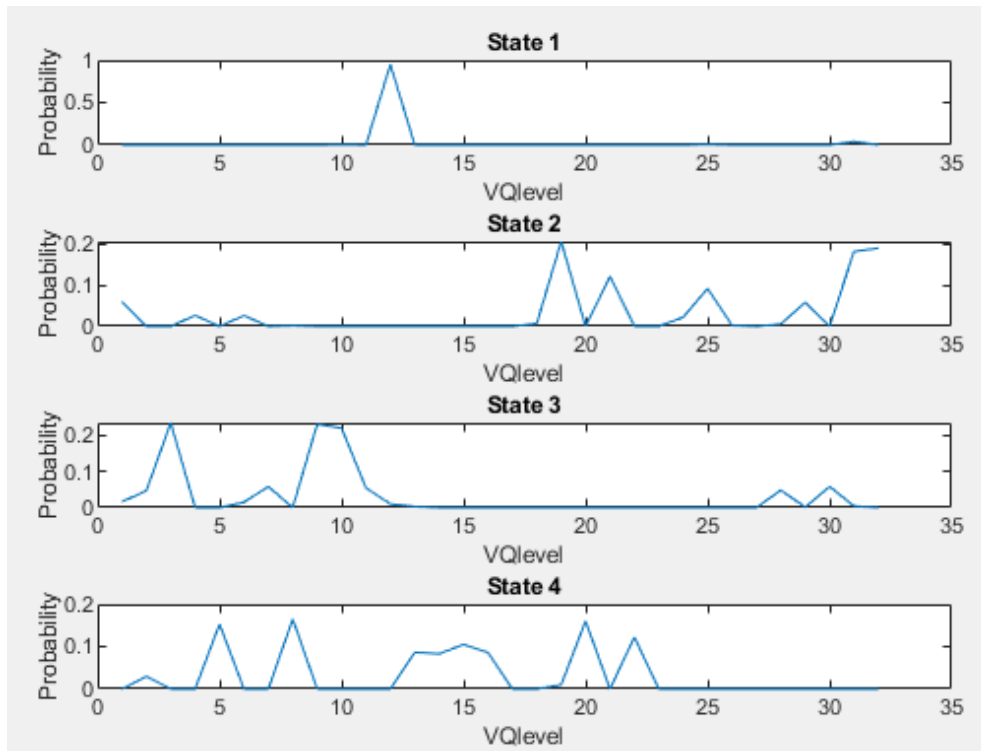


Figure 4.13 The optimal emission matrix for neutral emotion.

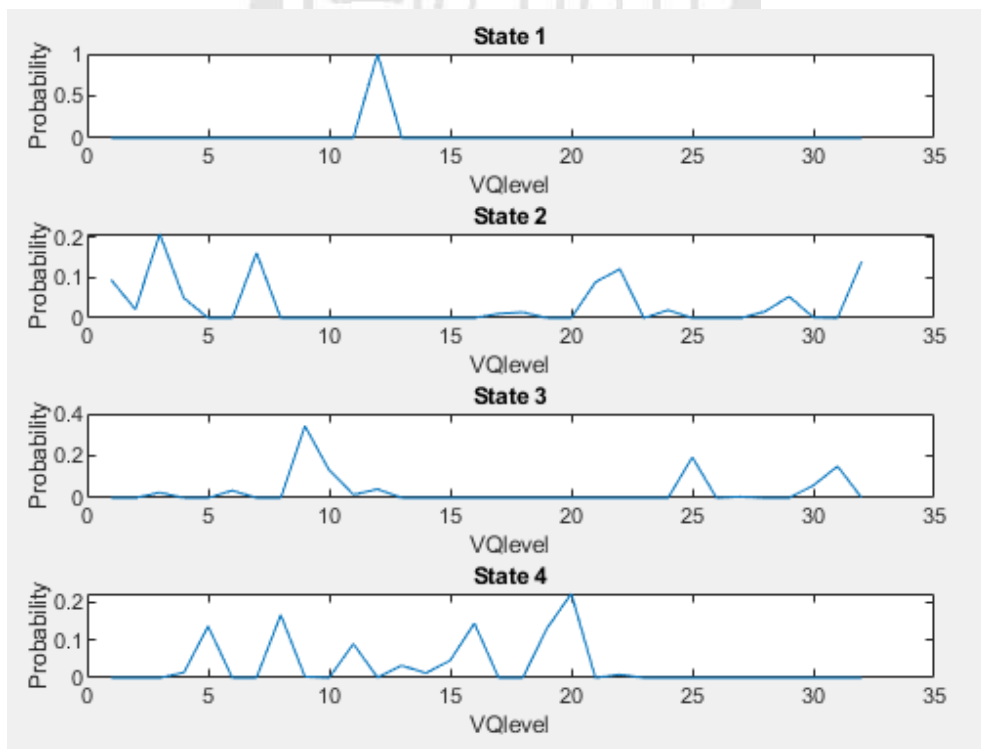


Figure 4.14 The optimal emission matrix for sadness emotion.

Here's a problem, if there is the particular index observation in the practical testing sample, it makes the forward variable zero and so the rest forward variable. That means that the discrete HMM recognition system considers that this testing sample is not right emotion even though one impossible observation occurs. This problem usually happens in practical case, so we use a very small value instead of the zero probability. Then the system becomes functional and the accuracy result is shown as Fig. 4.15. The highest accuracy is 58.53% for speaker DC at two states. Highest average accuracy is 46.01% at two states.

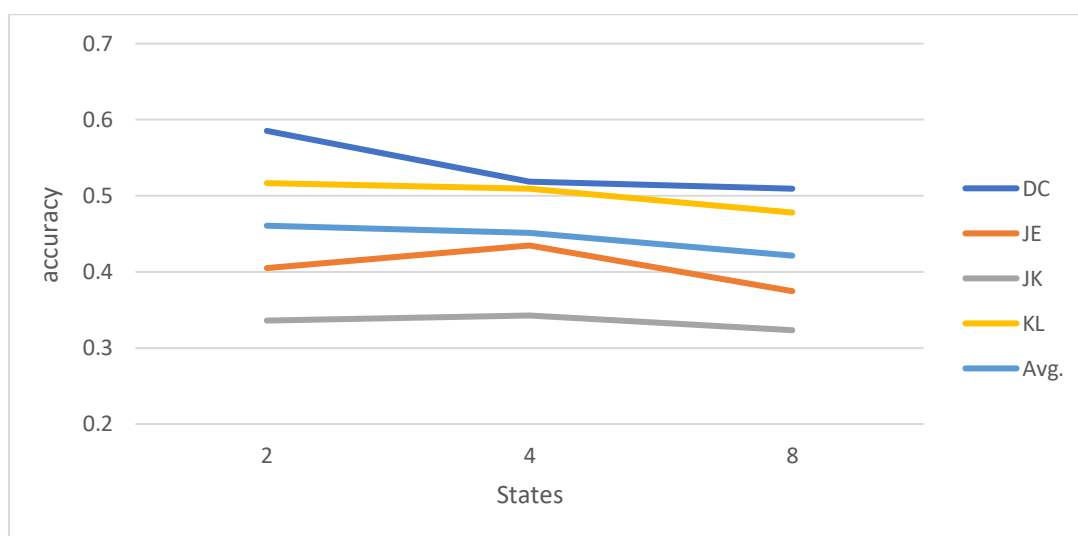


Figure 4.15 Accuracy results for different number of states.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	52%	0%	8%	40%	0%	0%
Disgust	0%	62%	14%	12%	0%	12%
Fear	0%	10%	64%	26%	0%	0%
Happy	16%	2%	38%	44%	0%	0%
Neutral	0%	10.4%	0%	0%	71.2%	18.4%
Sadness	0%	30%	2%	0%	12%	58%

Table 4.8 Confusion table with two states for speaker DC using discrete HMM classifier system.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	64%	0%	6%	30%	0%	0%
Disgust	4%	48%	20%	18%	0%	10%
Fear	0%	14%	56%	30%	0%	0%
Happy	28%	2%	42%	28%	0%	0%
Neutral	0%	12%	1.6%	0.8%	63.2%	22.4%
Sadness	0%	32%	8%	2%	6%	52%

Table 4.9 Confusion table with four states for speaker DC using discrete HMM classifier system.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	76%	0%	2%	22%	0%	0%
Disgust	2%	72%	6%	4%	0%	16%
Fear	18%	18%	24%	38%	0%	2%
Happy	40%	6%	24%	30%	0%	0%
Neutral	0%	15.2%	0%	0%	65.6%	19.2%
Sadness	0%	42%	2%	0%	18%	38%

Table 4.10 Confusion table with eight states for speaker DC using discrete HMM classifier system.

However, if we take a look for the 128-quantization level comparison, the optimal eight-state emission matrix for neutral emotion has too many zero emission probability at all states as Fig. 4.16 shown. Even use a very small value instead of the zero probability, it seriously affects the recognition processing due to the bad optimal emission matrix. After that, abandon the discrete HMM classifier recognition system with too many numbers of states.

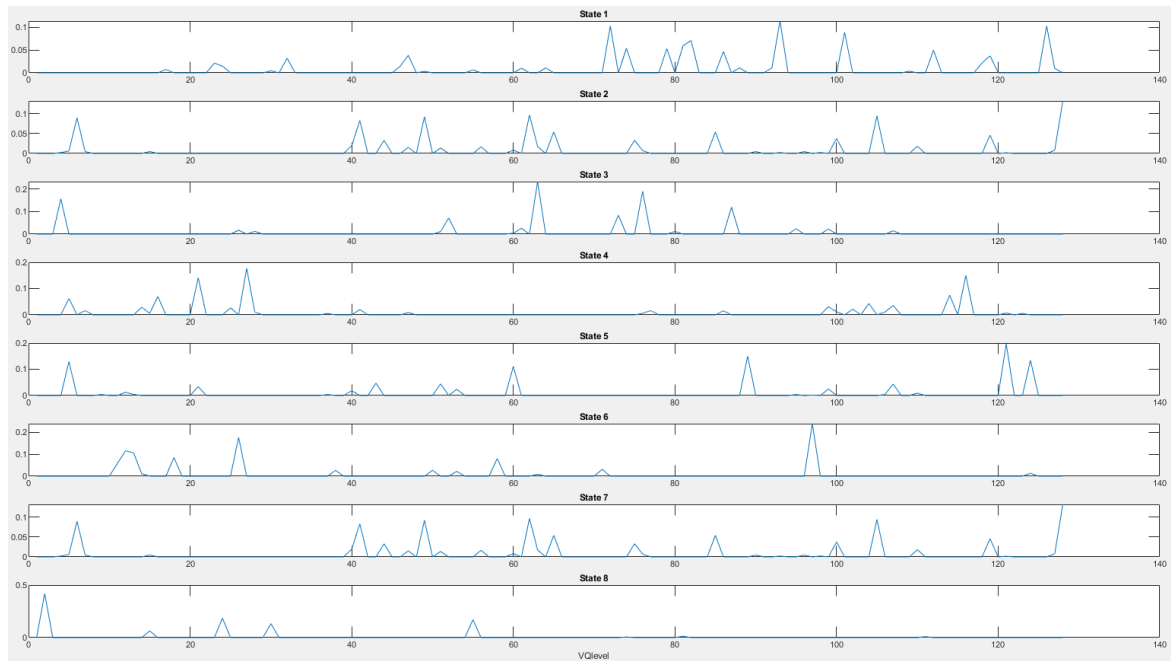


Figure 4.16 The optimal neutral emotion emission matrix for 128 VQ level.

4.3.2 Comparisons for Vector Quantization Levels

As described previous subsection, choose two number of states for this comparison. Compare among 32, 64, 128 quantization level with the initial settings as similar as previous comparison. The optimal neutral emotion emission matrix for 32 and 128 quantization level are shown as Fig. 4.17 and Fig. 4.18. As can be seen, the probability distributes more dispersedly within wider levels while using higher quantization levels.

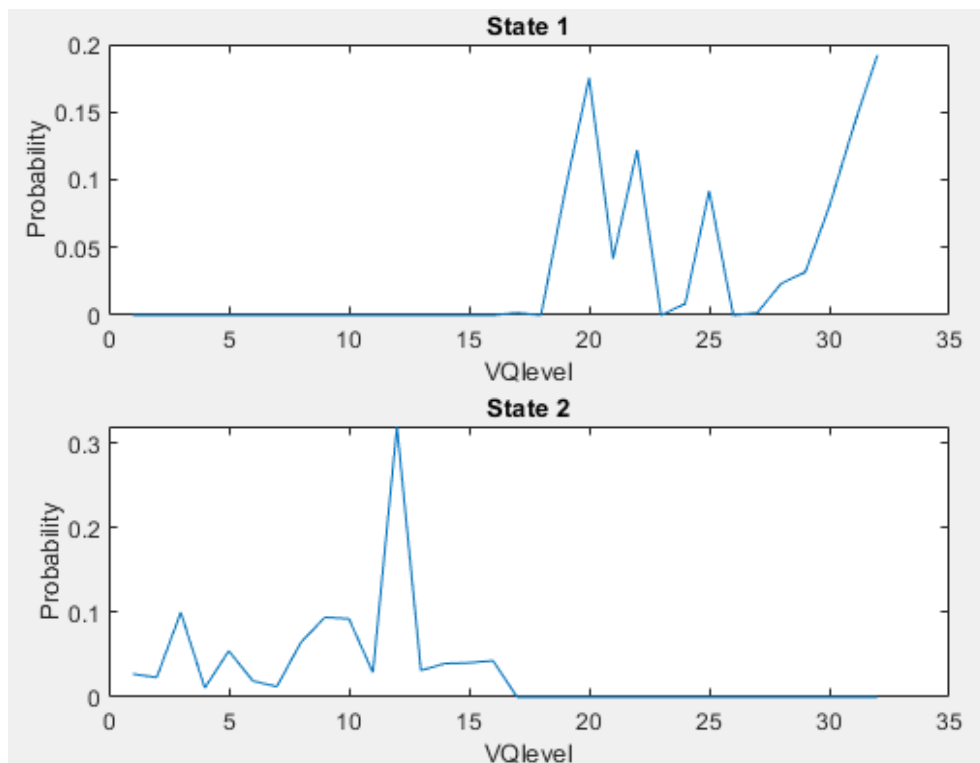


Figure 4.17 The optimal neutral emotion emission matrix for 32 VQ level.

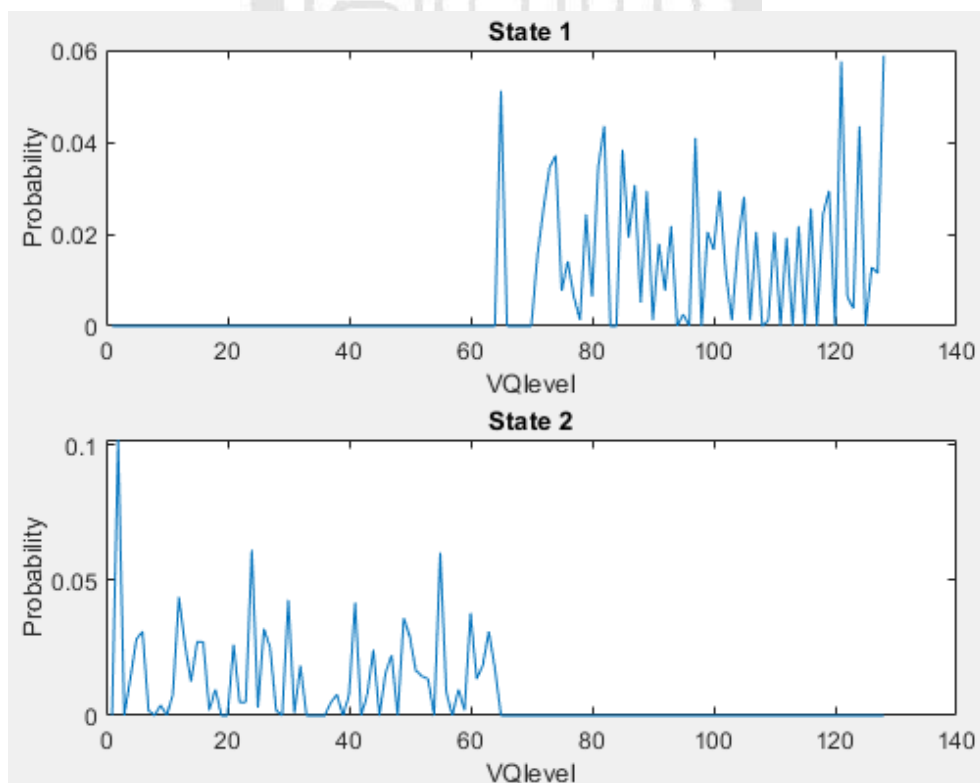


Figure 4.18 The optimal neutral emotion emission matrix for 128 VQ level.

For the difference of optimal emission matrix, the discrete HMM with 128 level can usually better recognize each observation sequence as its correct emotion. Now, Fig. 4.19 shows the accuracy comparison among the different quantization level. Highest accuracy is 65.67% for speaker DC at 64 level. Highest average accuracy is 53.2% at 128 levels. Moreover, most accuracy results show that 128 level are better than less level except there's 1.74% less than 64 level for speaker DC.

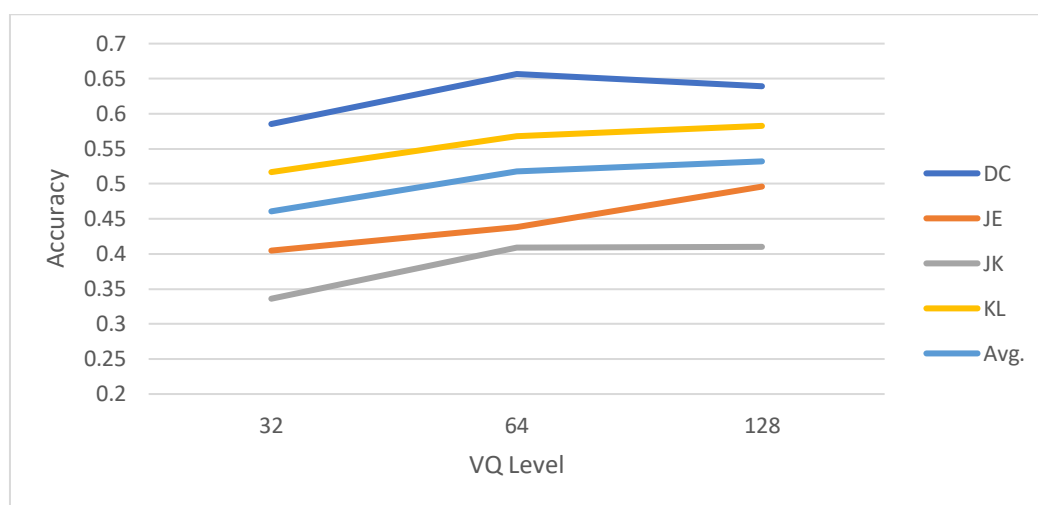


Figure 4.19 Accuracy results for different quantization levels.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	52%	0%	8%	40%	0%	0%
Disgust	0%	62%	14%	12%	0%	12%
Fear	0%	10%	64%	26%	0%	0%
Happy	16%	2%	38%	44%	0%	0%
Neutral	0%	10.4%	0%	0%	71.2%	18.4%
Sadness	0%	30%	2%	0%	12%	58%

Table 4.11 Confusion table with 32 VQ levels for speaker DC using discrete HMM classifier system.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	52%	0%	18%	30%	0%	0%
Disgust	0%	58%	18%	18%	0%	6%
Fear	2%	10%	60%	28%	0%	0%
Happy	14%	2%	24%	60%	0%	0%
Neutral	0%	7.2%	0.8%	3.2%	84%	4.8%
Sadness	0%	12%	2%	2%	4%	80%

Table 4.12 Confusion table with 64 VQ levels for speaker DC using discrete HMM classifier system.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	70%	0%	4%	26%	0%	0%
Disgust	6%	44%	18%	6%	6%	2%
Fear	0%	6%	46%	48%	0%	0%
Happy	28%	0%	22%	50%	0%	0%
Neutral	0%	0%	0%	0%	89.6%	10.4%
Sadness	0%	2%	0%	0%	14%	84%

Table 4.13 Confusion table with 128 VQ levels for speaker DC using discrete HMM classifier system.

4.4 Comparisons for Continuous HMM Classifier System

The recognition system with continuous HMM classifier is much more complex than GMM classifier and discrete HMM classifier. Moreover, we also propose the initial hypothesis approaches according to the property of EM algorithm s [23] [29]. In this section, compare the recognition system with continuous HMM classifier by the number of states, number of GMM components, the GMM constraint range and different approaches of initial

state hypothesis.

Number of states is choose two, four, eight to compare, number of components is choose two, four, eight, the constraint range η is choose 0.1, 0.2, 0.5, 0.8 as similar as above comparison and compare the uniformly segmented hypothesis and clustering method hypothesis.

4.4.1 Comparisons for Number of States

For this comparison, set two number of components, constraint range $\eta = 0.2$ and use uniformly segmented initial state hypothesis, then compare two, four, eight number of states. The two states initial model and optimal model for neutral emotion is shown as Fig. 4.20. For the initial model, the two circles at upper left are belonged to state one which is hypothesized by uniformly segmented, at upper right are belonged to state two. After optimization, the gaussian mixture density becomes concentrated for the state one circle. For the state two circle, it becomes to include very large range and another component include lower area. Next, the four states optimal model for neutral emotion is shown as Fig. 4.21. After optimization, the gaussian mixture density concentrate on the left position due to the probability of observation, others include within large range with at least three components but small probability.

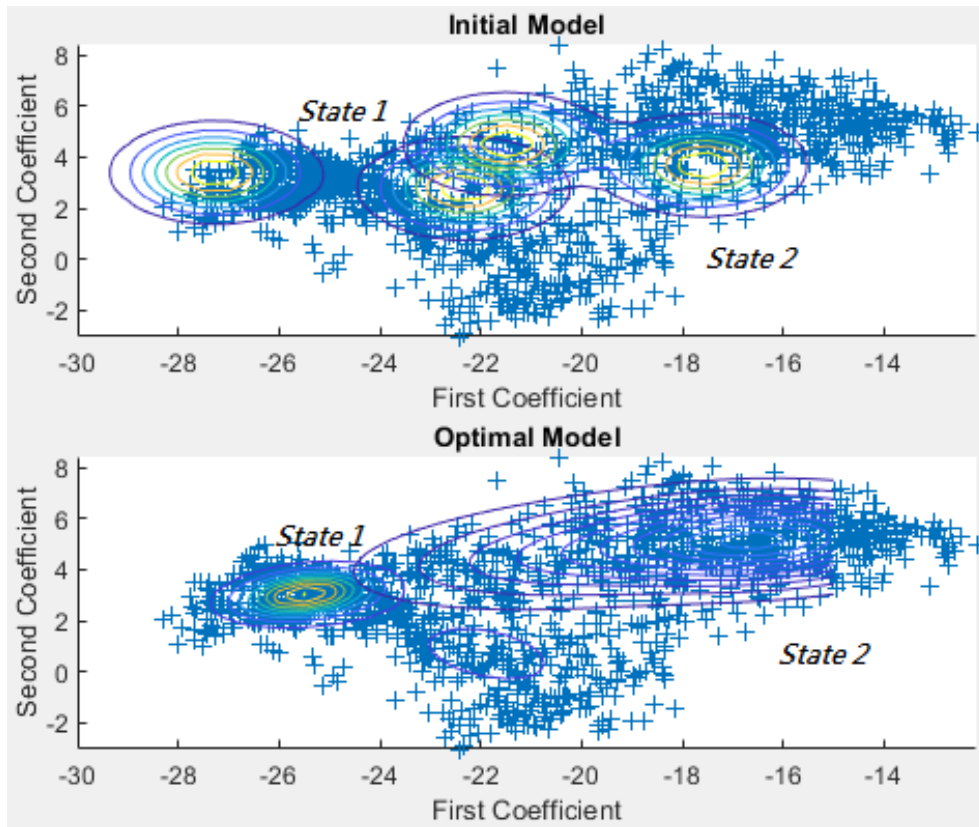


Figure 4.20 The two states initial model and optimal model for neutral emotion.

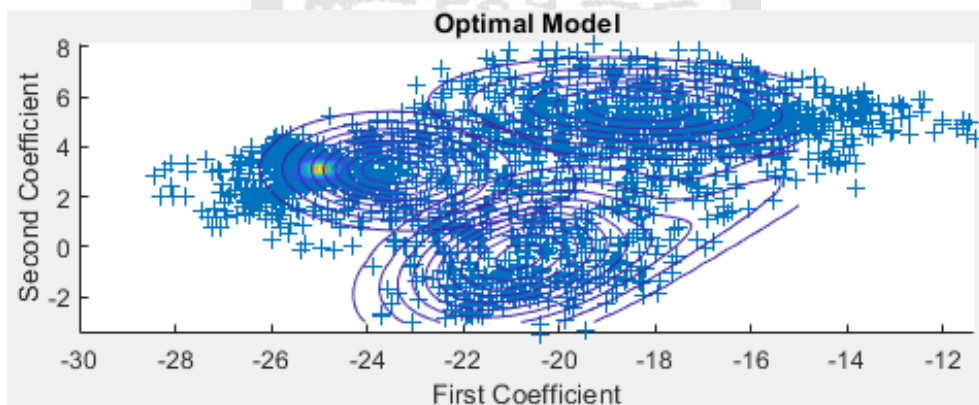


Figure 4.21 The four states optimal model for neutral emotion.

After showing the mid-processing of continuous HMM classifier recognition system, Fig. 4.22 shows the accuracy results comparison for different number of states. Highest accuracy is 81.47% for speaker KL at two states. Highest average accuracy is 68.02% at four states. The number of states affects just a little bit.

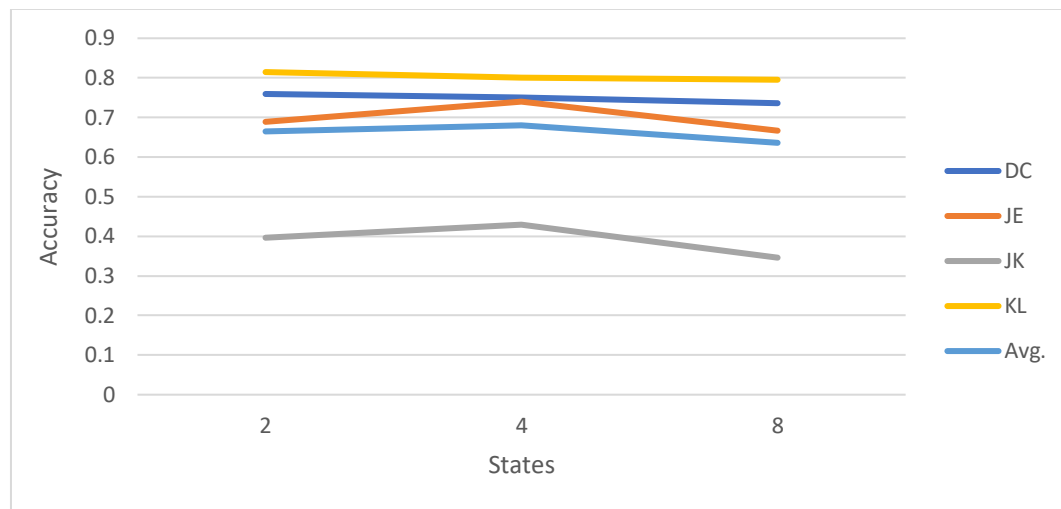


Figure 4.22 Accuracy results for different number of states.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	76%	4%	2%	18%	0%	0%
Disgust	2%	64%	10%	6%	4%	14%
Fear	0%	2%	54%	40%	0%	4%
Happy	10%	2%	14%	74%	0%	0%
Neutral	0%	0%	0%	0%	89.6%	10.4%
Sadness	0%	2%	0%	0%	2%	98%

Table 4.14 Confusion table with two states for speaker DC using continuous HMM classifier system.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	66%	6%	0%	28%	0%	0%
Disgust	6%	60%	16%	2%	4%	12%
Fear	2%	2%	66%	24%	0%	6%
Happy	10%	20	8%	82%	0%	0%
Neutral	0%	4%	0%	0%	80%	16%
Sadness	0%	0%	0%	0%	4%	96%

Table 4.15 Confusion table with four states for speaker DC using continuous HMM classifier system.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	66%	6%	0%	28%	0%	0%
Disgust	6%	60%	16%	2%	4%	12%
Fear	2%	2%	66%	24%	0%	6%
Happy	10%	20	8%	82%	0%	0%
Neutral	0%	4%	0%	0%	80%	16%
Sadness	0%	0%	0%	0%	4%	96%

Table 4.16 Confusion table with eight states for speaker DC using continuous HMM classifier system.

4.4.2 Comparisons for Number of Components

Now, choose two number of states and set constraint range $\eta = 0.2$, uniformly segmented initial state hypothesis as the settings above. The four components initial model and optimal model for neutral emotion is shown as Fig. 4.23. For the initial model, the gaussian mixture density components are predicted at two position but partially overlapped. After optimization, most components at state one concentrate on left position, some include neighbor. However, the gaussian mixture density at state two include with large area at upper right. This optimal model is as similar as Fig. 4.20 which is shown the two number of components except some components are assigned to more area.

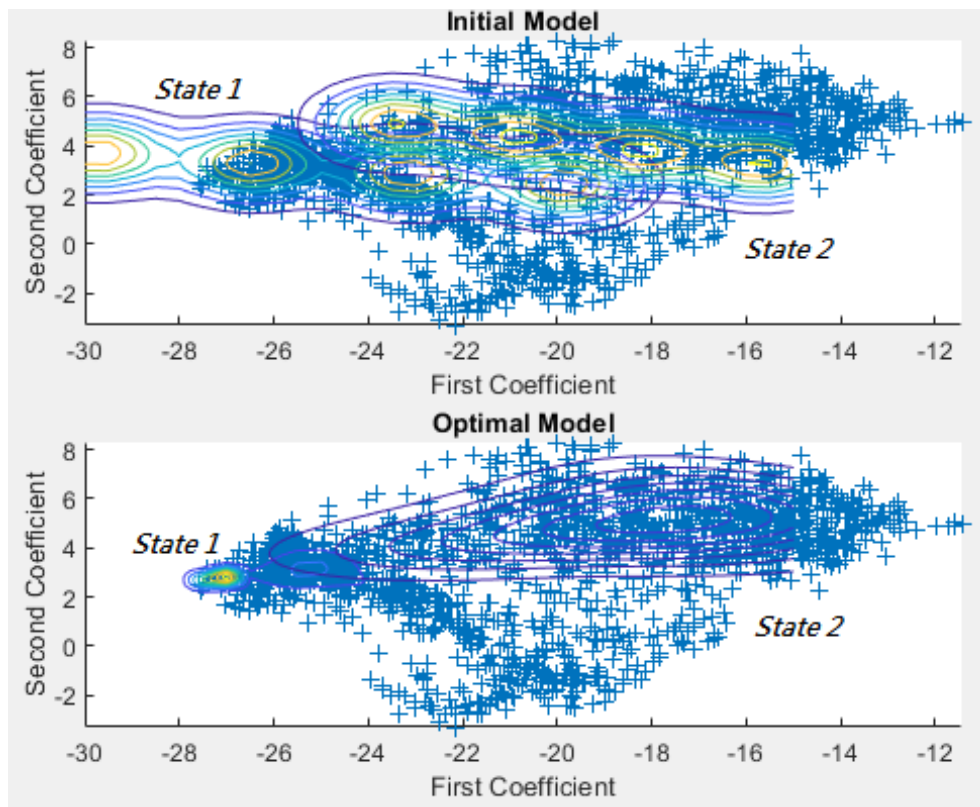


Figure 4.23 The four components initial model and optimal model for neutral emotion.

Following Fig. 4.24 shows the accuracy results comparison for different number of components. Highest accuracy is 84.53% for speaker KL at four components. Highest average accuracy is 66.48% at two components.

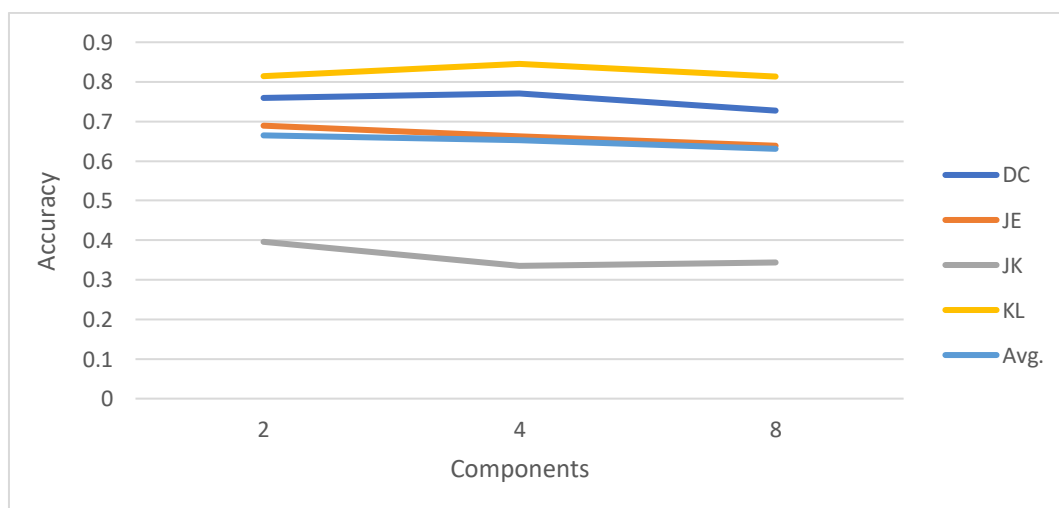


Figure 4.24 Accuracy results for different number of components.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	66%	6%	0%	28%	0%	0%
Disgust	6%	60%	16%	2%	4%	12%
Fear	2%	2%	66%	24%	0%	6%
Happy	10%	20	8%	82%	0%	0%
Neutral	0%	4%	0%	0%	80%	16%
Sadness	0%	0%	0%	0%	4%	96%

Table 4.17 Confusion table with two components for speaker DC using continuous HMM classifier system.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	66%	6%	0%	28%	0%	0%
Disgust	6%	60%	16%	2%	4%	12%
Fear	2%	2%	66%	24%	0%	6%
Happy	10%	20	8%	82%	0%	0%
Neutral	0%	4%	0%	0%	80%	16%
Sadness	0%	0%	0%	0%	4%	96%

Table 4.18 Confusion table with four components for speaker DC using continuous HMM classifier system.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	66%	6%	0%	28%	0%	0%
Disgust	6%	60%	16%	2%	4%	12%
Fear	2%	2%	66%	24%	0%	6%
Happy	10%	20	8%	82%	0%	0%
Neutral	0%	4%	0%	0%	80%	16%
Sadness	0%	0%	0%	0%	4%	96%

Table 4.19 Confusion table with eight components for speaker DC using continuous HMM classifier system.

4.4.3 Comparisons for GMM Constraint

After compared the number of states and number of components. Compare the constraint range at this subsection. Choose two number of states, two number of components and uniformly segmented initial state hypothesis as the settings above. Compare the constraint range η among 0.1, 0.2, 0.5, 0.8.

The two components initial model and optimal model for neutral emotion with $\eta = 0.2, 0.8$ are shown as Fig. 4.25 and Fig. 4.26. Fig. 4.25 shows the initial gaussian mixture density is separated with the constraint range for the initial model. When choose $\eta = 0.8$, the initial gaussian mixture density is separated with the very large constraint range for the initial model. That causes some problem because the initial gaussian mixture density is located far away samples. It produces zero probability for some observations when the gaussian density is too far away. Some weights of component become very small that almost lose the component and affects the optimization. Moreover, according to [23] [29], bad initial hypothesis makes bad optimization due to the property of EM algorithm.

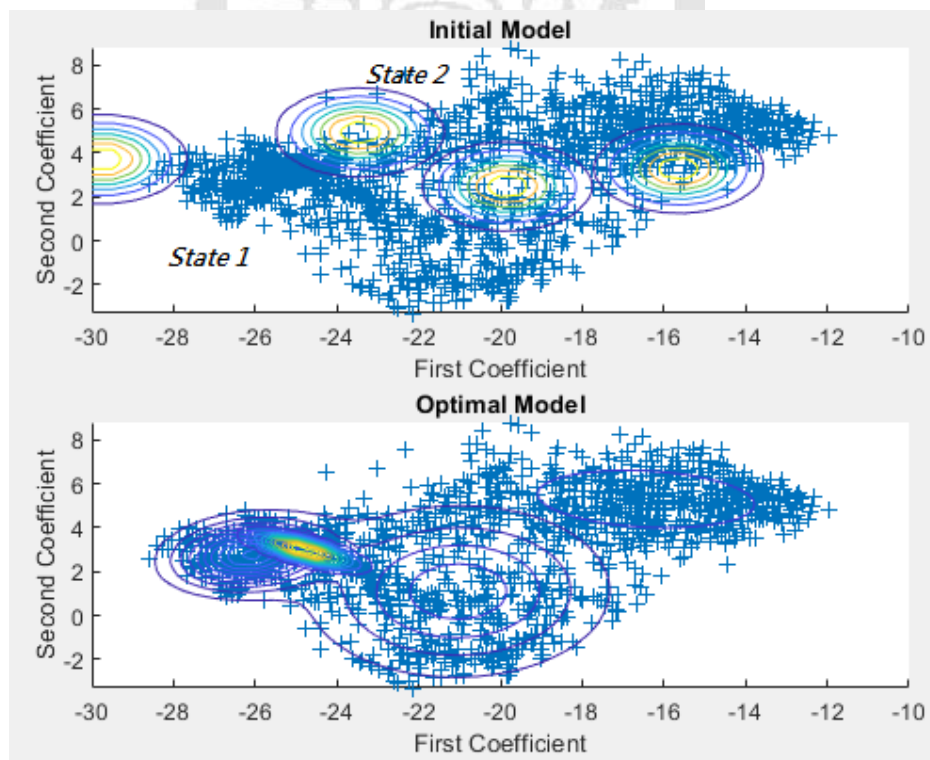


Figure 4.25 The two components initial model and optimal model for neutral emotion with $\eta = 0.2$.

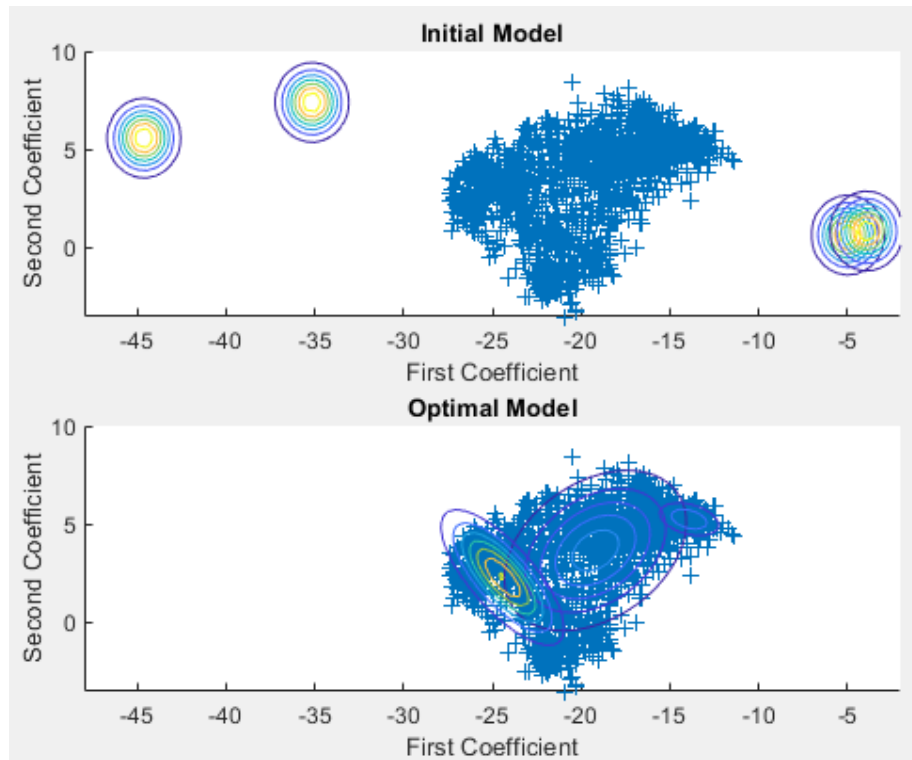


Figure 4.26 The two components initial model and optimal model for neutral emotion with $\eta = 0.8$.

Then, show the accuracy results comparison for constraint range η among 0.1, 0.2, 0.5, 0.8 as Fig. 4.27. Highest accuracy is 84.87% for speaker KL at $\eta = 0.1$. Highest average accuracy is 69.25% at $\eta = 0.1$. It should see the $\eta = 0.1$ is the best among them, for this range the initial model more fits the samples, so better the optimal model. The results support the argument that a good initial model makes better performance.

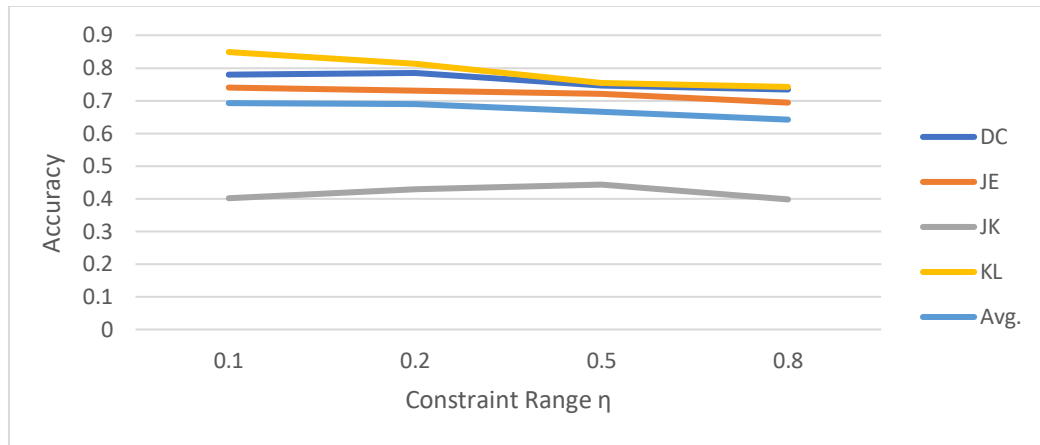


Figure 4.27 Accuracy results for different constraint range η among 0.1, 0.2, 0.5, 0.8.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	74%	10%	2.5%	13.5%	0%	0%
Disgust	10%	64%	10%	6%	0%	10%
Fear	0%	2.5%	60%	30%	0%	7.5%
Happy	4%	0%	20%	76%	0%	0%
Neutral	0%	0%	0%	0%	96%	4%
Sadness	0%	2%	0%	0%	0%	98%

Table 4.20 Confusion table with constraint range $\eta = 0.1$ for speaker DC using continuous HMM classifier system.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	68%	8%	2%	22%	0%	0%
Disgust	12%	66%	10%	0%	0%	12%
Fear	2%	2%	62%	24%	0%	10%
Happy	4%	0%	14%	82%	0%	0%
Neutral	0%	0%	0%	0%	92.8%	7.2%
Sadness	0%	0%	0%	0%	0%	100%

Table 4.21 Confusion table with constraint range $\eta = 0.2$ for speaker DC using continuous HMM classifier system.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	86%	0%	2%	12%	0%	0%
Disgust	6%	74%	6%	8%	0%	6%
Fear	8%	4%	50%	36%	0%	2%
Happy	26%	0%	16%	58%	0%	0%
Neutral	0%	2.4%	0%	0%	85.6%	12%
Sadness	0%	2%	2%	0%	0%	94%

Table 4.22 Confusion table with constraint range $\eta = 0.5$ for speaker DC using continuous HMM classifier system.

Emotion	Angry	Disgust	Fear	Happy	Neutral	Sadness
Angry	64%	8%	8%	20%	0%	0%
Disgust	0%	90%	4%	0%	0%	6%
Fear	2%	14%	54%	28%	0%	2%
Happy	22%	6%	20%	50%	0%	2%
Neutral	0%	0%	0%	0%	86.4%	13.6%
Sadness	0%	4%	0%	0%	0%	96%

Table 4.23 Confusion table with constraint range $\eta = 0.8$ for speaker DC using continuous HMM classifier system.

4.4.4 Comparisons for Initial State Hypothesis

At last, choose two number of states, two number of components, constraint range $\eta = 0.2$, compare different initial state hypothesis approaches which are uniformly segmented initial state hypothesis and clustering method initial state hypothesis.

Before showing the accuracy results, Fig. 4.28 and Fig. 4.29 show the initial and

optimal model with different initial state hypothesis approaches, four states for neutral emotion. Note that there are eight circles because of two components for each state. Fig. 4.29 shows that it is very different from the uniformly segmented approach, the initial gaussian mixture density are separated dispersedly and the optimal gaussian mixture density is more concentrated on the position which the observation occurred frequently, that means the optimal model is more fit the training samples for the clustering method initial state hypothesis approach.

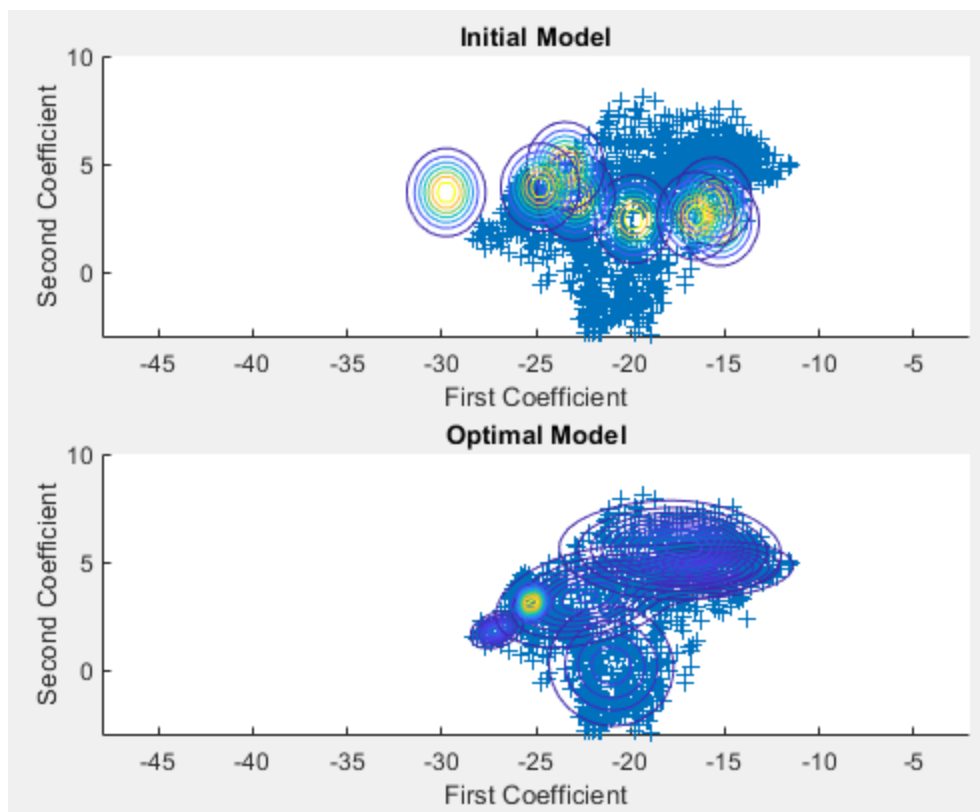


Figure 4.28 The four states, two components initial model and optimal model for neutral emotion with uniformly segmented initial state hypothesis.

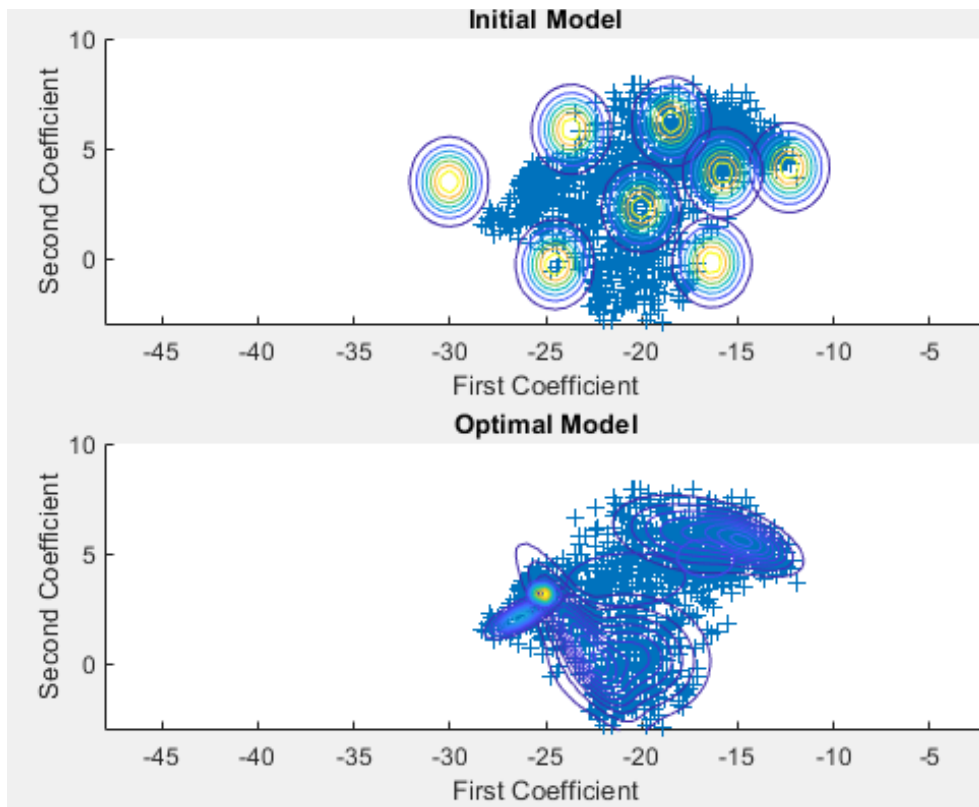


Figure 4.29 The four states, two components initial model and optimal model for neutral emotion with clustering method initial state hypothesis.

So, the reason for initial state hypothesis is according to the property of EM algorithm. For this accuracy results comparison, there adds additional experiment results with a bad initial state hypothesis, that we set first state at the mean of the beginning of utterance samples, but another state set as the times of the state vector. The last state would be far away from first state. Fig. 4.30 to Fig. 4.32 show the accuracy results comparison for different initial state hypothesis approaches and the bad hypothesis with two, four, eight number of states.

For two states, the highest average accuracy is 69.7% with the bad hypothesis, which slightly higher than clustering hypothesis about 0.6%. For four states, the highest average accuracy is 68.02% with the uniform segmented hypothesis, which is very close to clustering

method hypothesis. But average accuracy drops off to 58.6% for the bad hypothesis. Finally, the highest average accuracy is 69.35% with the clustering hypothesis. The average accuracy is just 15.7% for the bad hypothesis, the optimal model is almost trained to unrecognizable due to the bad initial model.



Figure 4.30 Accuracy results comparison for different initial state hypothesis approaches with two states.

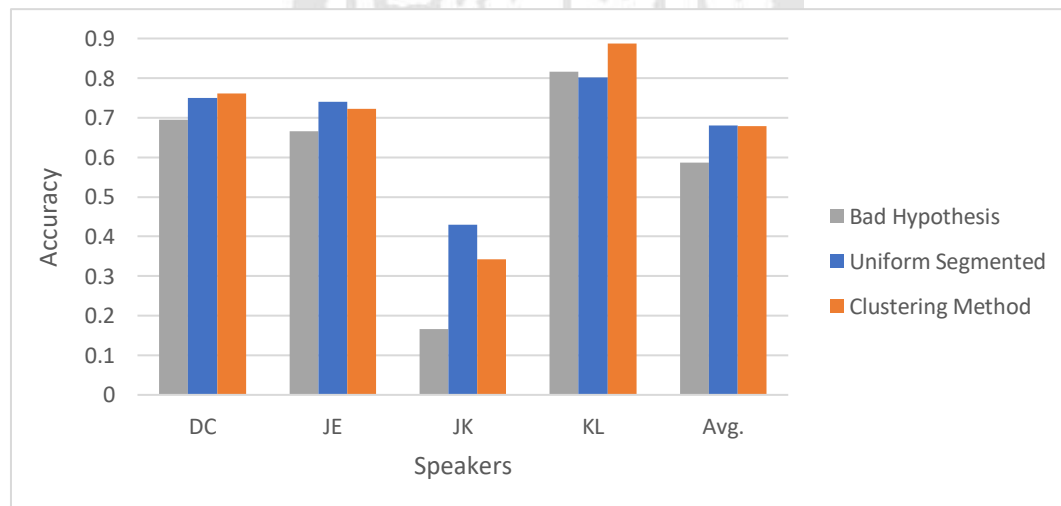


Figure 4.31 Accuracy results comparison for different initial state hypothesis approaches with four states.

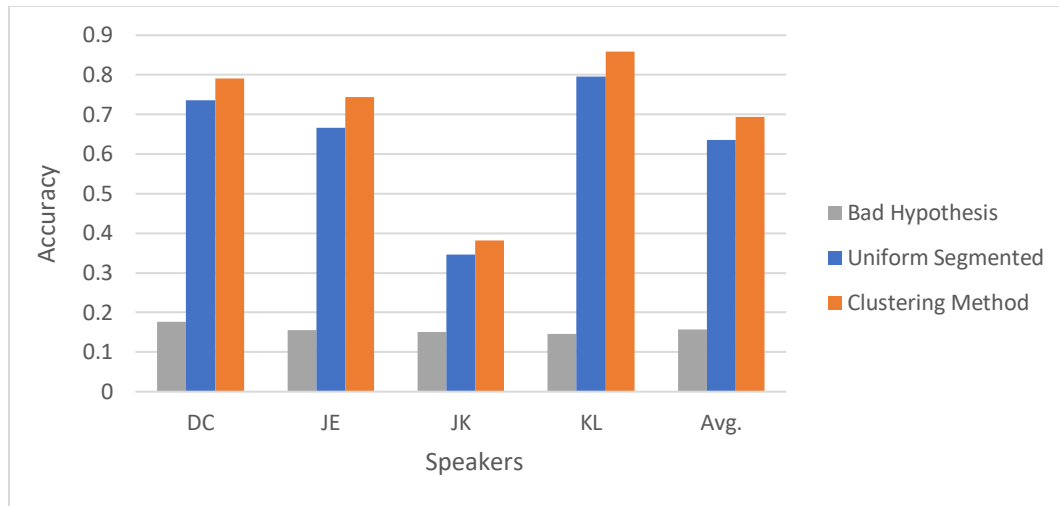


Figure 4.32 Accuracy results comparison for different initial state hypothesis approaches with eight states.

4.5 Comparisons for Different Systems with A Good Settings

In this section, compare all the recognition system that we interest, GMM classifier system, discrete HMM classifier system, continuous HMM classifier system with two different initial state hypothesis approaches. From now on, we have compared and discussed all the systems that we interest with the parameter settings, so we show the comparison among the systems with the good settings and show the best results as Fig. 4.33 shown. The best result of all settings reaches 89.20% for the continuous HMM with clustering method initial state hypothesis and set two states, eight components. The best average result of all settings is 70.15% for the continuous HMM with clustering method initial state hypothesis and also set two states, eight components. The proposed continuous HMM classifier system with two initial state hypothesis approaches are enhanced about 20% for the best result, about 15% for best average result.

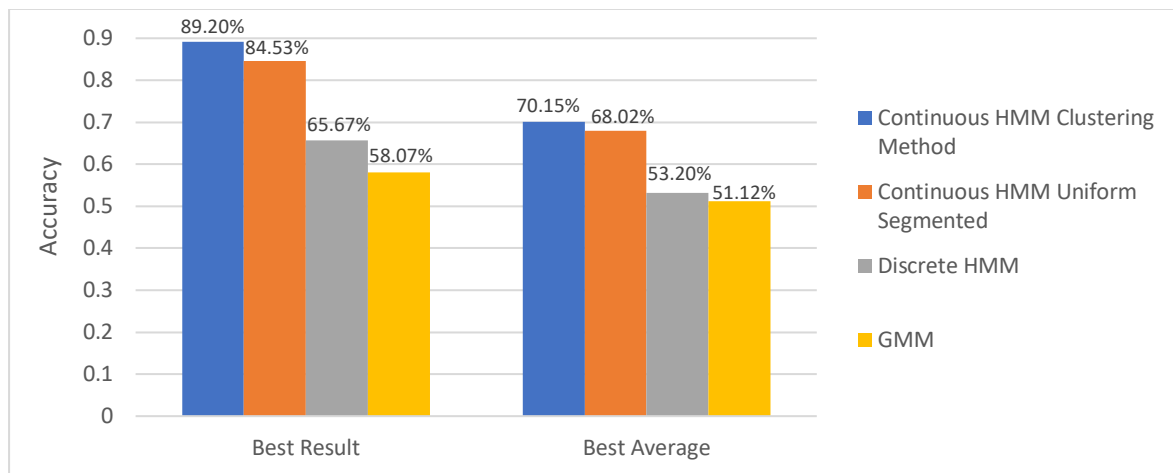


Figure 4.33 Accuracy results comparison among all interest recognition systems with a good setting.



Chapter 5

CONCLUSIONS AND FUTURE WORKS

This thesis compares the accuracy results of emotion recognition from speech signal among three different classifiers including GMM, discrete HMM and continuous HMM, shows the solutions to overcome underflow and singularity problems in those classifiers and proposes the initial state hypothesis approaches for the emotion recognition. The experimental results show that the accuracy using continuous HMM is the best among those classifiers, it supports the argument that it is better while considering temporary local features than global statistical features. The experimental results also show that using initial state hypothesis prior to training is a better idea which is figured out from the property of EM algorithm. The best accuracy result reaches 70.15% which outperforms discrete HMM classifier about 15%, GMM classifier about 18%.

However, there are still some research directions can be achieved. First, although the underflow problem is overcome completely by introducing scaling coefficients, the singularity problem is not solved completely in this work. To this problem, the deep belief network (DBN) has the potential to solve this problem even though the features are almost stationary. Second, there is a problem while using the Lloyd algorithm method for hypothesis of the states of HMM classifier. If there is any observation vector which is far from any center of cluster, i.e. hypothesized state, that means it doesn't hypothesize the particular observation vector. Third, an important research direction is the choice of the number of parameters setting. Although we have indicated that the model is trained to not good when the gaussian components in initial model are too far away the samples no matter too wider

perturbation range or too many states, the criterion of number of parameters should set clearer according to the collection of training samples. Another direction is at the front-end stage. The MFCCs are not a perfect speech emotion feature because it consists too many linguistic information and it is not robust to noise affections. The feature extracted by wavelets such as sub-band based cepstral coefficients (SBC) is preferred well.



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